

PET

Positron Emission Tomography, or PET, is an imaging technique that allows us to see how organs and tissues function inside the living body. It is based on the use of a radioactive substance, called a tracer, which emits tiny particles known as positrons. These tracers are usually molecules that participate in normal biological processes—such as glucose—so that their behavior in the body reflects real metabolic activity.

When the tracer is injected into the bloodstream, it travels to different parts of the body, accumulating in regions according to how active those tissues are. For example, in the case of fluorodeoxyglucose labelled with fluorine-18, areas that consume more glucose, such as the brain or a tumor, will collect more of the tracer.

After a short time, the radioactive atoms within the tracer begin to decay. Each decay releases a positron, which travels only a very short distance before it encounters an electron in the surrounding tissue. When a positron and an electron meet, they destroy one another in an event called annihilation, producing two photons of light that fly away in almost exactly opposite directions.

The PET scanner surrounds the patient with a circular array of detectors capable of sensing these pairs of photons. The system is designed to register only those pairs that arrive at two detectors at the same instant. When such a simultaneous detection occurs, the scanner knows that the annihilation—and therefore the source of the photons—must have been located somewhere along the straight line connecting those two detectors. By collecting millions of these coincidence events from all around the body, the computer reconstructs a detailed three-dimensional image (made of voxels) showing where the tracer is concentrated.

The brightness of each region in a PET image corresponds to how many annihilation events were detected there, which reflects the local biological activity of the tissue. Because of this, PET does not simply show anatomy—it reveals function: how cells are using energy, how blood is flowing, or how receptors are binding molecules.

Lesson 08/10/2025

With PET (positron emission tomography) we are able to get very specific and sensitive information, specifically targeting some compound of interest. In terms of brain activity, we can gather information about oxygen and glucose consumption, or blood flow supply or markers of synaptic activity. We can measure also other chemical compounds related to other chemical targets like neurotransmitters, not present in the brain tissue. This is possible knowing the metabolic fate of a tracer (^{18}F -Deoxyglucose).

PET imaging is important in determining neurodegenerative disorders. A possible valuation is a qualitative one, we primarily rely on static images, correlated to the concentration of a tracer, that is the number of annihilations. If we take in consideration glucose consumption, we can specifically consider this as a marker of neurons activity. ^{18}F -Deoxyglucose is better to use than anything else, for example water, because it is strictly related to the neuron, so that we recognize neurodegenerative disorders.

One of the painful aspects of dealing with the metabolic fate of the tracer is that these molecules are processed in different ways from different tissues. The brain takes about 45 minutes in order to have glucose equally distributed in all components, and for that time you must keep the brain

in the same mental state. Indeed, we can consider that as something that does not facilitate the investigation of mental function, whose time is much faster. Some tasks are very difficult to maintain for 45 minutes.

When you deal with aspects of synaptic activity that you like to refer to mental processes that are faster than the ones that you want to investigate, you can rely on markers that primarily are related to changes in blood flow supply, whose activity is much faster.

Example: H_2^{15}O half-life is 2 minutes, so in 10-15 minutes the activity of oxygen is disappeared, so you can monitor changes in blood flow that occur in few seconds. In some cases, oxygenated blood reaches a peak in the brain, that is compatible with several neural functions. Changes in blood flow supply are tightly coupled in changes in glucose and oxygen consumption (synaptic activities), so it can be used to represent synaptic activity, but you use tracers with different metabolic fate time.

When I'm dealing with an asymmetric distribution of glucose consumption and I'd like to compare these, you need to have absolute values for each of the voxels. Once you know the amount of dose that you have injected and the body mass of the participant, then you are able to reconstruct the curve, interpolating the values. Therefore, you are able to express Deoxyglucose as mg/100g tissue/minute (generally the range of value is 10-15 mg) or ml/100g tissue/minute (generally 50-70 ml).

Once you have the absolute value, we have the parametric or dynamic image, and at that point we can do any statistical operation with these images. When you take several volumes, you are trying to do an interpolation, so that you can have images to overlap. An important step is the spatial normalization, so that each we can move our brain into a standard reference state and have the possibility to compare different realizations (for example from different people). Each point is characterized by 3 coordinates. We use ATLAS (space of coordinates of reference), the first was the Talairach, obtained cutting the brain in slices and for many years it was the main reference for all brain studies.

In functional neuroimaging studies, cognitive subtraction refers to an aspect of experimental design involving the comparison of two conditions or brain states that are presumed to differ in only one discrete feature (the independent variable). Pure insertion asserts that there are no interactions among the cognitive components of a task

Donders' subtractive Methods

The main approach that we got derives from experimental psychology, that grew up from subtraction approaches.

Donders' work attempts to describe the processes going on in the mind, by analyzing cognitive activity into separate stages (many scientists assumed that responding to a stimulus occurred instantaneously!)

Donders was particularly interested in "timing the mind" and used a subtraction technique to time the different mental processes that the brain goes through when faced with different tasks.

Donders performed experiments using reaction time tasks in 1868:

1. a simple reaction time task - e.g. you are seated in front of a panel that contains a light bulb and a response button. When the light comes on, you must press the button. 190 ms

2. a discrimination reaction time task - e.g. you are seated in front of a panel with 5 light bulbs and one response button. When the target light goes on you must press the button - but not if the 4 other lights come on. 240 ms

These 50 ms of difference is the time that the mental function needs to discriminate the target.

3. a choice reaction time task - e.g. you are seated in front of 5 light bulbs, each with their own button. You must press the button corresponding to the appropriate light. 280 ms

Together with discrimination you have to select the correct motor action. The idea is that as soon as you add pieces, your reaction time is longer and you can build up experiments that allow you to add piece, that you can subtract from another condition.

Example: For visual stimulation, eyes open and white light or control condition and eyes closed. The only difference is the perception of the white light. You subtract statistically the PET images, and you obtain a group average response, and the mean difference image is something that tells you that perception of the white light relies on changes in activity in the occipital cortex.

Example: Chess as a case of study. Why chess?

- Chess problems are complex.
- The ability of the chess player is measurable.
- There are multiple studies in which the cognitive psychology of chess is studied.

In this study they decided to split the chess game in 4 different mental stages that we could compare:

- B/W discrimination.

Control baseline condition. The idea was to define specifically what the brain is doing during the tasks. (Visual perception, discrimination of colors, reading, recalling motor command, decision, ocular movement, thumb movement).

- Spatial discrimination (is the closest piece to the X white?).

All the things we said before but with an additional condition, spatial analysis of distance. (Spatial attention, static spatial analysis). Subtracting the PET images, we are able to see the different areas of the brain working in the new task. (Precuneus, occipital lobe, superior parietal lobule, medium temporal gyrus, superior premotor cortex).

- Rules (can the white queen catch the black pawn).

You have to discriminate the value of the pieces and recall the rules, so you must activate the memory. (Value discrimination, rules recall, dynamic spatial analysis). The areas of the brain working are different (left hippocamp, left temporal lobe, left superior temporal gyrus, left medium temporal gyrus, left fusiform gyrus, right post-central gyrus, cerebellum).

- Checkmate (can the white player win in one move).

(Recall of the game rules for all the pieces, consequence valuation for every move, generation of moves sequence). This was a 1994 study.

Example: Four different experimental conditions, in which order we should make the patient do the different tasks.

- Visual
- Spatial
- Rules (memory)
- Strategy

The first thing we would say is to make him do them in this exact order, but we have to take in consideration the increase of difficulty for the different tasks. How can we discriminate between the areas of the brain that are working trying to solve a more difficult task (involving emotional harassment) and the ones specifically working for the task itself. How to manage task effect? They also learn to do the tasks, so the tasks become easier and brain response is somehow reduced, or you can become tired and habituated. We can't present them in casual order because we would have to compare different patients that had the tasks presented in different order.

You teach the patients the tasks, until they reach maybe 90% accuracy, and then you can have your experiment putting them in the machine. In this way you avoid training effect.

Face perception

Example (How to build a PET imaging experiment)

Which are the variables that we have to take into account to determine our patients?

- Gender (sexual orientation)
- Ethnicity (cross-race effect)
- Age
- Education
- Health status
- Drug-free
- Psychometric characterization
- Possible selection bias (mostly students)
- Sampling methodology
- Calculating sample size

Cross-race effect: when we are asked to recognize the identity of people that we are not familiar with (Chinese, afro Americans), our performance is decreased in respect to recognizing Caucasians for example.

Experimental paradigm (How can we define the stimuli?)

The stimuli allow to study what is the GESTALT, psychology that focuses on how humans naturally perceive patterns and organize sensory information into meaningful wholes, rather than just as collections of separate parts.

One element is important; we have to discriminate low-level features of vision (the first kinds of information that our visual system extracts from the raw input that hits the retina). You want to have a stimulus that allows you to compare two stimuli with the same features (contrast, luminosity), so that the only difference is only the lack of a face.

We can have different regions of the brain involved with respect to the stimuli that is presented:

- Faces vs houses → Semantic
- Active vs passive → Attention
- Picture vs scrambled → Mid-level

Scrambled: a visual stimulus that has been disorganized meaning its low-level features (like color, brightness, or texture) are preserved, but its meaningful structure or recognizable shape is destroyed.

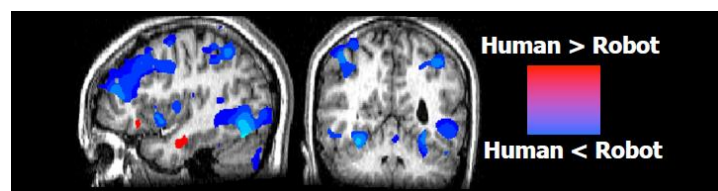
Everything that is in common is specific to face recognition (junction analysis), in this case the fusiform (one of the gyrus of the ventral surface of the temporal lobe) face area. One thing that you want to be aware of is that there may be something that biases the representation of faces, difference in contrast or luminosity, the brain catches low-level features of vision.

Pharmacokinetics is how the drug enters the body, the pharmacodynamics is different, what receptor the drug reaches or which neurotransmitter it stimulates. Serotonin receptors are primarily located in the pre-frontal cortex; dopamine is primarily present in the striate cortex and Parkinson disease is characterized by the reduction of dopamine levels (detectable with PET images).

Perception and sensation

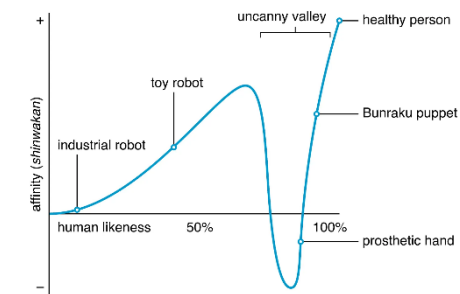
In the years we were able to see how visual perception affects the brain, how the occipital works.

It's important to study face perception not only in humans, but also in machines. An example is the project of Takanishi Laboratory.



People have been working hard trying to have precise details, but we struggle to say that this could be a face. Also, our brain reacts in different ways, there are two regions (one in the temporal lobe and one in the frontal cortex), associated with social cognition and interaction.

The Uncanny Valley is a phenomenon where something that looks almost human, but not quite perfectly human, evokes feelings of unease, discomfort, or eeriness in observers.

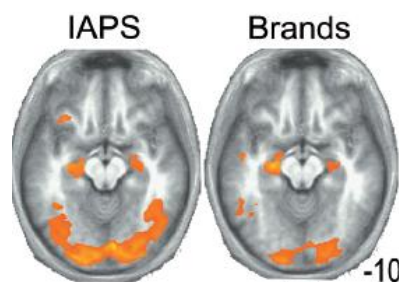


Source: Masahiro Mori, "The Uncanny Valley," *IEEE Robotics & Automation Magazine*, 19(2):98-100 (June 6, 2012).

Uncanny Valley phenomenon curve.

The International Affective Picture System (IAPS) is a database of pictures designed to provide a standardized set of pictures for studying emotion and attention that has been widely used in psychological research. They got images and asked how positive or negative these scores are.

Each picture was shown every 2 seconds (positive, negative and brand logos), what people was asked to do was to pay attention to the stimuli and then press the orange button (Oddball task). Looking at the brain's images, we see the visual perception regions activated, but also the amygdala, labelled as emotional sentinel. Even when you recognize something that you like, like a logo, your brain activates a pleasant score as a reward.



Perception should get piece of information just to reproduce and to survive, not to get a replica of the world. For instance, the frog has a great ability to discriminate between different tonalities of few colors (blue and green), but is totally blind to others, because they live in pods. Sensory modalities evolved to guide animal behavioral choices, necessary to reproduce/survive.

Our brain has to alter the kind of perception (change the colors or the proximity), just like top-down modulation, that tells the brain to focus and adapt to the perception.

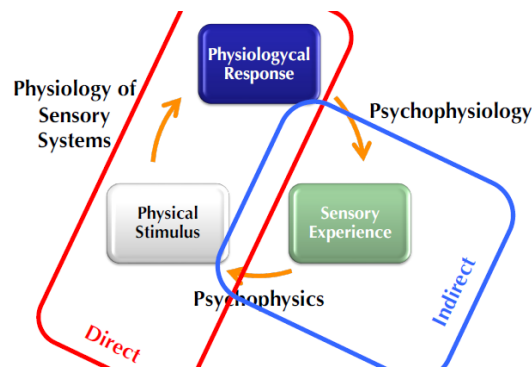
- Top-down perception is when your brain uses what you already know — your experiences, expectations, and context — to understand what you sense.
- Bottom-up perception is when understanding starts with the raw information from your senses, and your brain builds meaning from that data.

The difference between the way we interpret, and elaborate information is what we refer to as perception.

The way we see and represent colors is individual, but the wavelength of the color is the same for everybody. This is telling us that we can share some of this subjective representation but it remains different for each of us.

We have more than five senses, but all of them share 3 common stats, there is a physical stimulus that arrives, a second event transforms this stimulus into a nerve impulse (action potential), and

then there is a response in the form of a sensation or conscious experience of sensation. These are 3 steps that are acknowledged.



Early findings in psychophysics and sensory physiology exposed a weakness in the empiricist argument: the mind is not a tabula rasa. Our perceptions differ qualitatively from physical properties — colours, tones, and smells are mental creations constructed by the brain from sensory experience.

Nonetheless, our perceptions correspond to the physical properties of stimuli. Knowledge, therefore, is based not simply on sensory experience but also on the brain's inherent properties that organize that experience — echoing Kant's idea of a priori knowledge.

Since the first studies each stimulus could be characterized with 4 features:

- **Modality:** general class of stimulus, determined by the type of energy transmitted by the stimulus (light, heat), and it is defined by the receptors that capture it.
- **Location:** the set of sensory receptors that are active, and that are actually telling me that specific energy derives from a certain source.
- **Intensity:** the response amplitude, primarily related to the stimulus energy and the transformation of this energy into nervous impulses. The higher energy the higher the nervous transduction is.
- **Timing:** when stimulation starts and stops.

In each sensory system the initial contact with the external world occurs to specialize structures that we call receptors. The sensory receptors are the first cells of the sensory process that transform the stimulus energy into an electrical signal, this is called stimulus transduction. Each receptor specifically responds to one form of energy.

All receptors have different receiving fields, smaller or broader, determining the ability (together with the receptor density) of discriminating two stimuli and separate in space and time. The density of receptors defines the resolution across different body parts (fingertips or knee).

This type of labelled code line is not something that is equal across all different sensory modalities, there is also that all signal from sensors has to reach the brain, and the detail of this action potential tell the brain which type of energy was received, the timing and the location. All sensor modalities have central processing, from the thalamus to the cortex. All the information we gather from the thalami is called primary sensory areas. Only olfaction does not pass through the thalamus before reaching the cortex.

We are talking about sensation, that comes from the periphery, able to create a population code, a sequence of connection for example along the skin and the spinal cord, or the retina for the

eyes, a hierarchical organization and extraction of information to convey to the brain and then process the information.

Several factors operate to shape and sometimes distort perception. These factors can reside in the perceiver, in the object or target being perceived, or in the context of the situation in which the perception is made. This happens for example at the level of the retina for some type of illusions.

We see things as we are, and this is what we really call perception. It's not just the process of receiving the information but also making sense of this piece of information, this is how experience works. Perception can be influenced by several things, like physiological state (food when I'm hungry) or the context (crosta o crumble di pane). It's plenty of things that can alter our perception of things (just like blue risotto).

Organization is the process of placing selected stimuli in a framework, providing categorical recognition.

The f-MRI

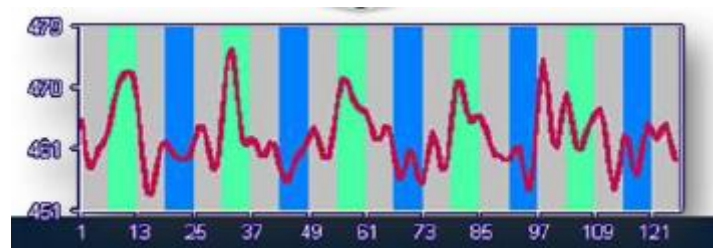
MRI studies brain anatomy, f-MRI studies the functioning. MRI is primarily using one image with high resolution, for f-MRI we acquire different images in time (every 2 sec for 5 minutes) with low resolution. Comparing it with PET imaging we are adding the temporal component.

Nuclear magnetic resonance is based the electromagnetic stimulation that deviates, flips the proton, and once the stimulus is over you let it realign with the magnetic field with a specific relaxation time, that influences different aspects.

We are dealing with endogenous contrast, the signal that we are measuring is the BOLD signal (**Blood Oxygenation Level Dependent signal**). When you have stimulation, you have a higher consumption of glucose and higher blood flow, but the flux of oxygen is not proportional to the amount of oxygen that the brain receives (luxury perfusion, it arrives more blood than needed).

Oxygen and carbon dioxide are bonded to one protein in the red blood cells, oxyhemoglobin, that has different magnetic properties, is slightly diamagnetic (attractive) while deoxyhemoglobin is paramagnetic. So, you have an indigenous tracer that has two different behaviors, and the T2 (relative to the interaction with hemoglobin) is reducing the signal when there is a higher concentration of deoxyhemoglobin (slightly attractive), so that in resting condition you have an unbalance between the arterial side and the venous side.

When I have a stimulus that activates the brain, I have a lot of oxygenated blood (more blood flow than oxygen consumption), and while in the arterial side there are no changes in the venous side something happens. This is because some oxygenated blood gets into the venous side, and becomes more diamagnetic, increasing the signal also in this side.



Cleaned signal in which we can see the stimulus.

BOLD signal is 12000-15000 ms, not related to action potential. The phenomenon I want to record is delayed by 4-6 seconds in respect to the moment the phenomenon occurs, because you must wait for the blood flow.

While in PET (8-10 volumes per subject) you are creating a difference between two responses (higher or lower) to some stimulus, in f-MRI (800-1000 volumes per subject) we are contrasting two temporal correlations to the stimulus you are providing. 2-3 ml, so higher number of voxels, higher number of information we can process.

There can be different strategies for neural recruitment. You may have very solid responses (block responses) for instance, but after few seconds you get bored, so you can play with different intervals. Example:

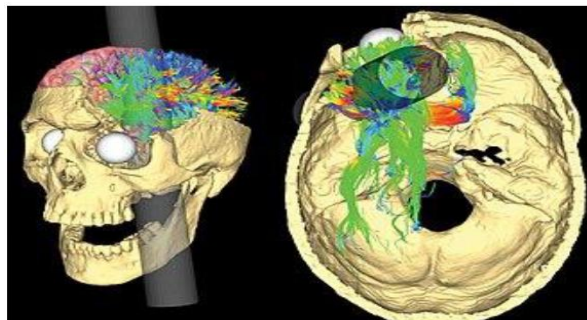
- Yellow → Faces
- Blu → Fruits



Lesson 15/10/2025 (The study of the brain and brain function)

We have many different techniques able to study the brain. When we study humans we use non-invasive ones, always considering that we are dealing with patients, the most used are the electroencephalography (the cost depends on the # of electrodes, max 256, generally cheap) and f-MRI, they are the cheapest and most accessible.

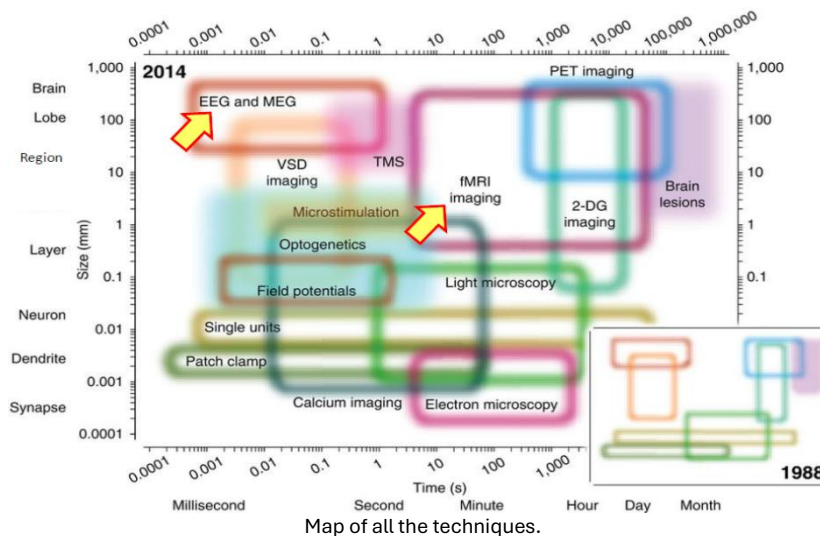
Example: The case of Phineas Gage. Lesion in the frontal cortex and a critical change in the behavior (no control and rage problem).



The studies on brain lesions are not good in general.

- Brain lesions are rarely selective, when you have a brain lesion it often involves different parts of the brain at the same time (inflammatory reactions, edema around the lesion, affect other lesions).
- Symptoms can change with time, after the damage you can recover, the edema decreases, the plastic processes that take place in the brain allow it.
- Sometimes the deficit is not identifiable: there are patients that have damage in the brain and can lose the ability to perceive colors, but depending on the type of lesion that they have they may not realize it.

This is why brain imaging became so important. We can get a more precise idea of what happened in the brain without relying on lesions and we can use some manipulations of behavior or of the experimental conditions to see what changes in the brain of the participants.



In the y axis you have the spatial resolution of the imaging technique, in the x axis the temporal one. All the techniques in the bottom part of the y axis are not accessible to humans, only animals. In humans you can't have access to the individual neuron. Most of our techniques allow us to see in the level of regions (centimeters/millimeters), or the level of the brain overall and of the lobes.

EEG and MEG have comparable temporal and spatial resolution, with a very good temporal resolution (order of ms, detect neural activity real time), but they have very bad spatial resolution. You can't see where the changes come from.

f-MRI has a much better spatial resolution, does not track neuronal and brain activity, only blood flow, a consequence of brain activity. If you want, you can apport changes in oxygen and in blood flow that are not related to brain activity. In addition, the main component that causes neurovascular coupling is the astrocyte, not the neuron. This imaging technique has a low temporal resolution, you acquire an image of the brain in 2-3 seconds (really slow), good to get an idea of which regions responds to a certain task, but not to discover for example which region responds first.

If you want to have something that interacts with the patient you can use what is called neurofeedback. A type of protocol (software, program) which changes the stimulus in the participant based on the changes in brain activity (used to control a robotic hand, not possible with f-MRI with the temporal limitations).

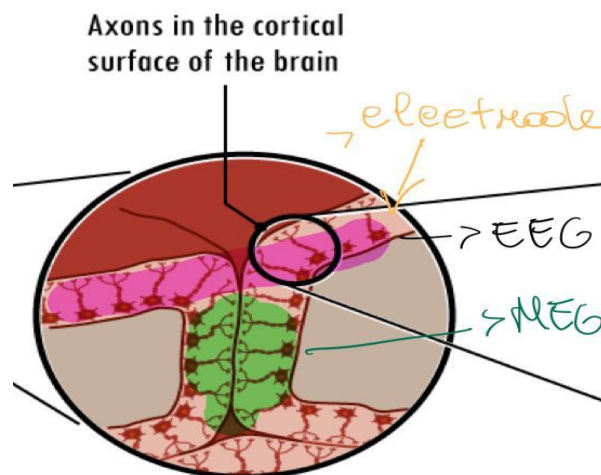
The EEG is different, the main point is that you are looking at the neural activity, you see changes related to pyramidal neurons activation, perpendicular to the surface of the cortex. The difference in potential related to asynchronous action potential is measured on the scalp. But the important thing about the EEG is that the electrical activity is volume conducted, you get a potential that is a difference between the body of the cell and the neurites. It represents a problem, because the electrical activity propagates through the tissue in all directions. Volume conduction:

- Good because regardless of where the activity is generated some electrons on the scalps will be able to get it, propagating through the tissues to the scalp.

- Bad because you get dispersion through the path and you have to amplify the signal.

The difference between MEG and EEG is that the last one is based on the detection of a signal that propagates in the same way as for the source, meaning that the signal that we detect is produced by the pyramidal neurons. For the MEG you are looking at the magnetic field produced by the current, and the magnetic field is perpendicular to the current that is produced, so if the neurons are perpendicular to the surface of the scalp, then the magnetic field will be directed in the perpendicular way, which means that the sensors that are exactly above the scalp will not detect the signal produced by that neurons.

MEG and EEG are related because they exploit the same kind of generator, but they could not necessarily produce the same signal, because they look at slightly different perspectives. EEG is a little better, because it can detect all the electrical activity produced by all the neurons.



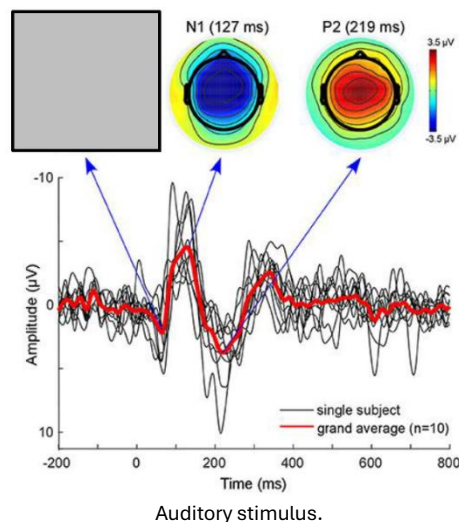
Example of signal detection, MEG can detect only the signals produced by sulcus neurons. EEG can detect all.

Some people prefer MEG, this because the spatial resolution is slightly better, because the signal is not distorted or propagated it is more precise. True based on models but not so much in reality. MEG is accessible by very few laboratories, less portable and less versatile in the use of the machine itself.

Different numbers of channels based on the use of the machine. For clinical views the laboratories use up to 21 electrodes, a number that is sufficient, but eventually for commercial EEG also only 7 sensors placed on the forehead. The problem with the EEG is that the contacts must have good contact with the scalps, you need lowest possible impedance (conductive gel, even 1h 30 just to prepare the subject), with 256 electrodes it could be difficult for a patient in an experiment. The lack of hair does not improve the conductivity but lowers it, because the scalps become thicker. The boundary between what is called standard EEG and High-Density EEG is the 64 electrodes configuration.

There could be also a net, immersed in water and soap, but also gel can be replaced with sponges. The subject in this way can be wet and once the sponge dries up the signal is lost. It's a good solution depending on what you do, there are cases where you have to be fast (children, people with disorders). Dry electrodes exist but their signal is very bad, noisy, good for piloting and testing, research is going towards implanted electrodes (many patients don't want the implant).

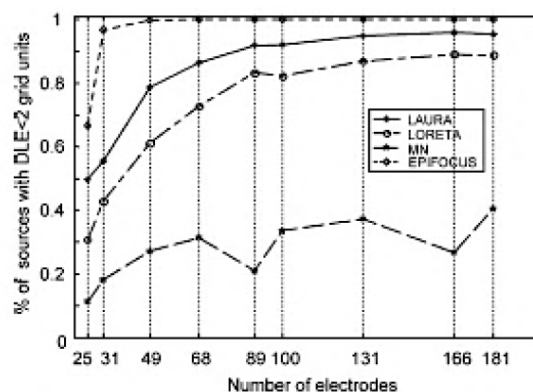
The EEG has bad spatial resolution, but very good temporal resolution.



We are not able to know where the activity started, we know what type of signal it is based on the form of it. We rely on the electrodes positioning and then a sum of the different signals.

The spatial resolution is constraint with some factor; one is the algorithm, reconstruction approaches applied can have slightly different assumptions and performances. You also need two other things, the MRI of the participant, to reconstruct the brain anatomy of the patient and have a segmentation of the tissues, to get a very accurate map of how the neural activity propagates in the brain. The second thing is that you need the exact position of the electrodes on the scalp, and then you have to align the electrodes position with the MRI. With these information's we can get a very accurate model of the brain, and consequently a great spatial resolution comparable to the one of MRI (mm).

The gain of having many electrodes, just like in the high-density EEG, is that if you want source reconstruction and you have a large coverage of the brain you have a more accurate reconstruction.



Accuracy of localization of a focus epilepticus (using source modelling)

We can see that error in source localization decreases with the addition of electrodes on the scalp. We don't use more than 256 electrodes because the time preparing the patients critically increases and the more electrodes you have there is the possibility that an electrode making contact with the gel of another one can create a bridge and the current would be the same for the two (you end up with a lower spatial resolution).

When you choose a technique to study the brain you need to think of what you want to know (spatial distribution of the processes in the brain?). EEG is cheaper and more accessible, also used in movable conditions even though there are a lot of artifacts. The main problem is that the subject is alive. Eventually you can program the use of a machine (ex. drone) relying on the eyes or muscle signal artifacts, not on the brain.

A lot of people prefer f-MRI, but the problem is what you need to get. This type of imaging technique is very fancy, if you need high temporal resolution you can't rely on it. Typically to have a more informative view you have to merge different techniques, also transcranial magnetic stimulation. There is no perfect technique, it all depends on what you want to get.

There is a position (near corona) in which there are no muscles, so no artifacts, but the speed of reaction is very low. With every heartbeat the EEG becomes very noisy, this is because the brain (the head) slightly moves inside the scanner, and the magnetic field produces a current in the EEG system.

You can get neural navigation (jump in the dark, you don't know where you are stimulating) by integrating EEG and TMS, getting the effect of the stimulation inducted, through the EEG.

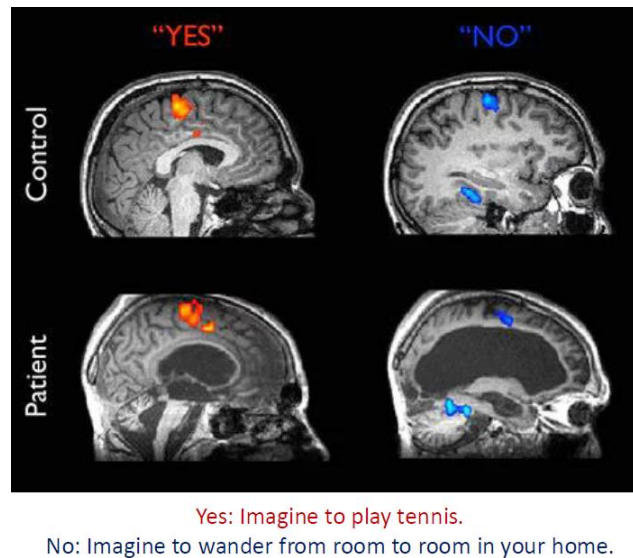
Important rule in neuroscience (Garbage In Garbage Out), meaning that if the data is collected in a bad way there is no method of analyzing that can improve the quality of the signal.

Consciousness (application of the methods described)

Consciousness is something that was not considered a science at first, because of a big debate on what parts of the brain are that makes us conscious. One theory says that any complex system can be conscious (the problem is only how conscious you are), the other says that without the frontal cortex we can't have consciousness (some animals have very small frontal cortex, or even newborns).



To say that someone is conscious you have to say what you mean by it, in this case she was conscious, having a dream. Is a person in vegetative state (coma) conscious? Seeing him you wouldn't say so, but actually he is so, and you can verify it by using the f-MRI making questions and seeing the answers.



Some studies verified that when the subject is conscious or more likely to gain consciousness and wake up, he has more complex responses, compared to when they are unconscious.

Consciousness: the ability to experience or to feel, having a sense of selfhood or soul, the fact that there is something "that it is like" to "have" or "be" it, and the executive control system of the mind, or the state or quality of awareness, or, of being aware of an external object or something within oneself. This is something that is totally inaccessible, everyone has their feelings, there is no quantitative measure to say if one person is conscious.

- **Consciousness:** intrinsic and integrated experience; it is what it "feels like" to be a system from the system's own perspective.
- **Awareness:** accessing or reporting information (*conscious access*); the cognitive ability to notice or respond to external or internal events.

Subtle but important difference. The information that something happens in the background may be processed in the brain, but I may not be aware of it. This is related to conscious access, the fact that some stimuli may be presented to the brain and be processed but I may not become conscious of this information until there is a trigger (driving a car, a lot of stuff is automatic).

We have the content (the level) and then we have attention. The process in which we direct our awareness, the attention, can be:

- **Bottom-up:** when a salient stimulus (e.g., a flash or a sound) capture our attention. This is not under voluntary control. In our brain there is a system that detects silent information and direct automatically the attention towards it. Comes from the perception.
- **Top-down:** attention willingly deployed toward a particular position in space, a particular feature, or a particular object. This is under voluntary control, when I direct attention towards something.

They are not necessarily in synergy; you may want to pay attention to something while at the same time there is noise outside and you get distracted. They influence each other. Awareness is the presence of conscious access; attention is a process that allows me to select what goes in the conscious access. Basically, the two things go together but not necessarily.

In blindsight the person has damage of the visual cortex of the brain, there is a stroke and the region of the brain is damaged (no visual consciousness), but is seen in this people that if you put them in front of a screen and you present some stimuli the person would say that he doesn't see it, but if you move the stimuli left or right and you ask him where, the person will get the right direction the majority of the time. Their ability to select the right response increases if they pay attention to the task and at the area of the screen that they don't actually see, they can guess more correctly.

Attention test → when you are focusing your attention to a very specific attention within the image you lose everything else. If I show this video while doing an f-MRI scan I can see that specific regions of the brain activate when the gorilla is present. Attention is what makes you select information.

Gist (the high-level semantic description of a scene) appears to be immune from inattentional blindness. Because gist is a property associated with the entire image, any process that locally enhances features, such as focal attention, is of limited use. When a photograph covering the entire background is briefly and unexpectedly flashed onto the screen, subjects can accurately report a summary of its content. The gist of a scene can be extracted in a period as short as 30 ms. A subject engaged in a task that requires focusing attention on the centre of the visual field, can still perceive the difference between a background scene containing an animal or a vehicle from one that does not. While the gist of a scene may be experienced vividly, it is also coarse. You can have conscious access even you don't pay attention to all the specific access.

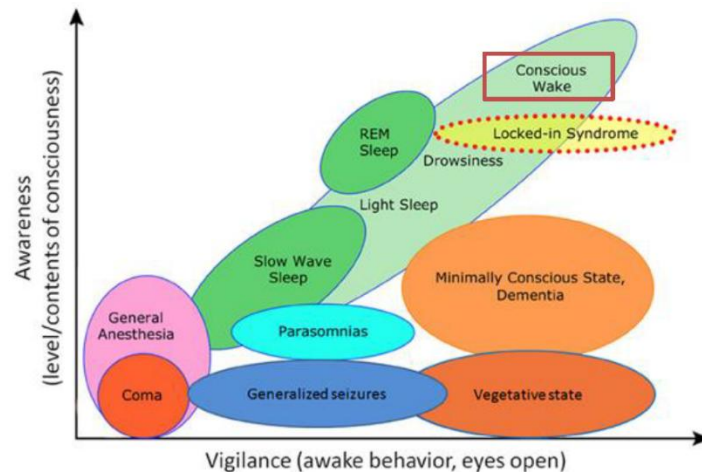
Attention it's a way to amplify information's so that you can be aware of it, really important for conscious access.

In normal conditions all of us is conscious. To study when I am or not we have to make contrast (someone that is awake vs. someone with brain damage). When I compare the two I will have a lot of differences, so it becomes really difficult to comprehend what is the thing that makes one person conscious. Each one of us has a different brain, so it would be better to compare the same brain in different conditions, but we can't, so we use other models (sleep or anaesthesia).

Under anaesthesia there is for sure an alteration in the level of consciousness and is in the same brain. Sleep is a little bit more interesting, because is a natural condition and it can be interrupted almost instantaneously with a big noise for example.

There is a condition in which consciousness is profoundly altered that is epilepsy, that is much more difficult to study (episode that last a short amount of time). There have been some studies but very few.

If you find a measure this should work for all the models. If you find that during sleep there is a brain region that goes off and it is important for consciousness, you should see the same behaviour for all the models.



This compares the content of consciousness (awareness) and the vigilance (the ability to interact with the external world). Parasomnia is a particular condition in which a person sleepwalks, it resides in that position because it is a condition where you may have a dissociated state (part of the brain awake and the other asleep), in this condition you are processing the information of the external world, but they live inside like a dream. Then lucid dream (a little above the REM sleep and close to conscious awake) is when you have a dream and you realize that you are in it, you are conscious and vigilant, aware of what happens (the frontal cortex lights up). Sleep paralysis is a condition in which you are blocked inside your body (in between REM sleep and conscious wake), in REM sleep your muscles are blocked and you become awake with a dream like experience. Brain is not all or nothing, is made of different structures the cooperate and communicate, but sometimes this communication is broken (even in physiological conditions).

A concept that is important talking about consciousness is the QUALE (sort of philosophical concept), the plural is QUALIA, and they are individual instances of subjective, conscious experience. The '*what it is like*' character of mental states, of an aspect or experience. The way it feels to have mental states such as pain, seeing red, smelling a rose, etc. We don't see all the world in the same way.



Example of different perception of something.

The reason is related to sleep regulation, our brain makes continuously assumption about what we see, we perceive a reconstruction of the world, and this is made of assumption (for example conditioned by the light). The perception of the colour of the dress may be affected by the time people go to sleep (different cardiac cycles).

Bat example:

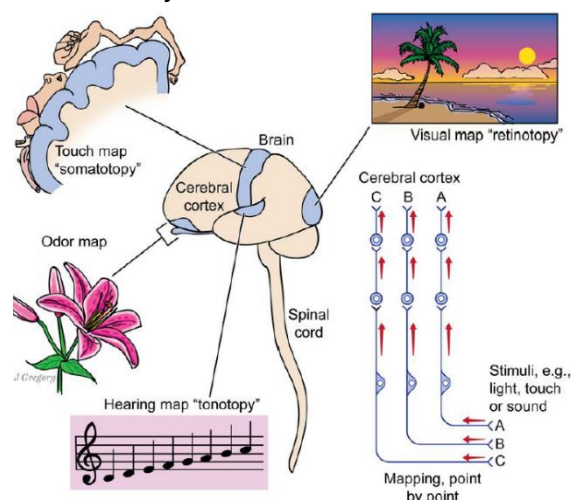
“The physical sciences can describe organisms like ourselves [...] but they cannot describe the subjective experiences of such organisms or how the world appears to their different particular points of view. There can be a purely physical description of the neurophysiological processes that give rise to an experience, and also of the physical behavior that is typically associated with it, but such a description, however complete, will leave out the subjective essence of the experience — how it is from the point of view of its subject — without which it would not be a conscious experience at all.”

Consciousness is one of the fundamental facts of human existence. Each of us is conscious. At least, one knows about his/her own consciousness directly. He/she can't be certain that other individuals are conscious. No matter how sophisticated someone's behavior is, behavior on its own is not sufficient to guarantee the presence of a conscious mind.

Turing test: proposed by Alan Turing in 1950, is a test designed to determine a machine's ability to exhibit intelligent behavior that is indistinguishable from that of a human. The test involves a human evaluator who engages in a natural language conversation with two participants: a human and a machine. The evaluator does not know which is which and must determine which participant is the human and which is the machine.

How do you determine if someone is conscious or not? You can't. This problem is known as hard problem of consciousness. The hard problem is understanding how a physical information is transformed into a subjective experience. If you believe that there is a mechanism that translate the one into the other the problem much more difficult, or you can think of it by associating this to the concept of soul.

The Neural Correlate(s) of Consciousness (NCC) is “the minimal set of neuronal events that gives rise to a specific aspect of a conscious percept”. The idea is “what do I need to get a conscious experience?”. I take the brain and start to remove pieces continuing to ask the same question until the brain doesn't sense the experience, then the last part removed was crucial for consciousness to exist and I can study it.



We have the cerebral cortex, connected to the spinal cord that is all the neurons that receive information from the periphery, we start from here. Can I cut it and continue to be conscious? Yes, but they don't have sensations and can't communicate with the external world (the person

is “locked-in”), they maintain eye movement. You can also have a complete locked-in Syndrome, where the eyes are paralyzed as well.

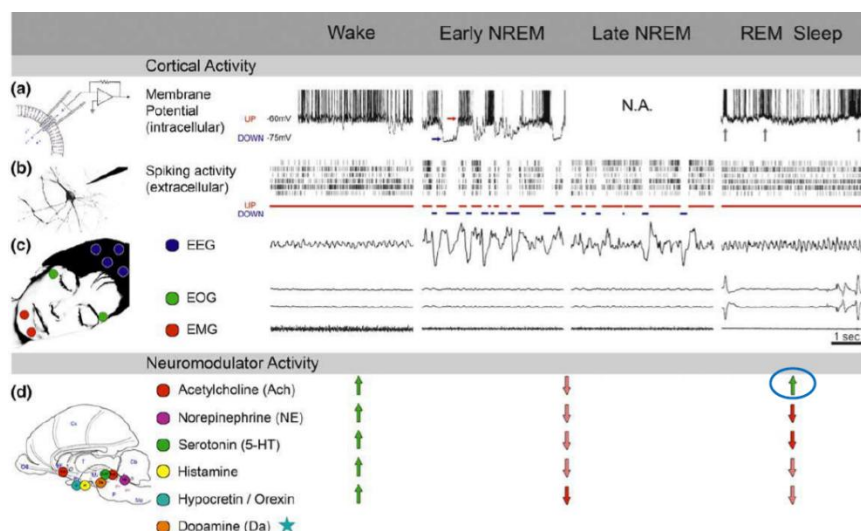
Another part of the brain that has been found to not be important for consciousness is the cerebellum, interesting because its structure (just below the brain) is crucial for acquisition of inputs, is important for learning, language. There are people that are born without it and don’t realize it’s not present, and they have some difficulties learning or with the movement.

The cerebellum contains a lot of neurons and connections (60 billion), and here they are connected in parallel, each neuron stay in a small room with some others and after the computation they send out the information that then arrives in the cerebellum, and it’s processed. So, they don’t communicate in an integrated way (with long-range communication and transversal one) like in the other parts of the brain.

Example: “A 24-year-old female patient was admitted to hospital complaining of dizziness and the inability to walk steadily for more than 20 years, and nausea and vomiting for ~1 month. She is married with a daughter, and her pregnancy and delivery were described as uneventful. [...] According to her mother, she was 4 years old before she could stand unassisted, and did not begin to walk unassisted until the age of 7, with a persistently unsteady gait. She never ran or jumped. Her speech was not intelligible until 6 years of age and she did not enter school. A neurological examination revealed she could cooperate and fully orientate. A verbal analysis test revealed her word comprehension and expression remained intact and she had no sign of aphasia, but mild to moderate signs of cerebellar dysarthria. [...] While she is able to walk unsteadily without support, her gait is moderately unsteady. The patient has evidence of tandem gait and moderately reduced gait speed. There is no focal paresis, but the muscle tone is mildly increased. Evaluation of the sensory system showed no abnormalities, no deformities of the fingers and toes were observed, and her complete blood count and urinalysis were normal.”

The cerebellum is not necessary for consciousness.

The Arousal System is a complex system, a combination of several structures. The thalamus is the structure that perceive sensor information from the periphery and send it to the cortex and it is part of the arousal system, that release neurotransmitters in the brain that keep us awake, just like noradrenaline, dopamine, histamine, glutamate, acetylcholine, serotonin, all these are attributed to different nucleus. Most of these systems are active when we are awake, and they keep up in this state.



The interesting part is that the default state of neurons is asleep (on-off state), they need the neurotransmitters to remain activate. Even without one of these systems we can remain conscious (noradrenaline-emotion danger, serotonin-emotions), these structures are needed for consciousness, if I remove these complete systems the brain is in the on-off state permanently. That is why we read articles of people waking up from long time vegetative states, because maybe the damage is not properly on the brain but in the arousal system that restart its work.

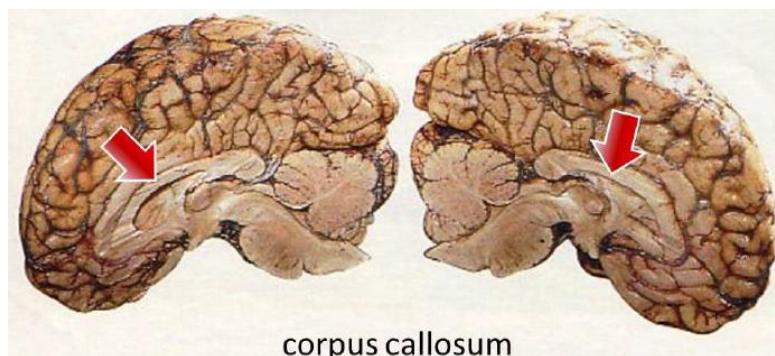
Now we are getting closer to the cortex, but the brain is composed on two hemispheres. The right hemisphere is much more literal in its recognition of images and events, and thereby often more accurate, in contrast with a left hemisphere that is constantly formulating hypotheses and trying to make sense of what happens. In addition to its superior language capabilities, the left hemisphere also retains normal problem-solving skills and reasoning abilities following callosotomy. The cognitive and linguistic abilities of the isolated left hemisphere are comparable to those of a healthy individual with an intact corpus callosum, suggesting that the left hemisphere retains a rich, human-like consciousness following split-brain surgery.

The right hemisphere's capacity for language is variable among split-brain patients and may gradually increase following surgery. The right hemisphere outperforms the left in several domains, such as perception (detection of motion, illusory contours), spatial processing (mental rotation, spatial matching, mirror image discrimination), facial recognition, and recognition memory.

What happen if you remove part or one of the two hemispheres?

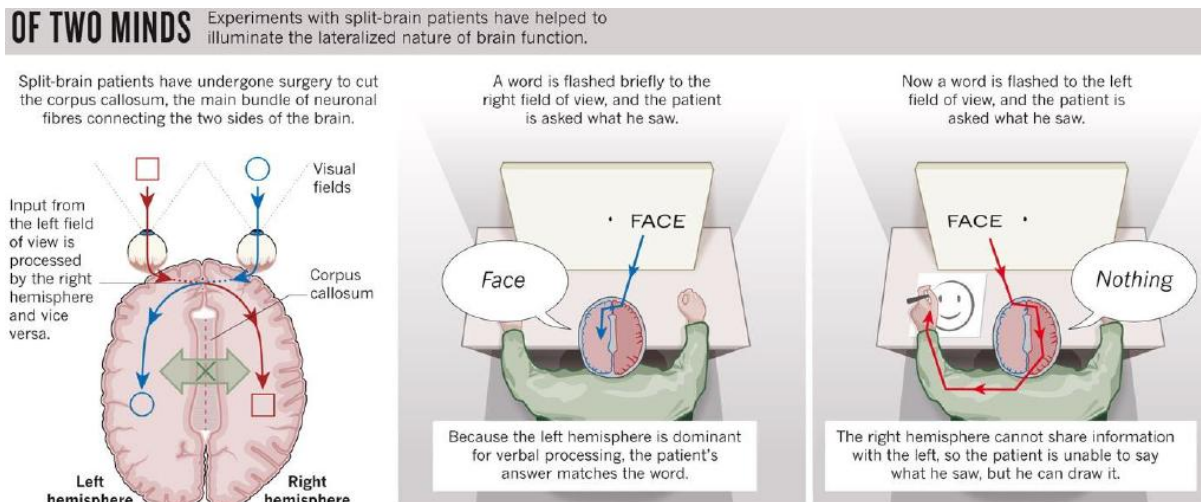
Example: a woman (neuroanatomist) got a stroke in the left hemisphere and then a complete recover. She was able to confirm all the ideas that people had about the hemisphere functions, she couldn't speak intelligently, she couldn't understand what people tell her, she was not able to think in words (internal speech). She felt a change on how she perceived the space around her, left hemisphere really important in space perception.

Example: callosotomy, the case in which the corpus callosum is cut. The arc is made of fibres that go from one hemisphere to the other, crucial for the brain to exchange information between the two hemispheres. Some people have a form of epilepsy in which seizures start from a part of the brain and propagates very fast to the other and this make the person fall of the ground (for security reasons you cut the corpus callosum). Not done anymore in almost all the world. The problem is: if one hemisphere can be conscious by itself, if I cut the connection I create two consciousness in the same person. Tricky.



Experiment: you put a person in front of a screen in which they can only see half of the screen. If I present a stimulus in the right side (perceived with the left brain), this person was able to say

that he saw a face (language is in the left brain). If you show the stimulus in the left side the person says nothing, but if you ask to draw the picture they can (right hemisphere controls drawing).

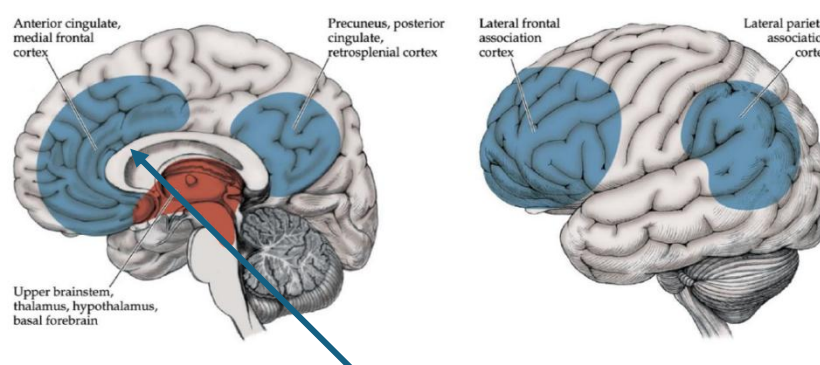


Each hemisphere of a split-brain patient cannot access the thoughts, intentions, or conscious experience of the opposite hemisphere, much as we cannot access the mental states of other individuals

Not so clearly this all is true (two consciousness in one mind), they have a really high delay. Most of them behave like child. Some of them can be smarter, maybe because the condition affected them from the birth. Now there are two theories, one says that is true (each hemisphere able to support an independent conscious self), the other one says that you don't have two separate consciousness but two separate perceptions. There was also a case in which a girl discovered to have only one hemisphere late in her life.

The brain always works to create new connections if some are lost, an example is language, even without connection with the left hemisphere the other one tries to adapt and learn how to speak (long process).

The conclusion is that the part in blue are the ones that are crucial for consciousness (present medially and laterally in the brain), the areas in red are part of the arousal system.



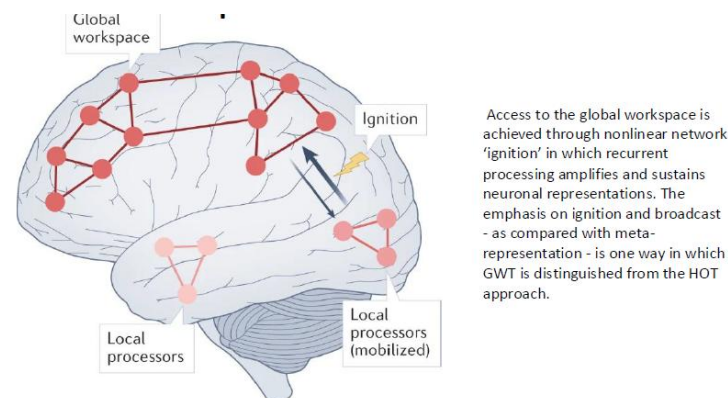
So, do we need all of the regions? The frontal cortex is crucial for decision making, interacting with the external world, producing a verbal response, to decide if I want to move or produce language (damage to this part of the brain can create a condition in which the person enters in a state of total apathy, total non-interaction, he is conscious but don't care). The role of this region is highly debated, if you pay attention, you can see that most of the top-down attention depends

on the involvement frontal cortex. If I need to pay attention to a stimulus, my frontal cortex will be activated, crucial. Is this really necessary for consciousness to happen or just necessary consciousness to be reported. This is why sleep and anaesthesia are interesting models, you don't have any voluntary control, you don't care and you forget it. Then the person only remember that they are in an experiment when they wake up (interesting to study the activation of frontal cortex while dreaming).

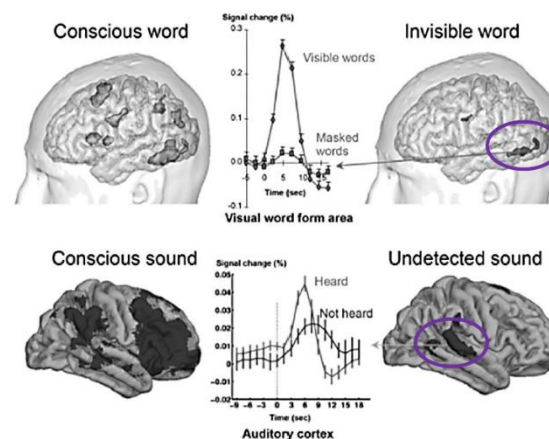
The theories of consciousness are different; many are philosophical and not scientific. If you have a theory that has no prediction and you don't test the predictions, this is not a theory.

Higher-Order theories → consciousness is mostly in the frontal cortex, able to transform experience in something that we feel. You are conscious because you reflect in you experiences.

Global workspace theories → originally proposed by Stanislas Dehaene, says that to become conscious we need also the frontal cortex. Information arrives to the posterior part of the and is processed, if this is strong enough it can reach the frontal cortex, that use an ignition, a feedback that potentiate the response (the brain creates a loop that increase the brain activation level).



There are scientific supports for this view. If I show you a subtle stimulus (embedded in noise) and it is not perceived there is an activity that is confined to the frontal cortex and does not include other regions of the brain. If you present the same stimulus but is perceived consciously you detect a much stronger activation. The ignition process is important to conscious access to happen. The information is spread through the brain and is needed for conscious access to happen. In most of the studies on these theories, you take a conscious person and present him different stimuli perceived or not and comparing them I see this type of ignition especially in the frontal cortex. These experiments can be manipulated, if in one eye the stimulus flickers then the brain will focus on the flickering stimulus, and the other one will be invisible.



Integrated information theory → everyone hates this theory because it is the most reported complete theory of consciousness (angry that their theory is not considered as much as this). There is a lot of effort by the supporters of this theory (a lot of mathematical test and prediction to try and attempt validating this theory). Any system can be conscious, depends on the connection that you have. Measure called integrated information (Φ), computed mathematically, but right now only in models, the moment you go on brain it doesn't work because it is too complicated. Φ cannot be measured on humans directly, related to entropy. The problem is that based on this idea I don't need the frontal cortex in order to be conscious, even a set of neurons that reach a certain degree of information integration can be conscious for themselves (main point of disagreement). The frontal cortex is only necessary if I need to report my experience, but then to be conscious I don't need it because the posterior regions of the brain have a sufficiently complex structure to support consciousness.

They tried to validate one of the theories as better than the others and no theory won. Only one got 2 out of 3 and the other ones 1.5 out of 3. A journalist said that Giulio Tononi's theory won (information integration) and so everyone else raised up to say that it was bullshit. The effect in which scientists are against each other is not good for research.

Information is high where there is integration and communication.

Lesson 22/10/2025 (Learning and memory)

What happens in the brain when we learn and acquire information?

Learning is the process that leads to the acquisition of new knowledge or skills, memory is the product of it (acquired knowledge and skills). Learning leads to some kind of changes in the brain that allows the brain to store the information acquired. Typically we go through 3 steps while learning:

- Encoding, that is the storage of information.
- Consolidation, most of it happens during sleep.
- Retrieval, the one in which you access the memory, and you use it in behaviour.

To acquire information, we need to rely in our senses, not only the main 5 but also through the interoceptive information processing, that acquires information from the state of the body. We learn to define emotional states based on what we feel in specific conditions (sweating, acceleration in the heartbeat, etc.).

Once that the information reaches the brain there is a sort of sensory memory, which is the most basic but also fleeting (few ms), that is the time of processing of the information that we acquire. The information stored somewhere in the brain, and this is called short-term memory, the other name is working memory. From this type of memory you can retrieve and use information, or send it in long time memory through consolidation, that continues for days, months or even years, at least few hours.

There can be also some loss of information, that can happen at all the steps, and it also may happen in all the moments in which we retrieve information from memory, and they become less accessible each time. If you learn something for an exam you see that memory weakens over time, in a point that you don't remember almost anything.

The terms short-term memory and working memory are strongly related (almost synonymous), even though the second one is more recent, introduced after realizing that when something is

stored in short-term memory it's not just there but is also used by the brain, and they started to have the idea that there could have been something more similar to running memory in computers, able to put something in memory to process it and then you can remove it or store it.

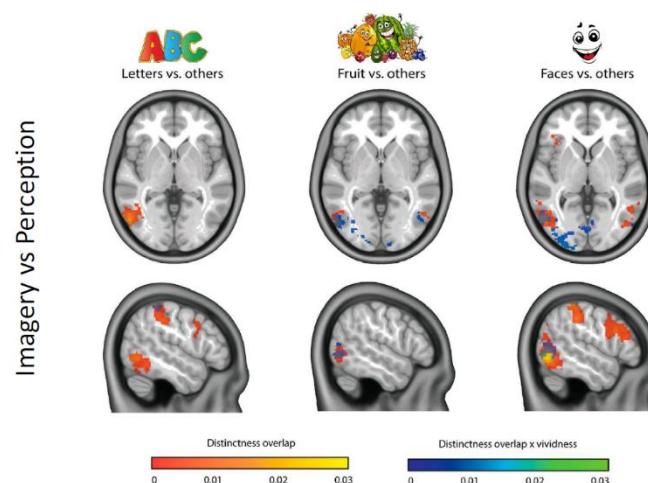
Short-term memory is the brain's temporary storage for a small amount of information that can be actively recalled for a brief period and is essential for tasks like holding a phone number or maintaining a conversation. Useful when you need to follow someone that is talking to you and maybe reconstruct the meaning of a sentence. The working memory can instead be manipulated. This type of memory has some characteristics:

- Is limited, if you try to memorize a number you can see there is a limit (4-7, found with studies).
- It has a brief duration, meaning that you typically store information from seconds to minutes, until it's lost or stored in long-term memory.
- The information is held in an active, readily available state for immediate use, contrary to long-term memory, in which we have information that does not need to be recalled in continuation and remains silent.
- It also serves as a temporary space for information before it is forgotten or transferred to long-term memory.

The encoding of memory during sleep is altered, but if something wakes you up in that moment you are able to remember the dream and encode it. The information can stay in the working memory for some time, and we can revisit and stabilize the memory of the brain or get distracted by the things we have to do and then lose it.

Memory is something that affects the brain development, usually those who become blind after the age of six have visual memory, so they are able to detect colours and remember images, while those who become blind before this age are unable to recall any visual experience. Furthermore, most of the memory you have of when you were 5-6 years old is probably false memory, something that you created based on what your parents told you or on photographs or video that you saw, but then after seven it stabilizes.

Some forms of imagery (create something from memory) can be considered also as a form of internal perception. When you do that you activate the same exact brain regions that you activate when you perceive, for example, an apple.

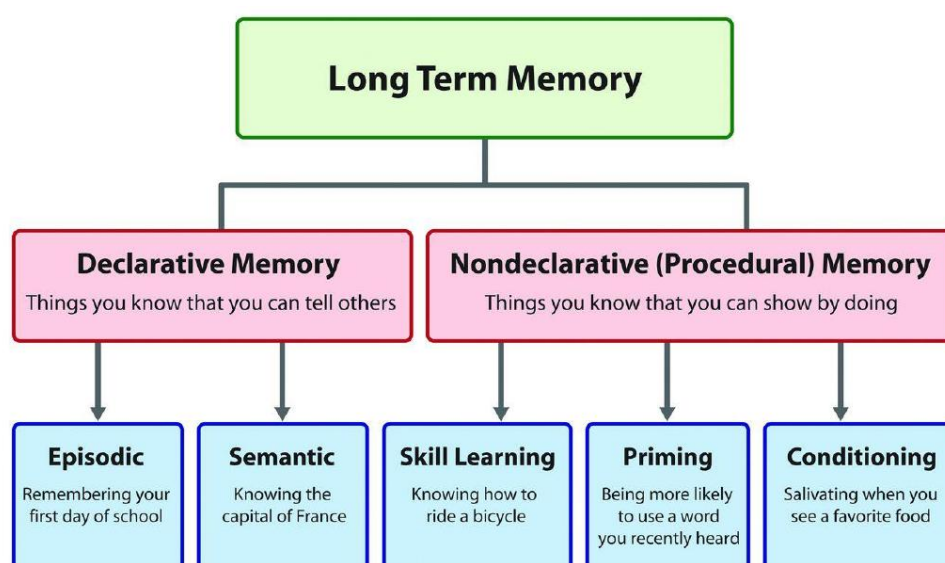


Here you can see the overlap of what you can see in the brain between the activation of the brain during a perception of a face and the imagery of a face. The region is situated in the inferior posterior part of the brain, that is specialized in perception of human faces.

What changes is that when you perceive something from the external world this happens because through the eyes you acquire information and there is a flow of information from the posterior part of the brain (occipital cortex) to the anterior part, so there is a distribution of activity across different regions. When you activate your memory (for instance of faces), the command starts from the anterior part of the brain, that retrieve the information about the face, for example of someone you know, and you have an activation in the cortex in the corresponding brain region. As long as you keep the information in mind you have activation in some frontal regions that are important to keep the information active in this top-down manner, and you have activation in the brain regions that are needed to process information.

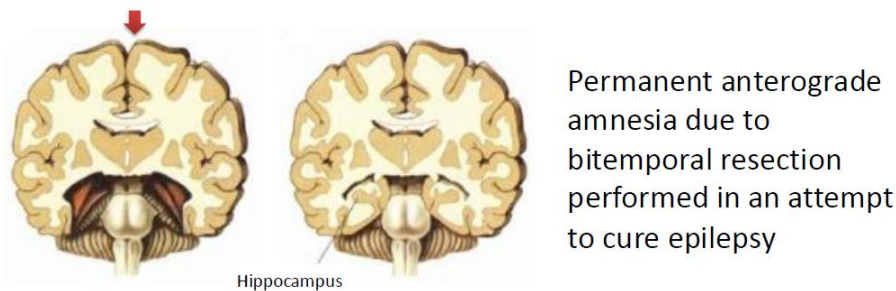
When we talk about memory we have different types of definitions. We focus on the one in which we divide memory:

- By duration. Saying that the shorter one is the sensory memory, related to the processing of the sensory information. Then we have the short-term memory and working memory (order of seconds-minutes). And finally, we have the long-term memory, that has potentially a duration of all life.
- By content. We specifically divide memory as declarative (explicit) or non-declarative (implicit). The difference is basic, meaning that implicit memory is for example learning how to ride a bike, acquiring skills in general that allows you to bike forever, and this is something that stabilize in the brain and helps you also with other activities. The declarative memory is instead the memory of facts, and it's the report of facts. These are partially dissociated in general; you can lose memory of facts but may maintain implicit memory, so they are supported by different parts of the brain.
- By process. Different phases of memory acquisition: encoding; consolidation; retrieval; reconsolidation. The last one is related to the process of retrieving the memory and stabilizing it and storing it again.



Some examples of different the different types.

If we talk about memory, we have to talk about Henry Gustav Molaison, that was known until he died as patient H.M. He had brain surgery for a seizure, and since it started from a specific region of the brain, that is the hippocampus, the doctor at the time thought they could just take out it. Removing the hippocampus (anterior-interior portion of the brain) he was not able to form new memory at all, not able to acquire any information. They even tried to give him cross-words and once completed and cancelled, the second time they presented it to him he thought that he had never seen it.

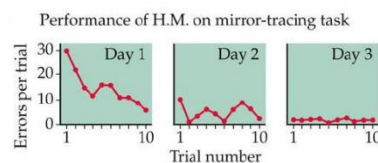
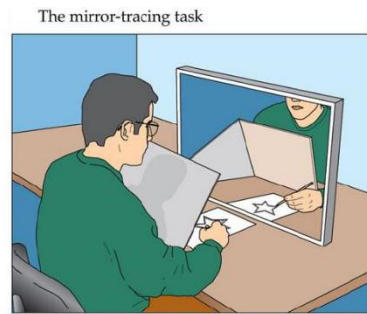


When this person died, the brain was donated to science, and analysing it was seen that the hippocampus was not completely destroyed but most importantly they cut the connections between it and the brain, so that even the remaining parts of the hippocampus were not able to communicate. This person was able to maintain working memory, but then when distracted his memory was lost again. The take on message is that the hippocampus is crucial for the brain to acquire and store information.

The hippocampus is close to other structures, like the amygdala (important for emotional processing), emotions are important in memory. It is also close to the olfactory cortex, in fact the smell of something can activate the retrieval of memory that we have in mind, because they have a direct access to the hippocampus. Furthermore, this structure is able to receive information both from the extern and the intern of the body, for example visual information is processed in what is called two streams, the dorsal stream processes the spatial context of the information (where something is in an environment), while the ventral stream processes the identity (the what of the object). There are two path that elaborate mostly different aspects of the information from the extern, then transferred to the hippocampus, but also internal information, for example from the amygdala, that processes the emotional context of what is acquired.

The idea about the context and the content is important, because short and long-term memories are both limited, the information that we can keep in the brain is not infinite. This become very clear when is asked to remember face from old famous people, because we tend to forget faces to create store for new ones.

H.M. was then asked to do another task, he had to draw in a paper that he couldn't see if not as the image on the mirror. You have to learn how to draw in a mirrored way, and measuring the number of errors he did they found that there was an improvement in the performance during the day that was also maintained the following days, but then when he was asked if he remembered that he had already did the task he would have said no. So there was an improvement in precudral memory, his skills. The two types of memory are separated, cerebellum for example is crucial for motor skills and also other structures, so that the motor cortex exchange information with them making the patient able to improve motor abilities. Motor memory doesn't rely on the hippocampus.



Preserved working memory and intellectual abilities.

Maintained ability to acquire new visuo-motor skills (procedural/implicit memory).



The patient lost the ability of creating new memory, but also other memories from few years before the incident. Some memories were actually not yet transferred in the long-term storage, they were still being consolidated.

Encoding

The memory is somehow stored in the brain, so that it changes in order to incorporate this type of information. This is an engram (not sure of what it is), a physical trace of memory in the brain, a neural correlates of memory, what happen in the brain corresponding to the storage of memory. This has 4 characteristics:

- An engram is a persistent change in the brain that results from a specific experience or event, that last forever;
- The engram should have the potential to influence the behaviour, brain change → change in behaviour (you put a finger in fire you don't do it anymore). This comes from the interaction with retrieval cue, you recall what you did and felt in life because the cue has this power.
- The content of the engram reflects what happens at the encoding and is reactivated during retrieval. Easy to understand if you think about the concept of imagery, those regions are also involved in the retrieval process.
- The engram is stored in a sort of silent way, unless you reactivate the engram voluntarily or there is a cue that does it, the engram is there and if not used it's just there.

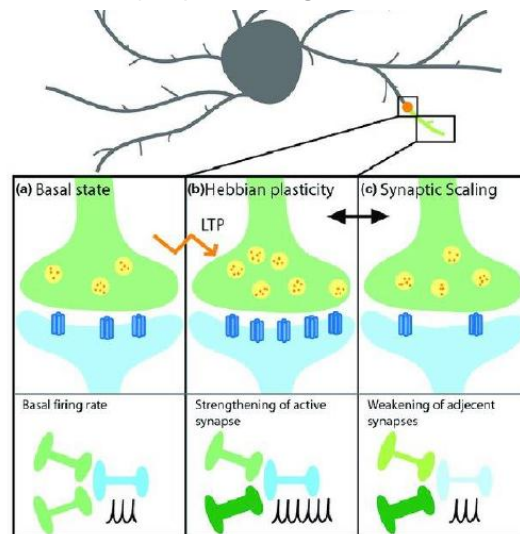
An engram provides the necessary physical conditions for a memory to emerge.

This trace in practice may be related to how neurons are connected, and the strengthening or weakening of connections (simplification). But there are other cells that for a long time were only considered as supporters of the neurons, just as astrocytes, that were discovered to do a lot of work in the brain, but this is just for the moment an idea. The problem is that neurons because of their electrical activity produce something that can be actually measured, but astrocytes do not produce a very clear signal (at least in humans). With f-MRI maybe we are also measuring their activity in the change of blood flow, caused by their request of oxygen, but from this type of technique we are not able to see which structure is demanding more blood flow.

The engram is been hypostasized to be mostly a change in the strength of connection between neurons, so when we get new information neurons will get the brain activated and this activation will generate some kind of interaction between neurons, with a strengthening of their connection. This is called Hebbian plasticity, saying that neurons that fire together wire together. There are

group of neurons that work together for the same task, for example when seeing an apple in the brain the neurons that activate for red objects, the ones for natural objects, will start to fire together to represent the apple in the mind, so those neurons fired together will start to connect among themselves strengthening it. If I see an apple only once in my life that image doesn't remain in the brain being useless, but if I see a lot of them this information become important for the brain and is consolidated in it.

This process of strengthening is called long-term potentiation, and it is a persistent strengthening of synapses based on recent patterns of activity that produce a long-lasting increase in signal transmission between two neurons. The opposite of LTP is long-term depression (LTD), which produces a long-lasting decrease in synaptic strength.



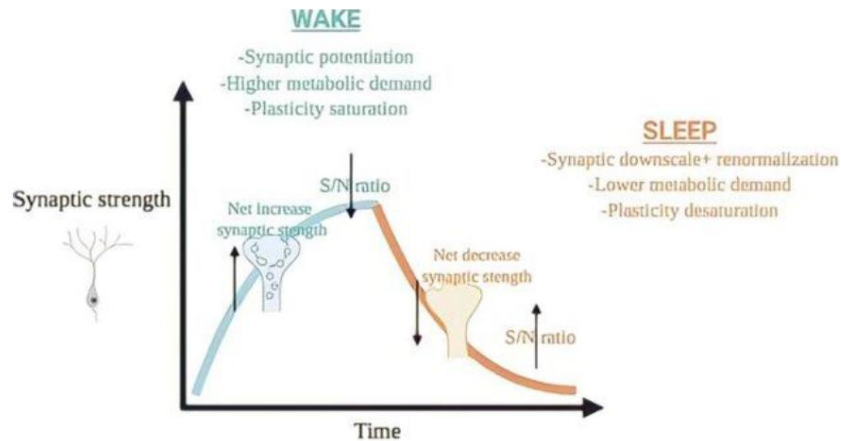
When strengthening it can happen that their synaptic button may become larger, so that I am able to release more chemicals and be more efficient. But it may also represent a growth in the number of chemicals stored in the pre-synaptic button, so that I can release more of them and be more efficient. But also, an increase in the number of receptors on the post-synaptic button. This type of event obviously requires time, even minutes, hour or even days.

The brain tends to select what is important for it, based on different things. Firstly, the number of times you see the element. Synaptic plasticity: ability of the brain to strengthen, weaken, destroy and create neural synapses as the basis for learning. The effect of a learning experience depends on the content of such an experience. Reactions that are favoured will be reinforced and those that are deemed unfavourable will be weakened.

- Short term: strengthening or weakening of a connection by modifying the preexisting proteins leading to a modification in synapse connection strength.
- Long term: new connections may form or the number of synapses at a connection may be increased or reduced.

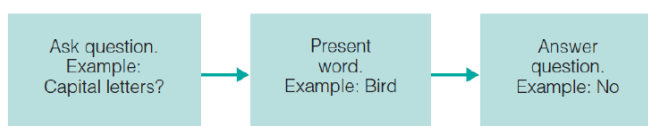
During the day is mostly activated by all the situations that happen around us, and many of our synapses are potentiated. This leads to a saturation of synapses and more energy needed by the brain resulting in a problem for it, since it is not sustainable on a long-term. For this reason a theory has been proposed, the synaptic homeostasis hypothesis (SHY), that tells that basically this type of increase is compensated during sleep, and we have some proof of this that in some regions the synapse strength decreases (downscale), but this downscale doesn't happen in all the synapses. Some synapses that are bigger and more important are not downscaled while others less important are weakened, leading to an increase of what is called Signal to Noise ratio. This

mechanism is what make us able to learn new things every day. Resuming synaptic potentiation during waking is connected to a homeostatic increase in slow-wave activity (SWA) during sleep, during which the systematic normalization of synaptic strength (synaptic downscaling) takes place. Larger and most potentiated synapses may remain unscaled. Weak connections may be eliminated, whereas the relative strength of the remaining connections is preserved.

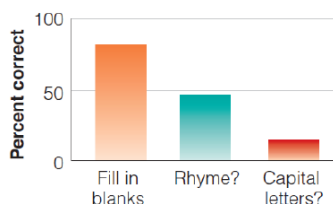


This type of studies were conducted on animals, because in humans the evidence is a bit more indirect, meaning that they relied for example in the potential that you can induce with TMS, so that if you have many neurons strongly connected and you sent a pulse, you get a response that propagates very efficiently, validating the theory of potentiation. If you do the same thing during sleep, you get a much smaller reaction, the brain is less efficient, and this may happen because the neurons are not strongly connected. Unfortunately, at the moment, we are not sure it really happens in all brain regions in the same way, for example in the motor cortex the evidence is clear, while in other, e.g. visual cortex this effect was not observed. A theory is that the visual cortex doesn't have to change that much during the day because our perception of the world it's the same, while the motor cortex has to do a lot of things.

Depending on if you want to store information, the depth at which you elaborate information can affect how much you are really able to store information.



(a)



(b)

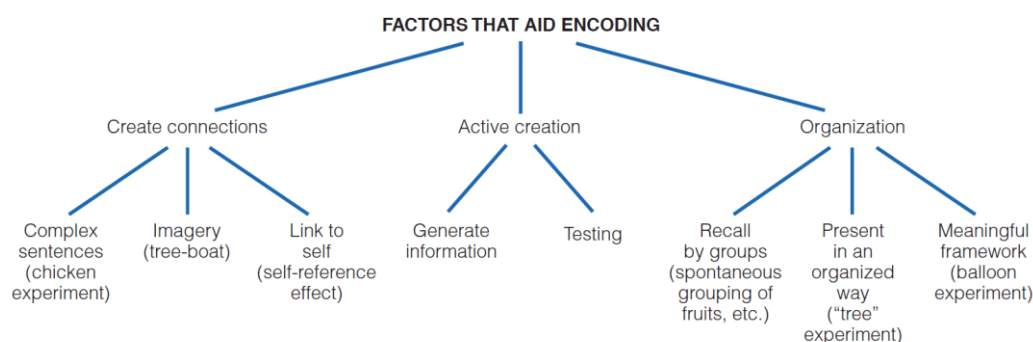
1. **Shallow** processing: A question about **physical features**
Question: Is the word printed in capital letters?
Word: bird
2. **Deeper** processing: A question about **rhyming**
Question: Does the word rhyme with train?
Word: pain
3. **Deepest** processing: A **fill-in-the-blanks** question
Question: Does the word fit into the sentence "He saw a on the street"?
Word: car

Experiment: participants had to memorize the larger number of words possible and then they had to rely on these different levels of processing. What came up was that if you have the deepest level of processing (reflection on the meaning of the word) you are much better at remembering the words compared to a more superficial listing of the word. So, the level of processing affects the capability of storage of the word.

Another study performed considered the effect of imagery. If I associate imagery, can I potentiate the storage of information? It was found that associating a visual corresponding to the word, you are much better at memorize the word. Imagery potentiates the storage of information in the brain.

The other aspect is the relevance for the person, same task of before (the level of processing) adding another condition in which they asked the patient to try and focus on the relevance of the word for him, imaging in some way a connection between it and himself. It was found that it potentiates the storage of information. For example, when you study something that you found interesting.

The other thing is the structure of what you have to learn. If you have a list to memorize and you try and learn in a total unstructured way, it's much more difficult than trying to separate the words in clusters and find connection between the elements.



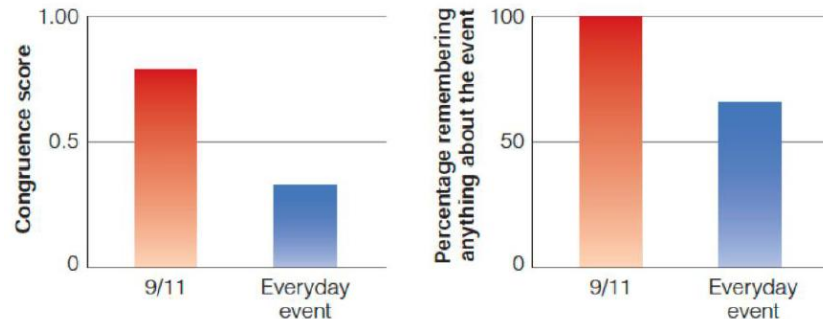
One thing that is also more important is the emotional connection. If something that has an emotional charge for the patient is shown to him this will potentiate the storage of information. If pictures or words that are associated with an emotional content are presented, they are much easier to remember. And this is why you have post-traumatic stress disorders, anxiety disorder that comes from a traumatic event. The strong emotional involvement makes the memories associated with it basically stamped in the brain in such a way that you can even revive the event vivid in the mind. You are in a condition of strong anxiety because this event can intrude spontaneously in your memory or caused by a cue. PTSD is the only case in which you dream about an event as it really had happened, and not an elaboration. One way to treat PTSD is to revive the memories, because when you reactivate a memory you can weaken it, and try to change what happened in the event.

We know that noradrenalin is crucial in the formation of new memories, and in case of strong emotional involvement we have a large release of noradrenaline and so this is really imprinted in the brain. Ketamine is not the best medicinal to treat this disease because it creates hallucinations.

Everyone that went through the 9/11 event remember what they were doing at that time. These types of memories are called flashbulb memories: vivid, detailed, and enduring memories of the circumstances in which someone first learned about a surprising, emotionally arousing, or consequential event. They have different key features:

- Vividness: people often recall memories with a strong sense of detail (where they were, who they were with, what they were doing).

- Emotional intensity: typically tied to highly emotional or shocking events.
- Confidence: individuals are usually very confident about the accuracy of these memories.
- Content: often includes contextual details (location, time of the day, ongoing activity, emotional reactions).



However, these memories are often not different from other memories, even if we have this vividness of the memory, we can always have an alteration of the memory. Example: for the challenger space shuttle explosion, in a time frame of two years a woman changed version of the facts reported because her memory changed. The memory is partially different, there is a lack of the first portion, maybe she deleted it.

Exactly for all the other memories, even this type goes down over time, but one thing that is interesting is that this memory is reported by the person with a great confidence. And this is an effect of the emotional involvement. Emotions are crucial component, whether something will be stamped in long-term memory and also affects the confidence that is in the person talking about it.

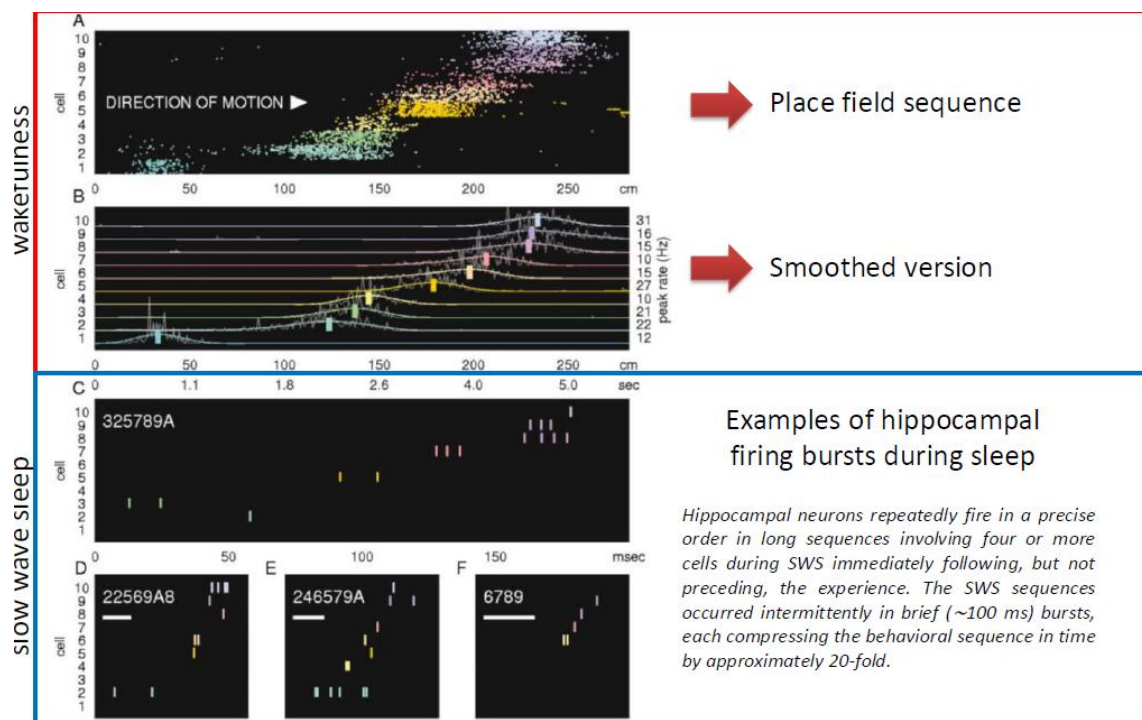
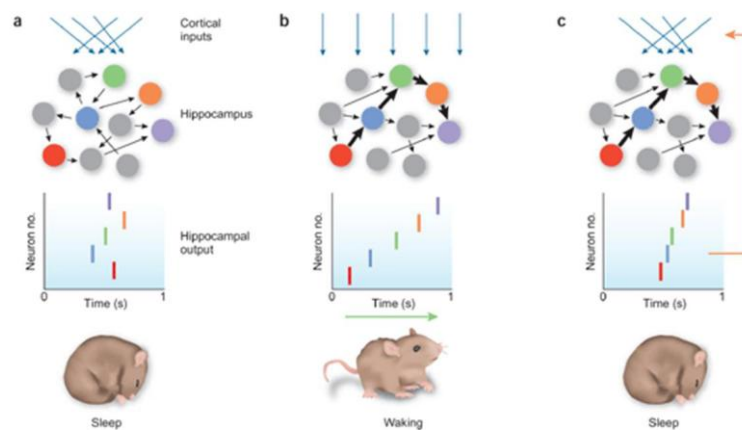
The memories can change over-time but at least at the beginning they are consolidated, and this process of consolidation happens every time we keep our brain at rest (having the time to re-elaborate stuff), especially during sleep. But how consolidation works?

- **Synaptic consolidation** (initial consolidation): Is achieved faster than systems consolidation (minutes-hours). Depends on Long-term potentiation (LTP). Writing in the brain, with the formation of the engram and the adaption of the connections between the neurons.
- **Systems consolidation** (how the memory is organized within the brain): It is a reorganization process in which memories from the hippocampal region, where memories are first encoded, are moved to the neo-cortex in a more permanent form of storage. Takes days to months.

Hippocampus is crucial for memories to be stored, but not for old memories, that must be stored somewhere else. This is what the consolidation process does, it stabilizes the memory and transfers in part the site of the memory. As we said there is a link between the neurons in the cortex and some neurons in the hippocampus, that basically are at least at the beginning controlling the memories, with no hippocampus this process will never become stable (no consolidation process). Then when I go to sleep the connections between the cortical and hippocampal modules weaken and at the very end (even years), the memory becomes independent from the hippocampus. Obviously we are talking about memories that depend on the hippocampus, not procedural memories that have different processes of consolidation.

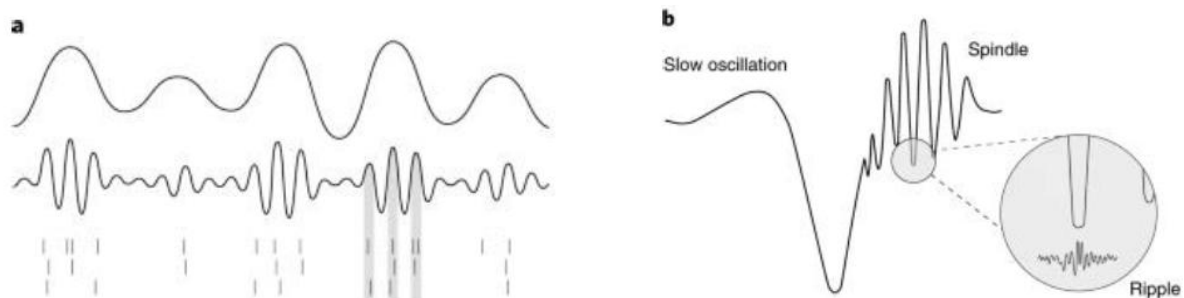
This is called Ribot's Law: a principle of retrograde amnesia, proposed by French psychologist Théodule Ribot in 1881, which states that recent memories are more vulnerable to loss than older, more remote memories (classical example is dementia, first lose recent memories and nothing of the day, no hippocampal mechanism). This temporal gradient in memory loss suggests that memories undergo a consolidation process over time, becoming more stable and resistant to disease or damage as they age.

What happens during sleep? Experiment: you create a labyrinth and implant electrodes in the hippocampus, where there are some cells that record and respond specifically to the spatial position and the direction of the animal, so when the animal starts to move there are some neurons the respond when it goes straight, left or right. You can give colours to these neurons and track the path of the mouse inside the labyrinth. Since the mouse learn the path, it is found that while sleep there is an activation of these neurons in the hippocampus in the exact same sequence, repeated several times (replay). Sometimes it is also reactivated backwards, several times.



We found evidence of a similar process also in humans, more indirect. During sleep when involved in this memory task there is an activation especially in the hippocampus, trying to replay the event. There are studies that suggest that these processes can also last for several days, and it's for sure something that needs some time to happen. Replay events in the V1 and HP were coordinated to reflect the same experience. These results imply simultaneous reactivation of coherent memory traces in the cortex and hippocampus during sleep that may contribute to or reflect the result of the memory consolidation process.

During sleep there is a crazy association of different oscillation with different parts of the brain, so basically in the hippocampus there are ripples, very fast oscillation, that reflect this sequence of reactivation, associated with spindles (oscillation at 40 Hz, generated by the thalamus and projected to the cortex) and slow wave at the cortical level. At the cortical level you have the slow wave, that at the end has the spindle, and during the spindle you have the ripple in the hippocampus. You have a coordination of events across the cortex, hippocampus and the thalamus that allows the network that has been activated during learning, to be reactivated e reprocess the memories, these sequence of events weakens the connection between the hippocampus and the cortex and strengthens the connection at the cortical level. This is something that happens mostly during sleep.

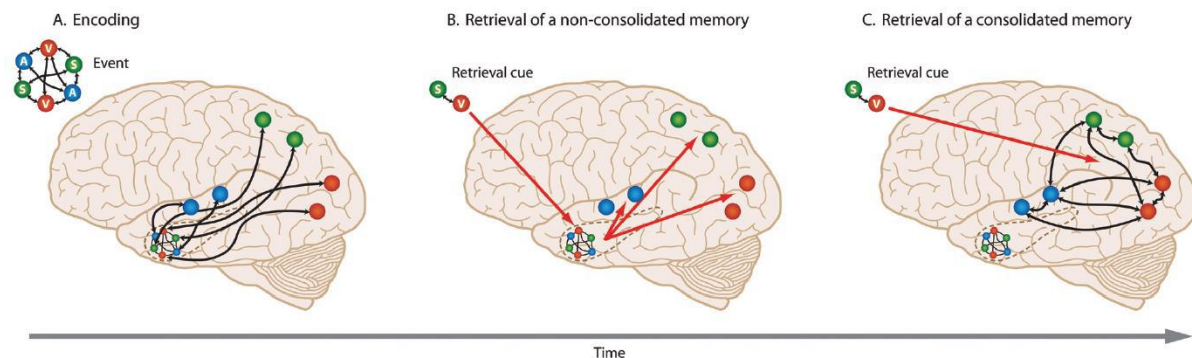


The relationship with dreams is not clear at all, but it's clear that dreams have something to do with these mechanisms of reprocessing. Right now, the only access that we have is the replay one, and in humans it is difficult to see the coordination in the hippocampus.

The retrieval is indeed the act of accessing information from memory. Retrieval can take the form of recall, in which one searches memory for a specific piece of information and is able to produce it. Retrieval is also involved in recognition, as when a person judges that they have seen or encountered something before. Retrieving information from memory is assumed to (partly) reactivate contextual features of the original experience, owing to the connections forged between the information and the surrounding context at the time of encoding. Similarly, being in an environment that shares features (cues) with the original encoding context can also facilitate retrieval of the information, this may need to mind wondering (memory starts to explore spontaneously the memories associated with something). Spontaneous retrieval is involuntary, triggered by cues with little effort, whereas voluntary retrieval is a deliberate, goal-directed process to recall past events. Spontaneous memories often arise unexpectedly, like from Proust's Madeleine story (there was a reactivation of a memory from a smell), and may feel more meaningful. Voluntary retrieval involves an intention to find a specific memory, often requiring conscious effort.

If I want to retrieve the memory when it is still not fully consolidated, B state, (from the cortex and the hippocampus) I must rely on the HC, that drives the retrieval of information. When I am in the

C state the memory is stored in the cortex, so the cue acts directly on the cortex, and that's why it is more stable.

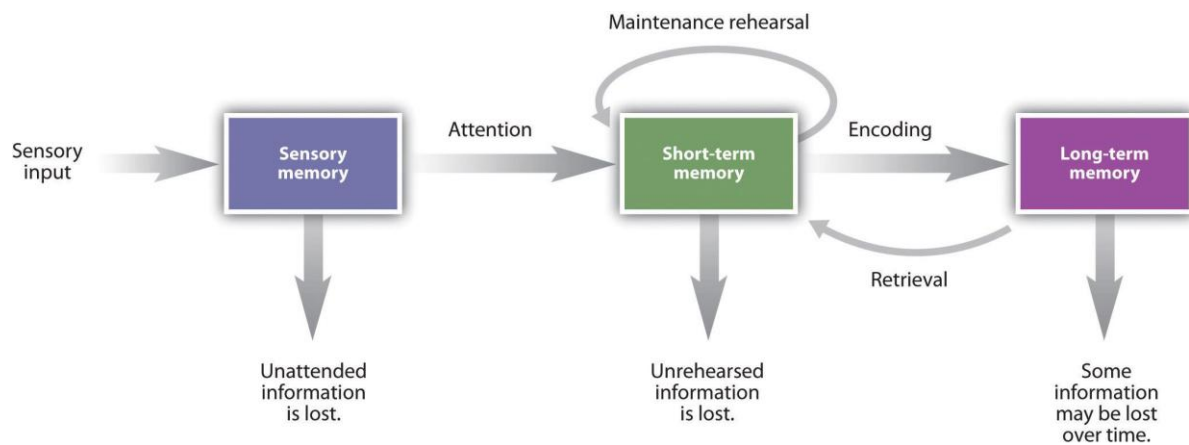


The memory can be influenced by external information to a process of competition. If I give you some images to study and you try to memorize them and then I ask you to reactivate this memory (retrieval) and in the same time I also show you other pictures, this process of retrieval will make you mix old and new images, but this doesn't happen if I don't ask you to do retrieval. Mixing is also more likely to happen if the images are acquired in the same context, that is a key element. The process of retrieval makes your memory more vulnerable to manipulation that could be voluntary or not. You should try to avoid conditions that can be an interference. Vulnerability can also be a positive thing, like in the case of traumatic memories when you try to get access to them and re-write them into something more acceptable, or in cases in which you want to improve your capability to memorize creating associations between the elements.

This applies to declarative memories, memories that are related to practice and emotional ones are more stable. Not all memories are vulnerable in the same ways; it depends on how strong the encoding and consolidation process is. Enhancing memory via retrieval. The act of accurately retrieving information at one point in time increases the likelihood that the information will be accurately retrieved at a later point in time. This phenomenon is referred to as the testing effect — because tests of memory require retrieval — or the retrieval practice effect. The retrieval practice effect is even more pronounced if participants in the retrieval condition are given feedback on their failed retrieval attempts.

If you want to make someone regain a movement, you can make them imagine the movement and this can help the process of training.

Information affected	Positive effects	Negative effects
Retrieved information	With accurate retrieval (recall or recognition), later retention is improved	The act of retrieval opens a window for possible changes in memory from both external and internal influences
Related information	In some circumstances, the act of retrieval can lead to retrieval of related information	The act of retrieval can inhibit recall of related information under other conditions
To-be-encoded information	The act of retrieval can facilitate encoding of additional information	The act of retrieval can facilitate encoding of misinformation

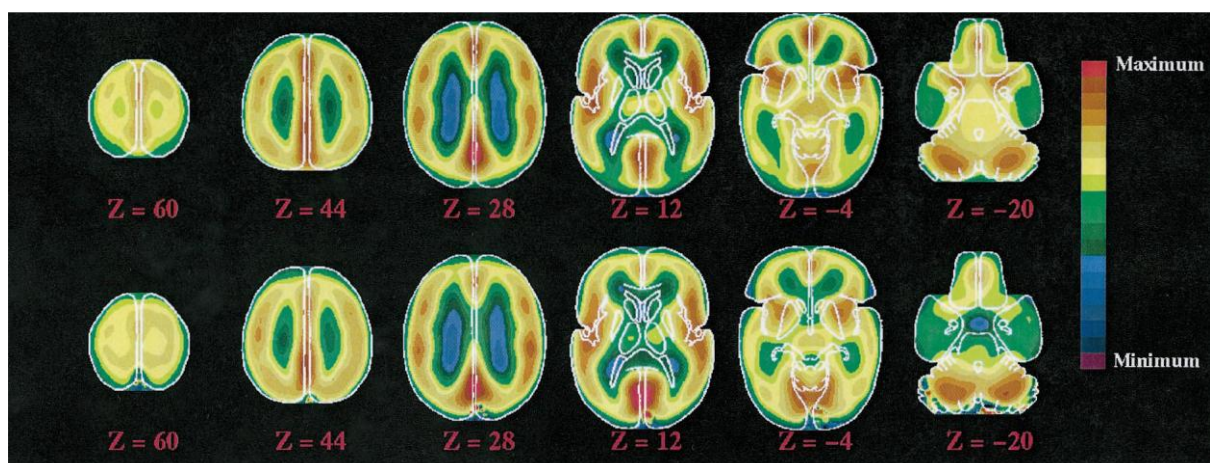


The brain at rest

We talk about how the brain works at rest because it has several implications for the study of the brain. Typically, we have a contrast in which we have a period of rest and a period in which we do a specific task, so we expect the regions that react to the specific stimulation to show a change in activity and then we compare the activity that we observe during the task to the images of brain at rest. So, the regions that show a response during stimulation are activated during it, involved in the task, and what we consider are the temporal and spatial correlates of the activation. What we were able to see is that while doing a task there are also regions that have a negative response, some parts get deactivated, and it may mean that during rest there are parts of the brain strongly activated. This led to the creation of one of the most widely used file dime in research with f-MRI that is called resting state, the most common type of data (simple way to detect it).

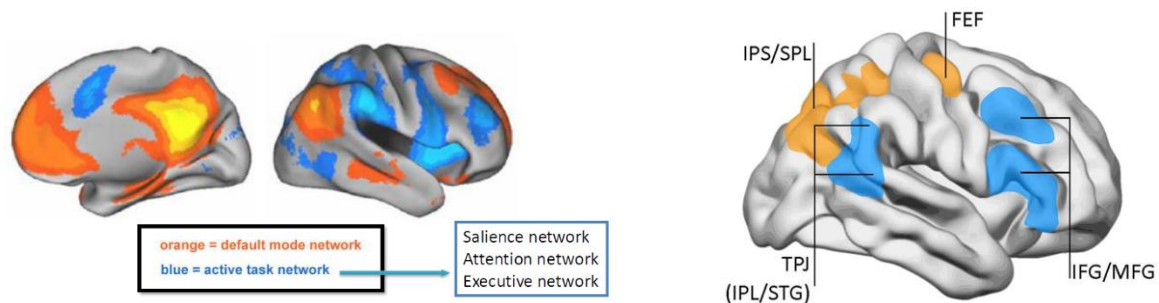
The idea at the very beginning was to collect a lot of data and see what would change in the brain for a pathological condition, and maybe even learn how to classify brains. It didn't really work out, because alterations in the brain can be found but are never specific in practice, we can have the same alterations in many pathological conditions.

There are two types of resting state, eyes open or eyes closed. The first one is typically preferred because you may fall asleep with eyes closed. The common view was that the resting state was a condition with minimal brain activity, but now the view is changed, and has been found that the increment in brain activity because of a task, it's very small (5% change in metabolic request), so even at rest the brain has a large energy consumption.



Quantitative maps of blood flow (Upper) and oxygen consumption (Lower) during quiet rest with eyes closed. Note the large variation in blood flow and oxygen consumption across regions of the brain. These vary most widely between grey and white matter. Despite this variation, blood flow and oxygen consumption are closely matched. There is a network of areas that are involved, and it is called “default mode network” because it works when you are at rest. These network includes the medial pre-frontal cortex, the medial parietal cortex, some parts on the sides, the superior frontal cortex, the lateral parietal cortex and also the lateral temporal cortex.

People started to do a lot of experiments to focus on what changed, also in negative terms, and they created some types of maps, representing the contrast between the task and the rest condition.

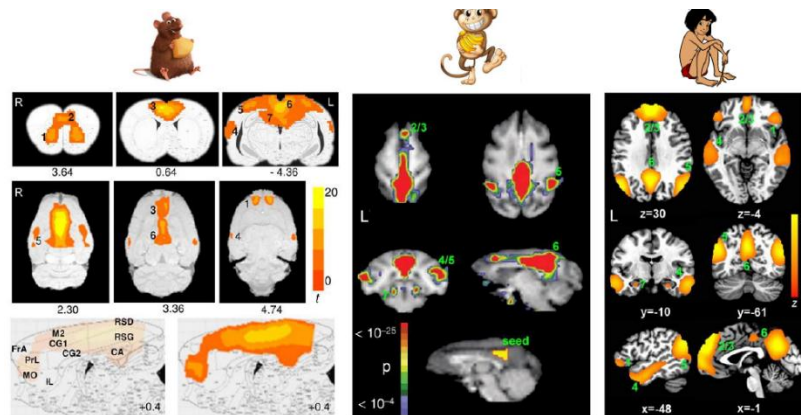


These types of images led to a view in which there are two main systems in the brain, one is the default mode network, and the other is the attention network (task positive network). In different tasks the patient will have activated some regions that are important for attention (right picture) toward the external world, top-down and bottom-up attention. In orange we have the dorsal system of the attention network, which includes the intraparietal sulcus (IPS) and the frontal eye field (FEF), that controls the movement of the eyes and so the direction of the attention, this is the voluntary top-down network for attention. The ventral system (in blue) includes indeed the right temporal parietal junction and the right ventral frontal cortex, the inferior frontal gyrus (IGF) and the medial frontal gyrus (MFG), and this is a system that activates when the attention is recalled spontaneously towards something. The moment you make your patient pay attention to something; the default mode network will decrease its activity.

The DMN oscillates spontaneously in activity every about ten-twenty seconds (increases and decreases, never 0). During the resting state the patient starts thinking about a lot of things, just like daydreaming, envisioning the future, retrieving memories, and gauging others' perspectives, and it appeared that DMN may be related to this process of thinking. Then they started to experiment what is called experience sampling (you keep the patient in the f-MRI and at a random moment you send an alarm and ask the patient what they felt, while scanning), so that you can have a report of what the patient was thinking about. It seems that this set of regions is related to something that is directed internally, a processing of internal experiences, everything that is unrelated to what happen around us. So, the moment people stop reasoning internally and pay attention to the external world the DMN is deactivated.

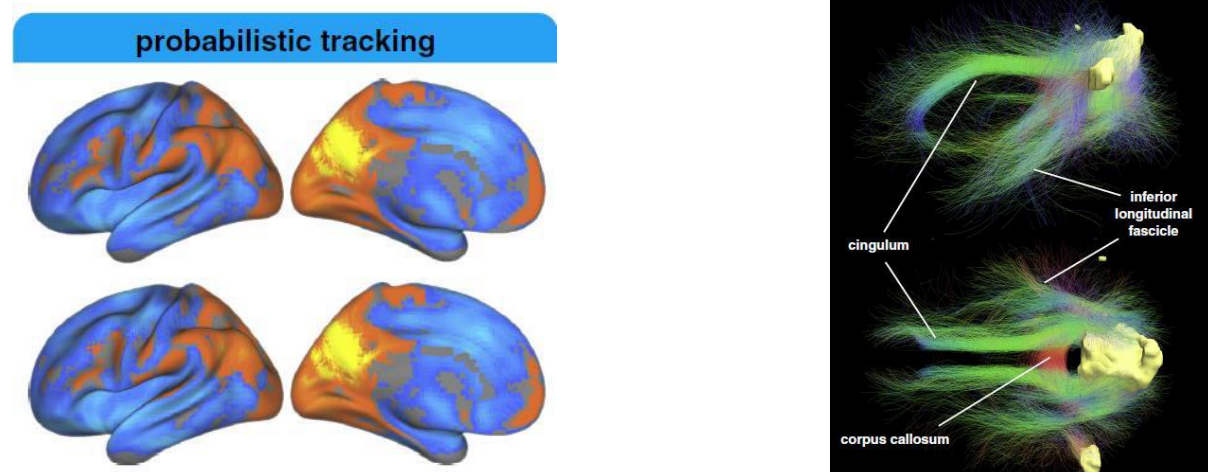
The function of the DMN is still unclear, and its role is still debated. The main school of thinking says that it is related to introspection and self-referential thoughts, when the direction is directed internally, spontaneous oscillation and continuous conflict between internal and external directing of thoughts. Some studies suggested indeed that the oscillation of DMN activity can influence our ability to pay attention to external elements, if I present you a stimulus in a moment in which the DMN is strongly activated it's very likely that you will miss it.

It's normal for the brain to shift from a task to another, but multitasking in the sense that we think about it like something in which our consciousness is divided across multiple things does not exist, we can only focus on one thing at a time. And here is the same thing, we can only focus internally or externally, not both at the same time.



Rats and Monkeys possess a DMN that is broadly similar to the DMNs of humans. A well-organized, intrinsically coherent DMN appears to be a fundamental feature in the mammalian brain.

What is interesting about DMN is also the fact that it seems to be a sort of a backbone for the brain, meaning that if you look to the structural connectivity, you see that they the strongest connected regions in the brain. Tractography is a technique (particular application of f-MRI) in which you use a signal that is sensitive to water movement in the brain, so water can move inside the cells only along the axons and if you look at the direction of the dominant movement in a specific voxel in the brain, you can track fibres of axons of the neurons.



The green things are all connections, and the region in yellow is the posterior part of the default mode network (praeuncus). As we can see there are strong connections between this area and the frontal region, the anterior part of DMN. These are the strongest connections in our brain. In children apparently is not exactly the same, the activity is not in the same regions than in adults, the visual cortex is strongly activated, and also in the motor cortex but not in the other regions, contrary to adults. In children most of the connections are local, in adults there are many long-range connections, and then during the growth process there is a reorganization of the brain.

The DMN is mainly studied with f-MRI.

One type of study is based on the view of DMN function depending on the condition of the patient or whether the DMN changes depending on the state of the patient. Even if you ask to meditate there is a change in DMN activity. Drugs and sleep also affect the DMN, during sleep particularly it gets stronger. Many different activities can have effects on the connectivity of DMN.

- **Attention to external stimuli:** A shift of attention toward specific stimuli or tasks is associated with a reduced connectivity within the DMN.
- **Meditation:** Long-term meditation practitioners may show reduced activation and reduced functional connectivity in the DMN.
- **Psychedelic drugs:** Psilocybin has been shown to be associated with a reduction of the blood flow to areas of the DMN (posterior cingulate cortex and medial prefrontal cortex).
- **Sleep:** Sleep deepening is associated with a decrease in connectivity between the posterior cingulate cortex and medial prefrontal cortex.
- **Sleep deprivation:** Sleep deprivation results in a decrease in connectivity within the DMN. A decreased anti-correlation between the DMN and the TPN can be also observed.

It's impossible that one f-MRI scan can determine if I have a pathological condition, it's not specific. The DMN overlaps areas involved in major brain disorders. In fact, alterations of the DMN have been observed in several pathological conditions. Discerning precisely which aspects of the network are affected by different disorders may provide new diagnostic tools. For example:

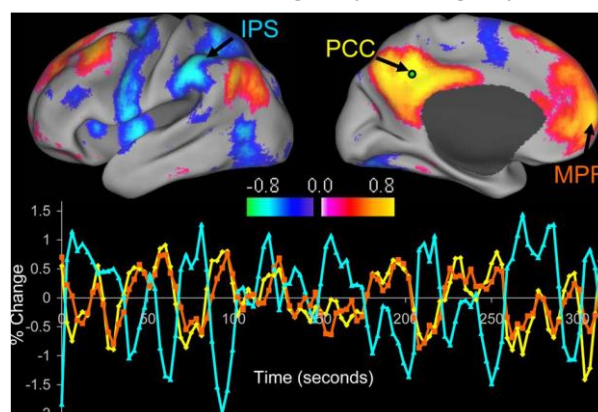
- **Alzheimer's Disease:** Brain areas that show marked atrophy in Alzheimer's overlap very closely with major centres of the DMN.
- **Depression:** Hyperconnectivity in the DMN seems associated with rumination (always thinking about bad things) in Depression.
- **Autism:** Patients may fail to reduce DMN activity during cognitively demanding tasks.

The methods used to study resting state functional MRI data, are mainly two (map-based methods):

- Seed-based functional connectivity (FC) analysis, hypothesis driven method (bottom-up).
- Independent component analysis (ICA), data driven method (top-down).

But also, a graph-based method, considering connectivity matrices and graph analysis.

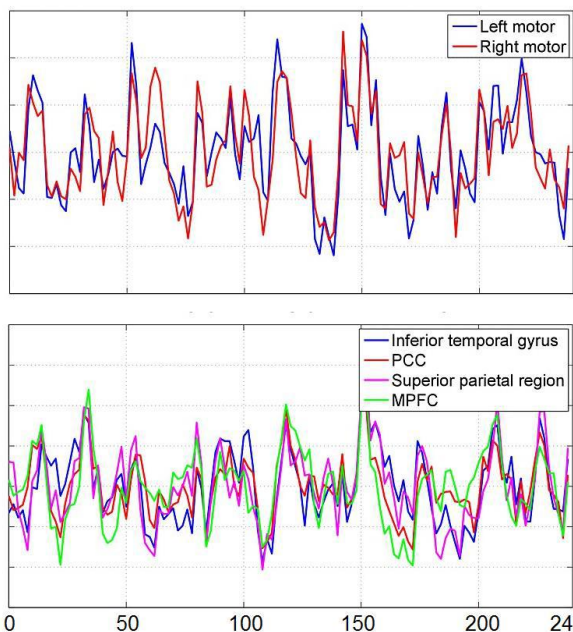
In seed-based correlation mapping you just need to decide a seed region, a target region that you use to look in the brain for other regions that correlate with the seed. I take a group of voxels in the brain, and I extract the time-series of the signal (f-MRI signal).



Example.

In this example we can see the region in yellow (PCC), which has a signal in yellow (BOLD signal) that records the hemodynamic changes over-time, through f-MRI. Then I take this time-series, and I calculate an instantaneous correlation (Pearson correlation coefficients) between this region and all the other processes in the brain. The correlation coefficient is close to 0 if there is no similarity at all, 1 if the time-series are identical, -1 if they go in opposite directions. Finally, you can apply some statistical thresholding and what you get is a map in which you can see in yellow or orange the regions correlated to your seed and in blue the regions that show an anti-correlation. Looking at this is clear that you have in orange the default mode network and in blue the regions that correspond to the attention network.

This kind of approach is called functional connectivity, that historically has been used to distinguish between the instantaneous correlation and lagged correlation. Basically, is used in contrast to the other one that is effective connectivity, that was originally (before 2015) used to indicate causal correlation. The idea was that seeing a change in activity in a region and then a corresponding change in activity in another region but after some time that would mean that the first region was causing the change to the secondary. But we cannot say this, in fact now effective connectivity is just used to indicate lagged connectivity.



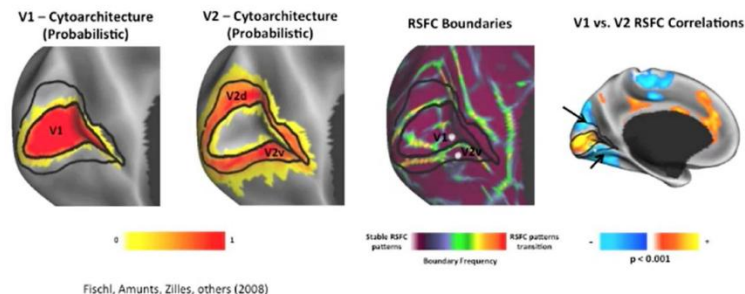
Then researchers realized something. If you place the seed in another part of the brain that is not part of DMN, you see that other networks pop-up. For example, if you place the seed (red one) on the right motor cortex and you explore the correlation between this time series and all other voxels in the brain, you see that at some point you will find regions that show similarity in activity. In the case at the left, for example the left motor cortex. They communicate even if the subject is not doing anything, only because they are strongly connected.

Hence, using this type of approach you can explore multiple networks in the brain, just moving the seed around the brain.

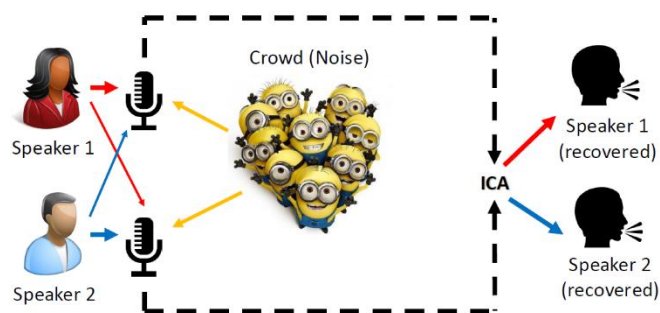
If you start to move around you can identify multiple networks and points for which if you move outside the area you start to see that the network changes, in this way you can map which part of the brain belong to which network and create functional atlases of the brain. These atlases are quietly available now to download and there are atlases that include 1000 regions, segmentation of very small regions, or 7 big areas, identification of larger networks.

The regions that you see at rest are exactly showing the same type of interaction that they show when they actually interact actively, so when you do this kind of approach in the visual cortex, and you start to move around within the visual area V1, when you move outside the border of this region you start to see a change in connectivity and you can move around and see that it shows a pattern of connectivity that reflects the secondary visual cortex (V2d, V2v). So basically these are

regions that have a specific functional meaning in terms of brain function, the same has been seen for the motor cortex or even in people with the eyes closed and blind people, especially for this last case you can see moving around in the visual cortex that the organization of it is still partially preserved.



The independent component analysis is a data driven approach, a method for separating multivariate signals into subcomponents.

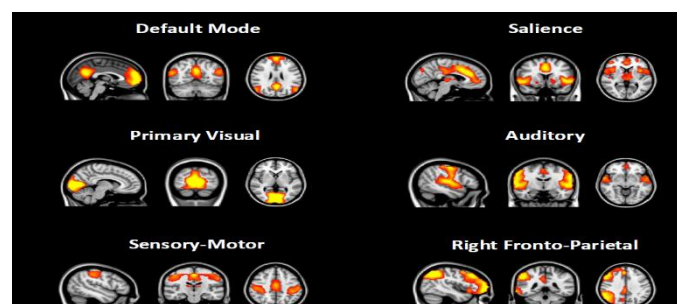


In this example you have the two speakers and some noise. What you recover in the end is that the signal sums up in the sensors (microphone), and with this method you can separate the sources assuming that they are independent and that each source has a non-gaussian distribution (blind-source separation).

This can be done also with f-MRI, but is not exactly optimal for this method, except for certain adaptations of the algorithm. The reason is that f-MRI has very low temporal resolution, so even if you acquire 10 minutes of data you get few 100s time points, considering that on the spatial site you have several thousands voxels, this disparity is not optimal.

ICA separates a signal into non-overlapping spatial and time components and for this reason can be used in the detection of resting state networks. It is highly data-driven and allows for better removal of noisy components of the signal (motion, scanner drift, etc). It also has been shown to reliably extract default mode network as well as many other networks with very high consistency.

Through this approach you are able to detect noise, just like head motion, heart pulsation, respiration, but they also found several networks. The most dominate one is obviously the default mode network, but you can separate for example a network that corresponds to the visual cortex (you have bilaterally the two visual regions), the sensory motor network (with the two motor cortices), the auditory network (involvement of the temporal cortex).



These networks can be studied in terms of not only functioning but also anatomical connections, because the functioning reflects the anatomy of the brain. If you do tractography (able to track the direction of axons), you realize that these functional networks actually correspond very well to anatomical connections in the brain. Sometimes you can find regions that show strong functional connectivity even though they are not anatomical connected.

By using this approach several networks have been identified, and they are independent from each other.



Resting-state networks consist of anatomically separated but functionally connected regions displaying a high level of correlated BOLD signal activity. These networks are found to be quite consistent across studies, despite differences in the data acquisition and analysis techniques. Most of these resting-state components represent functional networks of regions that are known to share and support specific cognitive functions. This organization can be affected in different ways by neurological disorders and brain lesions; the grade of damage can be seen by comparing the signal in the lesioned person to an atlas using mathematical transformations that can modify the shape of the brain that may be changed by the disorder.

One approach used for processing data is taking the anatomical and functional images and putting data in the space of the atlas (standard space, average brain anatomy), and then you apply brain parcellation to the data, then you extract the average time series for each of the regions of interest that you want to analyse so that finally you can use this data for example to calculate how much the activity of the DMN is similar or different to the activity of another network. You can also calculate matrices of correlation, calculating for each region the correlation with all the other regions of the brain, in which the diagonal corresponds to the identity (connectivity between the region and itself), and in each row you can see the connectivity between the region and all the other regions in the brain. Once you have this kind of matrix you can perform different types of analysis, for example you can build some graphs that represent connectivity within the brain.



In this case each dot represents a region of the brain, and the lines represent the strength of the connectivity (correlation between them). Once you have a graph like this you can extract many parameters.

- **Global efficiency:** Is a measure of how efficiently a network exchanges information. It is the average inverse shortest path length in the network and is inversely related to the characteristic path length.
- **Clustering coefficient:** Is a measure of the degree to which nodes in a graph tend to cluster together. It is the fraction of triangles around a node and is equivalent to the fraction of node's neighbours that are neighbours of each other.
- **Characteristic path length:** the average shortest path length in the network (lengths of shortest paths between all pairs of nodes).

Resting-state f-MRI confounds (not so important)

In the scanner there are different artifacts, and some of them may be avoided or minimized, others can't be controlled. Some parts of the brain (e.g. brain stems) move a lot each time we breath, and they are very noisy for the scanner. One problem is when the patient changes the breathing rate or when he stops breathing holding it, this because f-MRI depends on oxygenated blood and changing the way of breathing can affect the BOLD signal.

Movement of the head causes the content of each voxel to change, and this will determine a change in the BOLD signal! This is particularly problematic at tissue interfaces and around large vessels.

Functional connectivity conclusions may be erroneous when motion artifacts have a differential effect on resting BOLD signals for between group comparisons! For Instance, Children and patient populations are usually more uncomfortable in the MR environment than young healthy controls, resulting in a greater amount of head motion.

Cardiac and respiratory cycles are relatively high frequency (~ 1 Hz and ~ 0.3 Hz, respectively) compared to the low-frequency (< 0.1 Hz) BOLD fluctuations. Bulk motion related to the cardiac and respiration cycles will lead to similar confounds as motion of the head itself. Pulsations of the vessels caused by cardiac-related pressure changes will generate small movements in and around large blood vessels.

Changes in arterial CO₂ concentration and blood pressure (both under sympathetic and parasympathetic nervous system control) can determine low-frequency BOLD fluctuations. Regional low frequency oscillations may artificially inflate functional connectivity measures or cause spurious connectivity patterns where BOLD connectivity is not present or low.

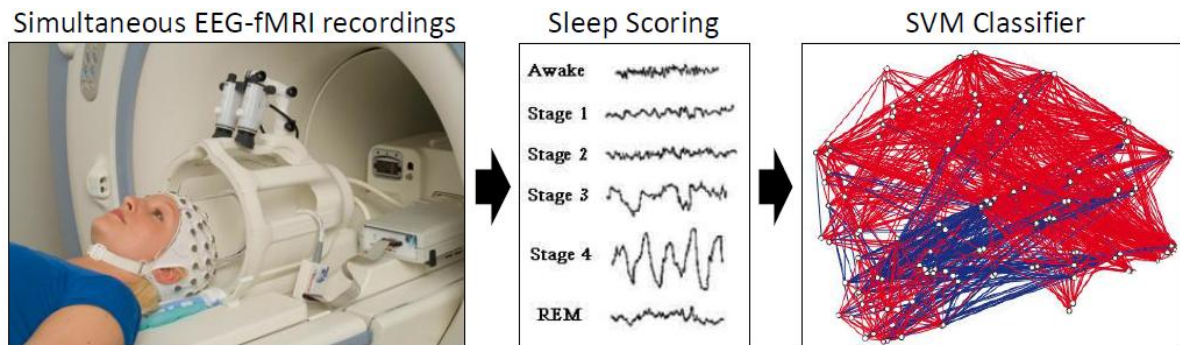
Hypercapnia arising from breath-holds increases BOLD signal whilst hypocapnia arising from hyperventilation reduces it. An increase in CO₂ concentration determines an increase in CBF as well as in the BOLD signal: the increased CBF reduces deoxyhaemoglobin concentration! The response of the vasculature to CO₂ is very sensitive, thus any changes in respiration rate and resulting CO₂ variations will have consequences for resting BOLD signal fluctuations.

Cerebral autoregulation is the intrinsic dynamic ability of cerebral vessels to maintain steady-state CBF despite fluctuations in arterial blood pressure. The arterial and arteriolar systems allow to maintain stable CBF by changing vascular tone through vasoconstriction and vasodilation processes. Blood pressure variations are a key source of CBF fluctuations due to the delay in dynamic autoregulation. Fluctuations in arterial blood pressure are likely to have a great influence on resting-state BOLD signal variations.

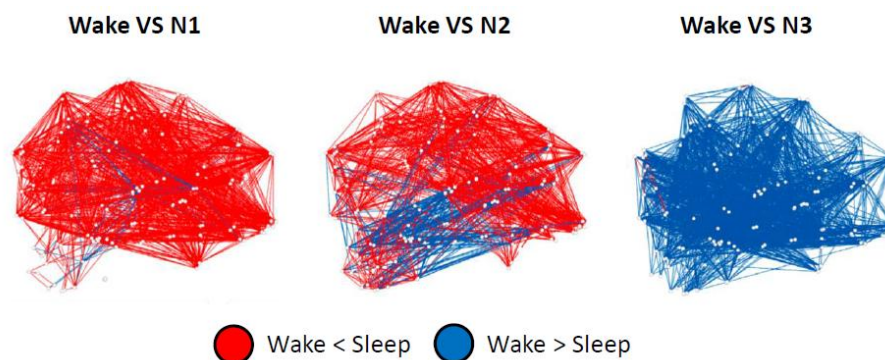
Implicit to the use of connectivity estimates over extended resting-state sessions is the assumption that the brain state of subjects will be homogeneous. Brain states associated with considerable neurophysiological changes should be monitored and avoided. Vigilance shifts indeed occur during rest; therefore, it is in the interest of researchers to maintain a steady brain state during fMRI acquisition (e.g., wakefulness).

Typically, a solution is keeping the experiment short.

The advantage of combining different techniques is that you can take the advantage of each of them. EEG with MRI is pretty basic, you just need an EEG that is MRI compatible (not by definition), especially the amplifier is not MRI compatible. Most importantly this combination allows to monitor the vigilance of the subject, how sleepy they are, because right now the only method we have to measure sleep is through EEG. The machine learning classifier is able to recognize the EEG stages based on the f-MRI connectivity.



What they found is that if you do an MRI scan of 10 minutes, 50% of the subjects will be asleep at the end.



The net reduction in connectivity observed during deep sleep reflect a reduction in brain interregional cross talk, i.e., a loss of global integration well in line with a phenotypically concomitant reduction in consciousness. Modularity, a spatial measure of functional segregation, generally increased across the brain with deepening sleep.

The issue if the subject fall asleep is that first of all data is not constant over-time, not uniform. But the biggest problem is that some fall asleep and others not, you can't compare them. People with social anxiety difficultly fall asleep knowing that someone is watching them. In some systems you can have eye-tracking, so that you can always know if the patient is still awake.

Vigilance and awareness

Sleep is important for memory processing, metabolism, tissue repair, memory processing, immune system, mood. When you sleep after something happen two things happen, the memory will be stabilized in the brain (increase of the signal to noise ratio), and sleep operates on mood, emotional regulation. When you sleep on it the reason that maybe angered you before sleeping it seems now softer, thanks to sleep.

When you have a patient that is in a vegetative state, they don't fall in a state of sleeping, it's when you see that they finally fall asleep that there can be possibilities for them to wake up and recover.

The brain has a sort of special system that is different from any other part of the body, the glymphatic system. In the brain there is no lymphatic system, but researchers saw that during sleep something was changing, the shape and position of the astrocytes, creating space from one cell to another, canals used to move around fluids that remove from the brain metabolized things, if you don't remove them the brain saturate. This system doesn't work properly in patients who develop Alzheimer disease.

In an experiment some people had to do some tasks, after sleeping their performance improved, with respect to the deepness of their sleep. Deeper sleep has more slow waves, and these slow waves were specifically located in the regions of the brain involved in the learning process.

With another experiment it was found that if the subject practice and then slept, he was much more likely to find the solution to the problem compared to the people that didn't.

Experiment: Randy Gardner

Randy Gardner (born 1947) is the holder of the scientifically documented record for the longest a human has intentionally gone without sleep not using stimulants of any kind. In 1964, Gardner (at the time high school student in San Diego, California) stayed awake for 264.4 hours (11 days 24 minutes). The SD led to serious cognitive and behavioural changes:

- Alterations of mood.
- Problems with concentration.
- Problems with short term memory.
- Paranoia.
- Hallucinations.

On the eleventh day, when he was asked to subtract seven repeatedly, starting with 100, he stopped at 65. When asked why he had stopped, he replied that he had forgotten what he was doing. It was seen also that the need of sleep put us in a negative mood.

For children the effect of not sleeping is worst, because they are in a period of development in which the brain really needs sleep. You can die after months without sleeping.

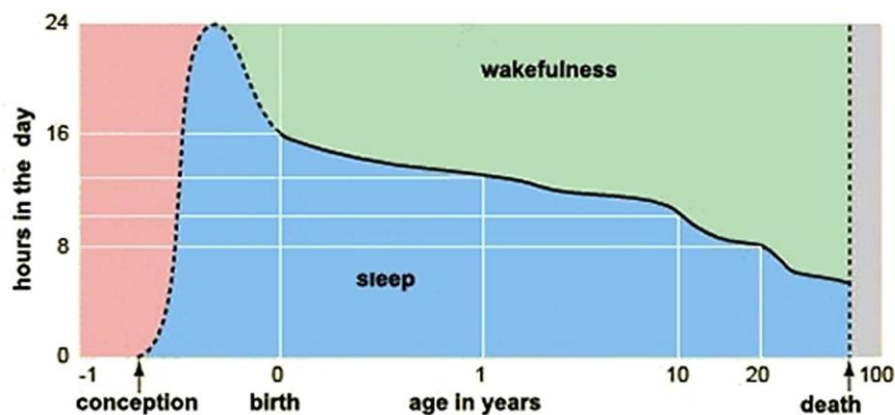
Fatal familial insomnia (FFI) is an extremely rare autosomal dominant inherited prion disease of the brain. It is usually caused by a mutation to the protein PrPC but can also develop spontaneously in patients with a non-inherited mutation variant called sporadic fatal insomnia (sFI). FFI has no known cure and involves progressively worsening insomnia, which leads to hallucinations, delirium, confusional states, and eventually, death. The average survival time for patients diagnosed with FFI after the onset of symptoms is 18 months.

We can live with also few hours of sleep, but there are going to be alterations in behaviour. Performances are affected by the need of sleep.

For older people the amount of sleep they are able to maintain decreases, but precisely because the brain can work with less sleep, but only because you are able to maintain sleep for lesser time.

Humans sleep about 8 hours a day, but it depends on the age. New births sleep also for 16 hours, very often non continuously, but then at some point sleep stabilizes (1-2 years of age) decreasing

and the reaching 8 hours around adolescence. When we are younger (from 10 to 20 years old) we need to sleep more, because our brain needs it for memory and plasticity to occur. In fact our brain becomes lesser plastic, the capability to learn and adapt to new environments decreases over-time.



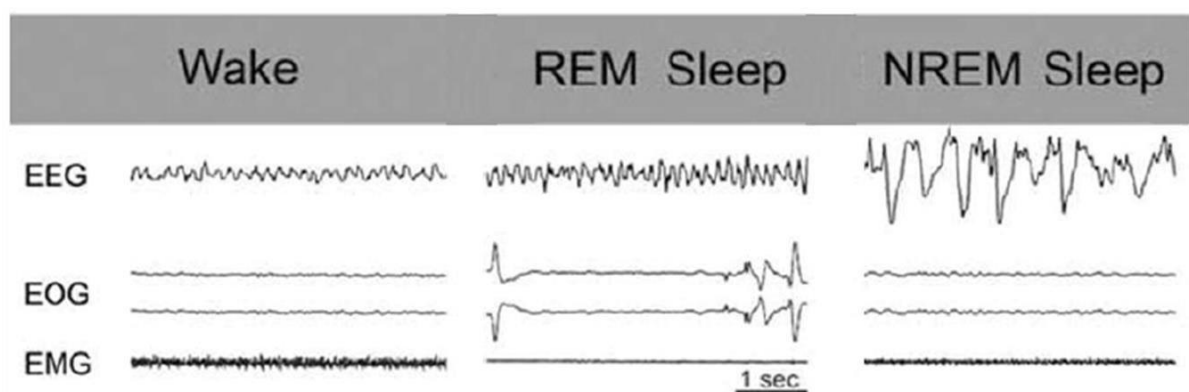
Sleeping too much (more than 10-12 hours per day) is also bad, if it happens multiple times. If you do that you have higher probability of developing some metabolic or other diseases.

Definition of sleep

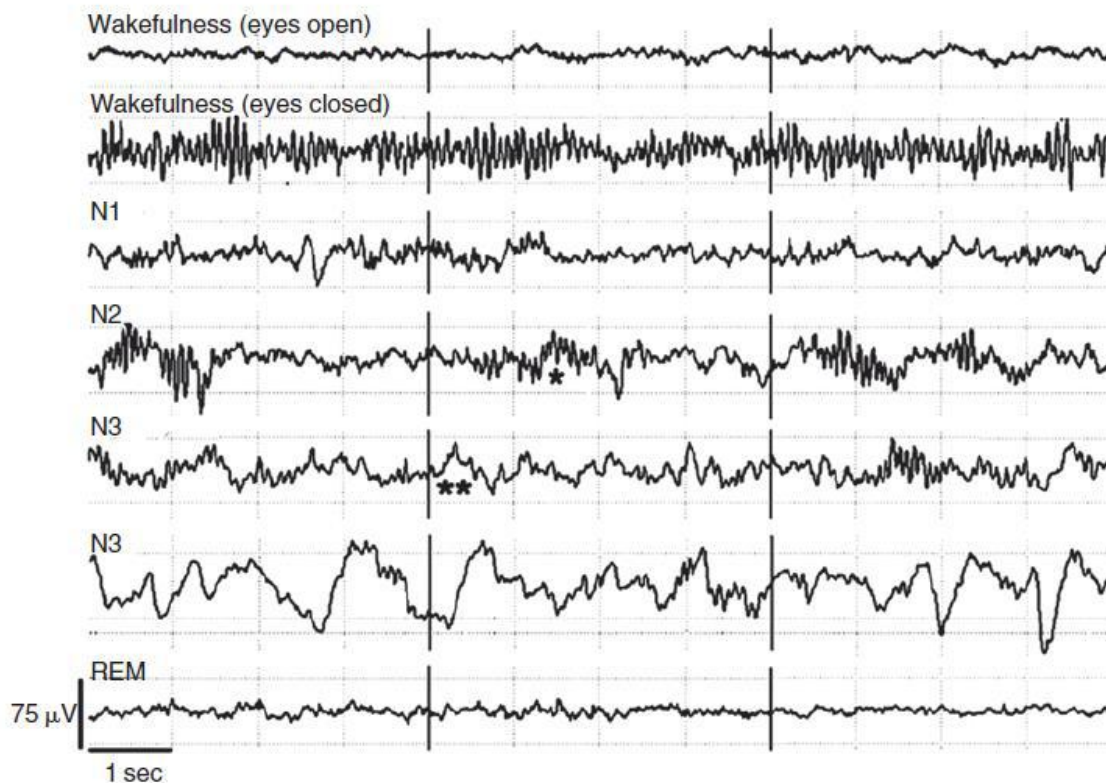
Behavioural: Behavioural state with characteristic immobile posture and diminished but readily reversible sensitivity to external stimuli.

But this definition is not so efficient, because the person may be in a vegetative or come state, not sleeping at all or just faking. So, we need a marker to define whether someone is sleeping or not and this comes with EEG. We can use also other measures, like temperature of the skin or heart rate, but the EEG is the most reliable.

Electrophysiological:



We typically use three signals, because we need to distinguish different conditions. The wake EEG comprehends very fast activity; the eyes movement are voluntarily controlled and there is high muscular tone. Then when I enter the REM sleep, the activity may be similar in terms of EEG, but in the EOG I get a non-voluntarily controlled eye movement, and the muscular tone is decreased. That is why I need these two additional signals. Non-REM sleep is recognizable because there is no eye movements, the muscular tone is intermediate, and the brain goes on-off all the time, completely different pattern of activity.



In the first line we can see that wakefulness with the eyes open is dominated by a fast activity, but then when I close my eyes α -activity appears, this is called the α -block and it happens because this activity is mainly produced by the occipital cortex, by the visual regions in the brain, and is produced when the visual regions are not receiving inputs. If the subject is tired, sleepy, his α -activity will be larger when he closes his eye and will also spread to more anterior regions (frontal).

Then at some point if I try to fall asleep this α -activity will disappear and the pattern of activity will slow down, going to another stage called non-Rem state (N1, N2, N3, N3 (N4 at first)). When you go from N1 to N3 basically the brain slows down, the oscillation is slower in frequency and bigger in amplitude. Then when you have REM sleep you go back to a faster activity.

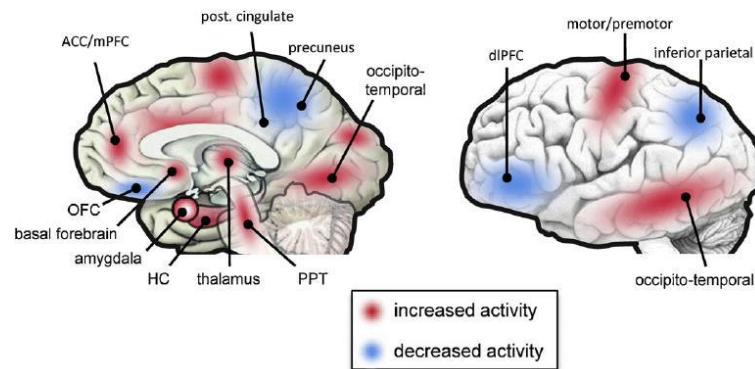
REM sleep is not a state that you reach immediately, typically happens after 90 minutes when you fall asleep, specially at night.

It seems that NREM is more important than REM from a functional perspective, right now the researchers are trying to amplify the number of slower waves, so that I can sleep less than 7 hours. Drugs and anaesthesia drugs don't work. Right now, sensory stimulation is being used, the reason is that basically when you are in sleep already, the brain is trying to maintain sleep, so if you give a stimulus not too strong, the brain will generate more big slow waves. Trying it the professor didn't find an advantage from it.

Changes in brain activity

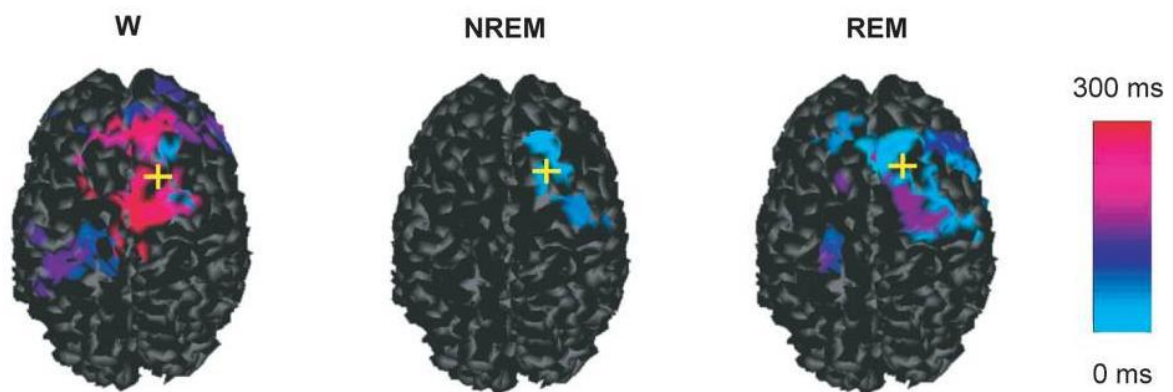
There are different arousal systems activated in the brain that all turn off when you fall asleep, with one exception, acetylcholine. During sleep there is a reduction in metabolism in the brain, but not generalized, in NREM sleep there is a reduction in metabolism in regions that overlap in

part the default mode network and in part the thalamus. Sleep is not a state of overall reduction of metabolism.



Overall, REM sleep seems to be in a more active state, there are a lot of things going on. In REM sleep there is a lot of activity in the limbic system, in the hippocampus and the amygdala, regions that process memories and emotions, sleep is important for memory consolidation and emotional processing. There are theories that say that REM sleep at least decreases the emotional components in memories.

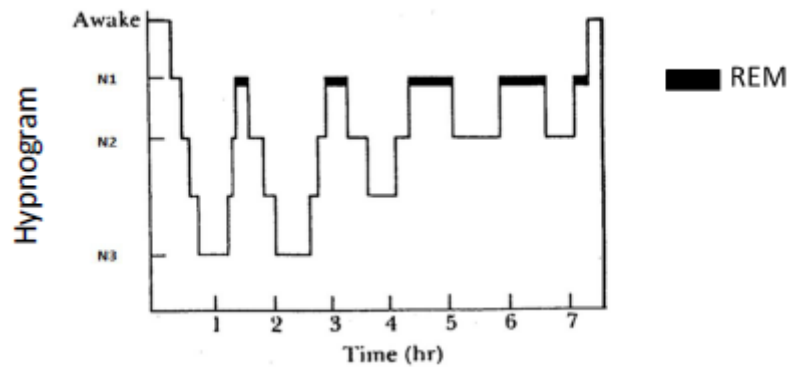
When we look at the states, we see that they have different characteristics in terms of complexity.



In this study they used TMS+EEG. The yellow cross is the site where they applied transcranial magnetic stimulation and the colour represents the response of the brain to the stimulation; red is late response. When the subject is awake the stimulus flows across the different regions forth and back in different directions, but if you do the same in NREM sleep you have a short lasting and very simple response, in which most of the brain response remains localized. And finally in REM sleep you have something that is in between, you get a more complex response.

Sleep cycle

The hypnogram represents the distribution of the different sleep stages during the night. On the x-axis you have the time since falling asleep and in the y-axis the time spent in the different stages. You start awake and then into N1, N2, N3, you spend some time into N3 sleep and then you go back to REM sleep, and this is repeated several times. Each time, the amount of time spent in N3 decreases and is especially strong at the beginning of the night. Young people are much more efficient than older ones, so if you take someone around 20-25 years old they will produce a lot of huge slow waves, so a lot of N3 sleep, but when you get older the amount of N3 sleep decreases.



The succession of NREM sleep stages followed by an episode of REM sleep is called a sleep cycle and lasts approximately 90–110 minutes in humans. There is a total of 4–5 cycles per night. Slow wave sleep (N3) is prominent early in the night, especially during the first sleep cycle, and diminishes as the night progresses. As slow wave sleep wanes, periods of REM sleep lengthen and show greater phasic activity. A healthy young adult typically spends about 5% of the sleep period in stage N1, about 50% in stage N2, 20–25% in stage N3, and 20–25% in REM sleep.

The first two cycles are especially important, because they contain most of the NREM N3 sleep, and this is why if you really need to sleep is better to sleep at least 2-3 cycles.

Sleep regulation

The less I sleep the more my brain accumulates sleep need. The homeostatic factor of sleep regulation just tells you that you need to sleep, and the amount of sleep you need depends on how much you have slept before.

The other concept that is crucial is the concept of circadian regulation, and these two regulatory factors interact with each other, to determine your sleep propensity during the day. Û

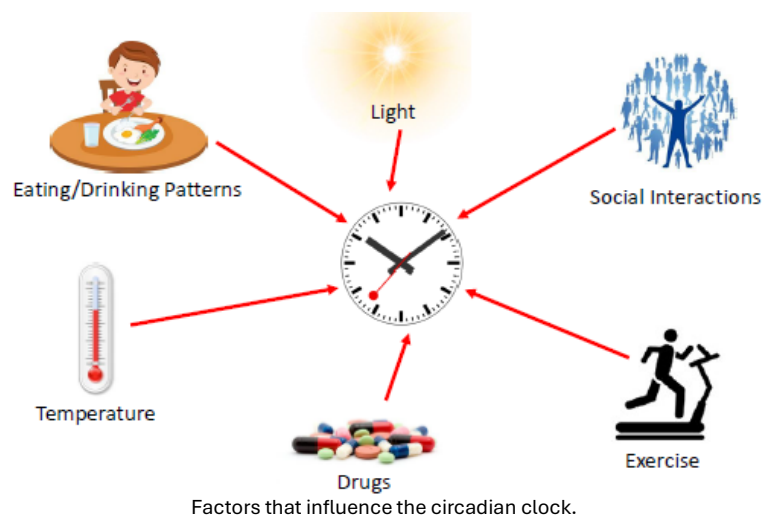
In sample terms circadian regulation is our internal clock. We have cells and systems that regulate the work of our body with an autonomy of 24 hours, but others can be also tuned in a slightly different rhythm, maybe of 24.5/25 hours. But then these genes and system can be entrained, being regulated by exposure to light.

In our retina there are specific receptors that are different from the receptors to see, and these are especially sensitive to blue light, and they send a signal to a nucleus (suprachiasmatic nucleus), located in the anterior hypothalamus. This nucleus regulates the release of a hormone called melatonin that is produced by the pineal gland. In normal conditions, after the beginning of the dark period you have an increase in release of melatonin, and its circulation signals the transition to sleep, determining drowsiness and a reduction in the core body temperature. Right now (this period with phone, tv etc.), the presence of light sources can affect and delay the release of melatonin, reducing the quality of our sleep.

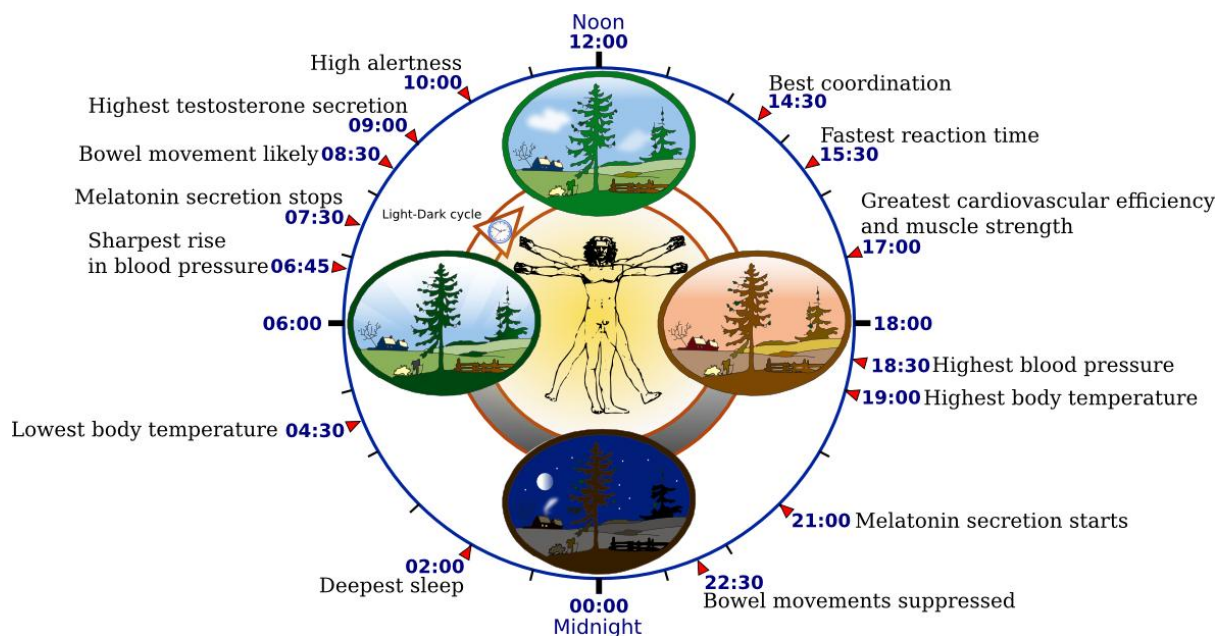
Exposure to light is a factor that regulates the release of melatonin, because our internal clock is not regulated exactly in 24 hours. Light is not the only factor, others contribute to align the circadian clock to the actual time, if we don't do this alignment our cycle of melatonin will follow the normal circadian clock (not normally aligned).

- The rhythm has an endogenous free-running period that lasts approximately 24 hours.
- The rhythms are entrainable (zeitgeber).

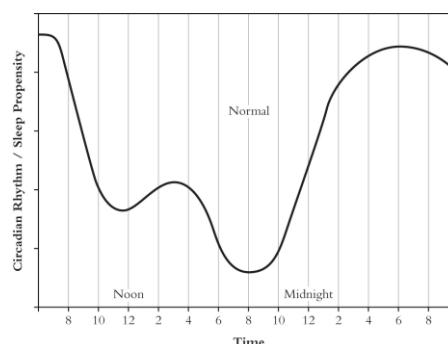
- The rhythms exhibit temperature compensation (maintained circadian periodicity).



All functions in the body are all affected by the circadian clock, they are not under our control.

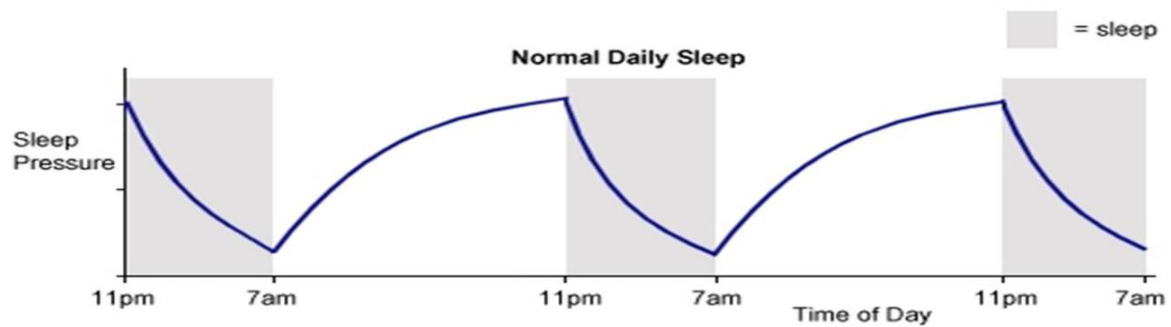


Sleep propensity is modulated by the circadian clock. Starting from midnight, is the moment in which sleep propensity is the lowest (non-sleep zone), and it then starts to increase reaching a peak around 3-5 in the morning, and then during the night it starts to decline again. During the day, after launch, sleep propensity starts again to increase, related also to the digestion process but is also something that is really inside our circadian system.

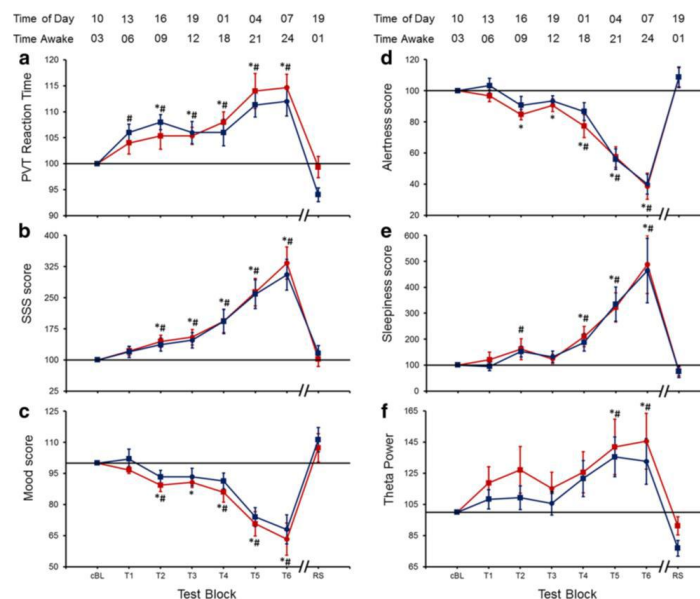


Homeostatic sleep regulation

The more we stay awake the more we accumulate sleep need. But what drives this homeostatic factor? We think it could be the accumulation of metabolized, that are produced by the brain activity. It has been seen that adenosine is associated with the increase of sleep need (produced by the energetic consumption), reaching specific receptors on cells and makes us sleepy. This is why coffee works, because it binds the same receptors and blocks these receptors, but you can't drink too many coffees, because adenosine will accumulate and then when you stop drinking coffee you will feel worst.



The homeostatic sleep need and the circadian regulation interact together.



In this example, in which the speed of reaction time was measured, you can see that it goes up and then down exactly in the time where the circadian propensity to sleep is at the lowest, so at 19 my performances improve, and then after the non-sleep zone passes my sleep need will increase again.

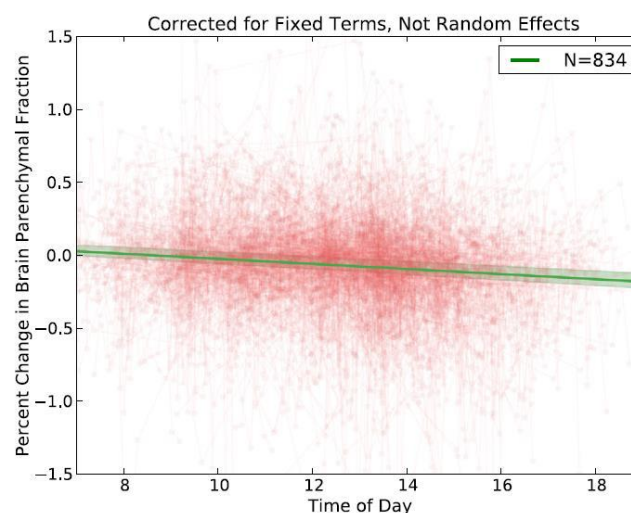
The same kind of pattern can be seen also for behaviour, mood. For the last one they asked people to rate their mood during the day (every 2 hours), and it tends to go down at the beginning of the task and then they recover after lunch, and finally it goes down again during sleep before recovering after sleeping. In this experiment they went sleeping in the morning, not the best because I am much less likely to sleep well. If I go to sleep outside the circadian optimal window the sleep will never be optimal.

Looking for the other graphs (alertness, sleepiness), they also follow the same pattern, they felt sleepy in the afternoon and then it decreases in the evening.

The last graph represents instead brain activity, from which we can see that it follows exactly the same pattern, in particular here we looked to low frequency activity, not exactly as low as sleep frequency, but slower than normal frequency. Delta activity (4-9 Hz).

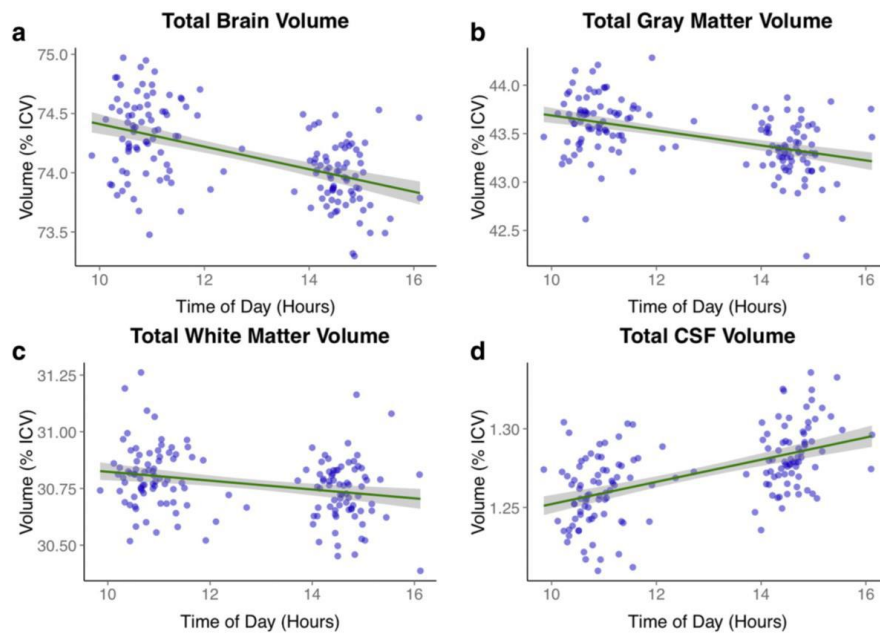
Brain activity changes during the day, if you measure it during the day, you will get different pattern of brain activity. Also apparently, in the brain, different regions have different sensitivity to the circadian regulation, showing a modulation of activity that is differently aligned with the circadian clock. Brain activity is not constant during the day.

A study published showed that the brain structure changes across the day, the measures of the brain structure that we have. What we get measuring is a magnetic resonance image, an image that reflect not only the characteristics of the brain tissue but also water distribution. The study took the data from a database, comprehending the time of the day, and looked at the brain size (volume), measured with the techniques that we normally use to analyse MRI data, across the day. What they found is that, if you correct for all possible variability across individuals, you seem to see a reduction in brain size across the day.



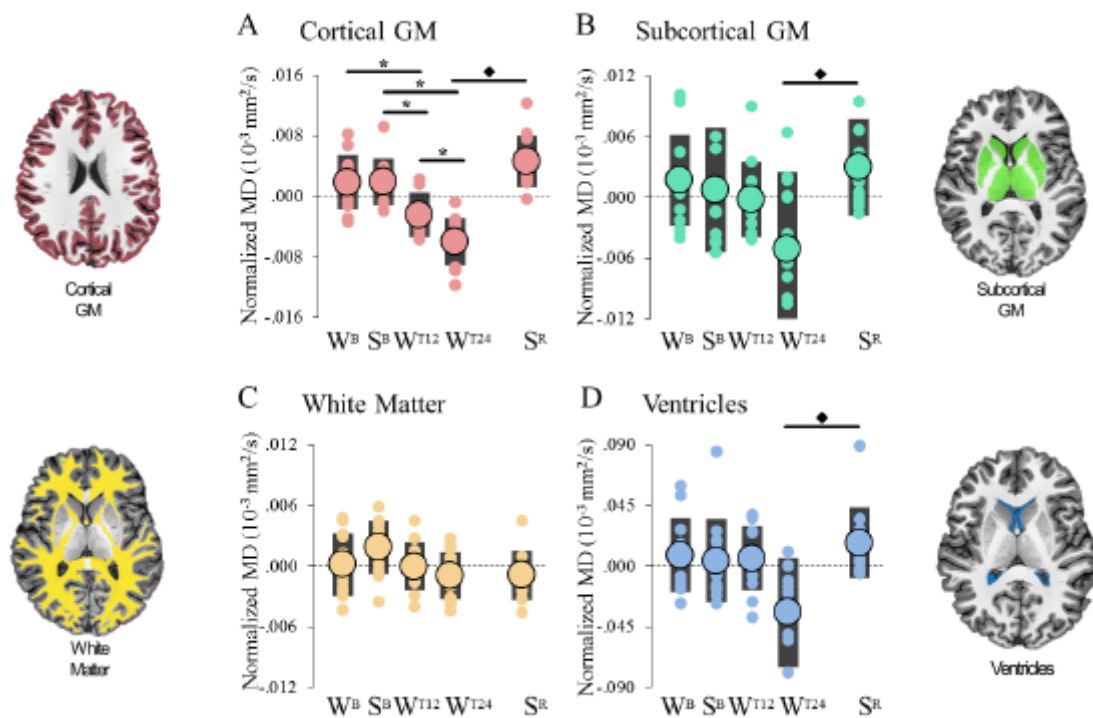
It was seen in another studies that if you practice with a task there is an increase in brain matter volume specifically in regions related to the task, so apparently these changes may not reflect just water distribution but real anatomical changes. So, we don't know actually, because we have only thus indirect measure. If you do an experiment, it's important that data are extracted at the same time of the day.

In other studies, it was shown that even with only 4 hours of difference from one scan to another, it is associated a difference in brain volume that you can measure. There is only one thing that increased, total CSF volume, intracranial volume for CSF (amount of volume), so in this case the changes that we see are not really anatomical but related to a redistribution of water in the brain. Water basically moves from inside the brain to around the brain, when we go from the morning to the evening.



For another study of the professor, MRI scans were taken from Friday evening and then every 12 hours. What you can see is that when the subject stays in the lab doing task for 24 hours the brain shrinks and their cortical grey matter decreases, increasing again after sleep.

Measures of how much water can move in the tissue that you are analysing were also taken for different regions. In the first case the decrease means that in the cortical grey matter water was freer to move, if you increase the number of synapsis or number of cells water will have less freedom of moving because there is less space. If you learn to do the task, more space in the tissue will be occupied by synapsis for instance and so water will have less movement.



The key point is that brain activity and brain structure measures changes across the day.

Implications for neuroimaging

All subjects should be examined at around the same time of day (If possible, in a time window not larger than 1-2 hours). It is good practice to report in your article at what time of day the data was collected and whether consistency across subjects and groups was ensured!

It's also important to choose the best window of the day in which collect my data, and that is in the morning around 10-13, when the subject is fully awake and alert. The circadian pressure is low and also the homeostatic pressure is relatively low, so I can expect the performance to be maximal. It depends also on the habits of the participants; you have to match the circadian alignment of the person.

Overall vigilance and sleep quality may be assessed using simple questionnaires (useful to exclude extreme cases). Actigraphy recordings may be used to ensure no between-group differences in sleep time before the experiment, or to exclude a lack of circadian alignment.

Another instrument that is very useful (present also in smartphones) is the actigraphy. It is an accelerometer that looks at the activity of the arm and tells you if the subject is still, sleeping or moving. You can measure the circadian rhythm of the subject just looking when they wake up and fall asleep. It can also control the exposure to the light, temperature, heart variability and say the different stages of sleep.

First you have to indicate the beginning and the end of the window in which the subject was more likely to be sleeping and then the software calculates, based on the amount and amplitude of movement, the amount of sleep that the subject has. With this measures you can see how stable the time is the subject wakes up and fall asleep, and you can measure the central point in this window that is an indication of the circadian alignment of your participants.

If group-level differences cannot be excluded, the best practice is to monitor the state of vigilance during the MRI session and discard subjects or segments including behavioural sleep. Though it is impossible to avoid the noise.

It can be helpful to ask the patient to keep the eyes open, you can do an EEG-fMRI co-registration for sleep scoring, and you can do shorter acquisitions.

The difference between wakeness and sleep is not just the time we spend in bed but sleep is part of wakeness. Sleep is regulated across it, our vigilance change in the day as well as our sleep need, and there are specific mechanisms that regulate it and these have an effect on behaviour, brain activity and even brain structural measures.

Somatosensory

We can think that vision is the main modality conveying information, but when we talk about all the different sensor modalities, we can imagine that all can be used for several tasks including aspects of processing. The somatosensory modality is the first modality that develops in the human brain, you can consider touch as a very basic aspect of perception.

When we talk about somatic sensibilities we should imagine that we have a variety of receptors that are distributed throughout the body that provide different pieces of information about not just what we typically call touch, that is something (at least in his active form) that we use to recognize size, shape, etc...; we have also proprioception, the sense of static position and movement of limbs and the body, something that is not just related to perceiving the effect of the movement of our levers but also to represent body per se. All elements that should be embodied, we have to think also of the perception of our body.

The bodily senses

Somatic sensibility arises from information provided by a variety of receptors distributed throughout the body. Somatic sensibility has four major modalities:

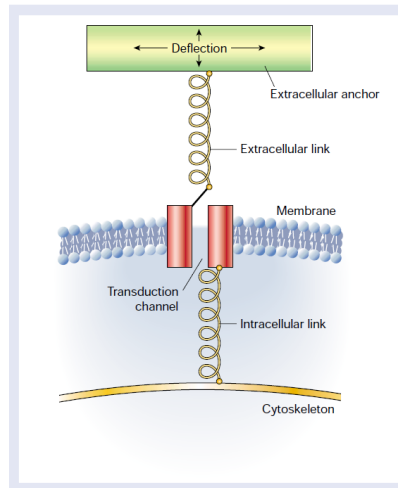
- discriminative touch (required to recognize the size, shape, and texture of objects and their movement across the skin).
- proprioception (the sense of static position and movement of the limbs and body).
- nociception (the signalling of tissue damage or chemical irritation, typically perceived as pain or itch), not only burning and stinging, but also chemical responses of the nerves to specific notch stimuli.
- temperature sense (warmth and cold).

These elements that are going to discriminate all this information relies on different population of receptors, typically capturing mechanic changings in mechanic forces, that are not just located on the skin but also located in the muscles, in the joints, etc.. All this piece of information from the periphery of the body are carried to the brain by different pathways, that run paralelly inside the spinal cord, then in the brain stem and then in the thalamus, reaching the primary somatosensory cortex in the post-central gyrus of the parietal lobe.

When we think about touch we primarily relies on the idea of skin, that have different components, including glands that produce liquids not only for regulating temperature but also to favour the interaction of the skin with objects. We divide the receptors on the skin in three groups:

1. Mechanoreceptors, able to capture mechanical forces.
2. Nociceptors, able to capture painful stimuli.
3. Thermoreceptors, able to find thermal information.

This division of function is integrated also by a division on the morphology, with receptors on the body surface could be free type or encapsulated type. Despite their variety, all these receptors are actually able to capture every piece of information getting to the skin, and then affecting the permeability of the membrane, allowing the sensory neuron to transduce the signal to the spinal cord and then to the brain.



The membrane gets typically anchored with the protein to the cytoskeleton structure, and then there are some kind of information getting and moving in the extracellular anchor, and once this is deflected than the channel opens, inducing a sodium depolarization. This is how typically we transduce mechanical forces.

All these kinds of receptors are then associated to specific fivers. The AB fivers are large and myelinic, meaning a higher conduction velocity, maintained by the myelin around the nerve. The non-encapsulated (free nerve endings), instead rely on C fivers or Ad fivers; these receptors appear to be very important for primarily conveying information about nociceptive and painful stimuli and thermal changes, but these C and Ad fivers are each amyelinated and poorly myelinated, so with a slower conductivity.

Table 22-1 Receptor Types Active in Somatic Sensation

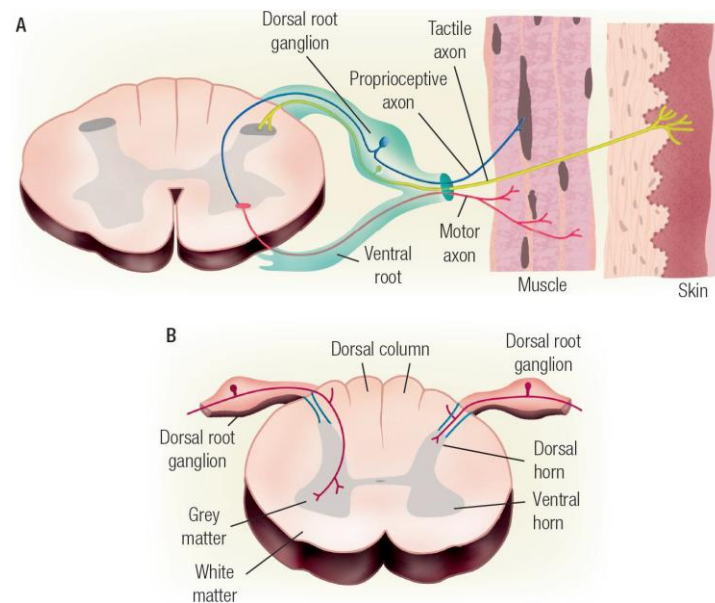
Receptor type	Fiber groups	Fiber name	Modality
Cutaneous and subcutaneous mechanoreceptors			Touch
Meissner's corpuscle	A α , β	RA	Stroking, fluttering
Merkel disk receptor	A α , β	SAI	Pressure, texture
Pacinian corpuscle	A α , β	PC	Vibration
Ruffini ending	A α , β	SAII	Skin stretch
Hair-tylotrich, hair-guard	A α , β	G1, G2	Stroking, fluttering
Hair-down	A δ	D	Light stroking
Field	A α , β	F	Skin stretch
Thermal receptors			Temperature
Cool receptors	A δ	III	Skin cooling (25°C)
Warm receptors	C	IV	Skin warming (41°C)
Heat nociceptors	A δ	III	Hot temperatures (>45°C)
Cold nociceptors	C	IV	Cold temperatures (<5°C)
Nociceptors			Pain
Mechanical	A δ	III	Sharp, pricking pain
Thermal-mechanical	A δ	III	Burning pain
Thermal-mechanical	C	IV	Freezing pain
Polymodal	C	IV	Slow, burning pain
Muscle and skeletal mechanoreceptors			Limb proprioception
Muscle spindle primary	A α	Ia	Muscle length and speed
Muscle spindle secondary	A β	II	Muscle stretch
Golgi tendon organ	A α	Ib	Muscle contraction
Joint capsule mechanoreceptors	A β	II	Joint angle
Stretch-sensitive free endings	A δ	III	Excess stretch or force

¹ See [Table 22-1](#). ² Pacinian corpuscles are also located in the mesentery, between layers of muscle, and on interosseous membranes.

These receptors at the level of the skin get back to the ganglia neurons, located dorsally closed to the vertebra, so once these neurons get information from the periphery they then project it to the spinal cord, so on the periphery there is a long peripheral branch that gets information from

the skin, transmitting the action potential to the soma and then to the synapses on the spinal cord. These neurons are called T-shaped, because they have two axons, one is a peripheral axon that gets to the skin and the internal organs, and the other goes towards the spinal cord. These neurons are primarily located along the vertebra, because closer to the vertebra there is an empty channel where the spinal cord stays and information from the periphery has to get there.

The Bell-Magendie Law says that the anterior portion of the grey matter in the spinal cord (ventral horn) is primarily conveying motor information to the muscles, and dorsal horn receive sensory afference from the skin and the different body parts.



Once the information gets from the skin to the dorsal ganglion route then it enters the spinal cord and goes up to the brain stem and then to the thalamus, and finally to the post-central gyrus in the cortex. In the spinal cord there is a fine organization of all the pieces of information getting up and here all these pathways (thermal, proprioceptive, tactile) flow parallel up to the brain, and also here there is a specific organization that mirrors topography of the body. In the spinal cord there is a primary organization of the information that goes up to the brain.

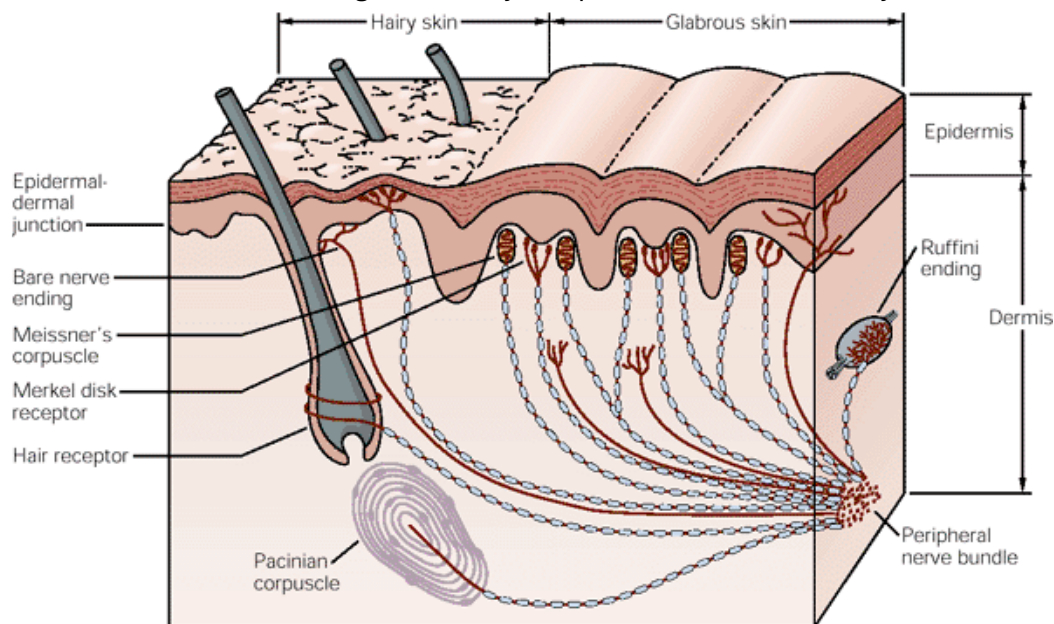
There are four major types of encapsulated mechanoreceptors, primarily specialized in providing information and capturing all mechanical information (pressure, vibration, tension), and all these receptors are collectively referred to as high sensitivity mechanoreceptors, because even weak mechanical stimulation of the skin induces them to produce action potentials. One aspect very typical of all sensor modalities. All low-threshold mechanoreceptors are innervated by relatively large, myelinated axons (type Aβ), ensuring the rapid central transmission of tactile information.

There are two attributes that refer to somatic sensations: epicritic and protopathic.

Epicritic sensations involve fine aspects of touch and are mediated by encapsulated receptors to: detect gentle contact of the skin and localize the position that is touched (topognosis); discern vibration and its frequency/amplitude; resolve by touch spatial detail, such as the texture of surfaces, and the spacing of two points; recognize the shape of objects grasped in the hand (stereognosis). You should consider tactile information as information that needs to be elaborated, because in this case we don't immediately have the whole information (like in vision) but we have a sequential acquisition of information that we have to merge.

Protopathic sensations involve pain and temperature senses (as well as itch and tickle) and are mediated by receptors with bare nerve endings.

If we want to know where these high sensitivity receptors are located it's very difficult.



The glabrous skin is made up of ridges that are actually able to better capture and discriminate texture, so they favour the capture of very small mechanical forces in the skin. Some encapsulated receptors are located immediately subcutaneously, where the ridges are, while others are encapsulated deeper, so that we distinguish cutaneous and subcutaneous receptors. In addition to the four encapsulated receptors, we also have free nerve endings that reach the surface of the skin, but there are also nerve endings getting around the hairs or directly on the hairy surface.

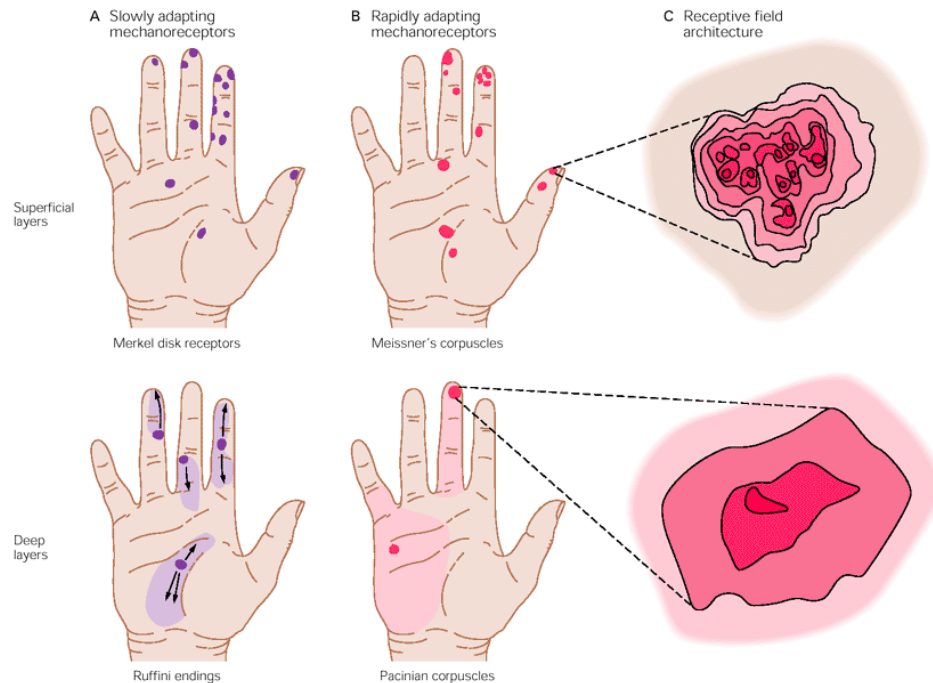
There is a 2 by 2 design, and an important aspect is the location, the Messner and the Merkel are located cutaneously, the Pacinian and the Ruffinian are located subcutaneously. But this design is not only related to location, but also to the kind of response. In order to define the time of response, receptors may act on two different ways: they can discharge at the beginning and then slowly adapt so the firing rate is decreasing at constant for all the time it takes to stimulate; or you can have rapidly adaptive responses and receptors that discharge at the beginning and at the end of the stimulus. The Messner and the Pacinian corpuscles are rapidly adapting and the other two are slowly adapting.

Talking about tactual modality, we are actually saying that all our body is spread by different kind of receptors that capture several pieces of information and energy that get into the body.

(2) Temporal

		Slow Adapting (SA) (< 3 Hz)	Fast Adapting (FA) (> 10 Hz)
(2) Spatial Scale	Local (I) (1 mm^2)	SA-I	FA-I
	Global (II) ($> 10 \text{ mm}^2$)	SA-II	FA-II

Each receptor has his own receptive field, and the deep layers layers receptors (like the Ruffini) have a broader one. Typically, the Ruffini is involved in monitoring and capturing movements over the skin, instead the receptive field architecture of superficial receptors offers a higher resolution than the deeper ones. This aspect of resolution is very important, because is determined by two elements, one is the individual receptive field, and the other is the density.



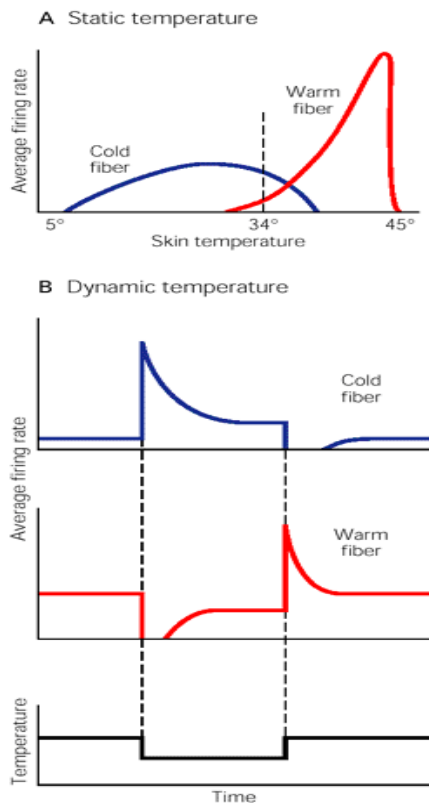
The deeper ones, like the Ruffini, are more present in the palm of the hand, while the others are primarily located in the fingertips, and this is where we rely for discriminating elements. It's important to distinguish areas that have high density of receptors or high resolution of discrimination. You can improve the experience of discriminating but not the physiological number of receptors.

Other parts of the body may not have any utility in discriminating objects or texture, so typically discrimination of two points is almost up to 4-5 cm, so two stimuli in our leg are not two points but just one is very close.

One thing that is also important to know is that each family of receptors may have a specificity in response, if you want to study different receptors you can just change the frequency of the vibro-tactile stimuli to probe response from different receptors. Touch detection thresholds vary with vibrotactile stimulus frequency and probe area consistent with the 4-channel model.

One of the things that we have to keep in mind is the population code, hardly we have one receptor activated when a stimulus arrives, most of the time we have several receptors working and responding at the same time, and the brain analyse it with what is called population code. It means that each receptor provides a piece of information and the brain has to merge it into a unique representation and synthesis.

If we go to observe other information, the very specific information we get to the skin is thermal perception, the other types of information (texture, shape) can be seen also by other perception modalities. Whether an object is cold or hot you have to touch it. When we talk about thermal sensation, the discrimination of the temperature is the difference between the air temperature



that we sense on our skin and the temperature of the object. Our thermal receptors work in a range from 5 to 45 °C, below or above there is no discrimination, only pain and damage. When we look at temperature we have two fivers, cold fivers and warm fivers, and these also act in two different ways. The cold fivers are increasing and going up to stimulate from 5 °C to a peak and then going down until body temperature (37-38 °C). Thermal receptors modulate the firing as a function of temperature. The warm fivers start responding around 30-32 °C and they go up exponentially with their firing rate until they reach 45 °C and then their response drop.

Around body temperature the discrimination abilities are much higher, because there are two fivers responding to the stimulus.

Proprioception

Proprioception is the sense of position and movement of our body. We acquire two pieces of information, the stationary position of our body and limbs (actually related to body representation and awareness), and the sense of limb movement (kinaesthesia). There are many disorders related to this perception modality, there are problems in how our brain represent and is aware of our body parts.

It relies in three families of receptors that are located in muscles and joints and are able to signal the spin and direction of the limb. These receptors are also located in the tendon or in the joint capsule, and they can sense the tension of the tissue and the flexion and extension of joints and muscles.

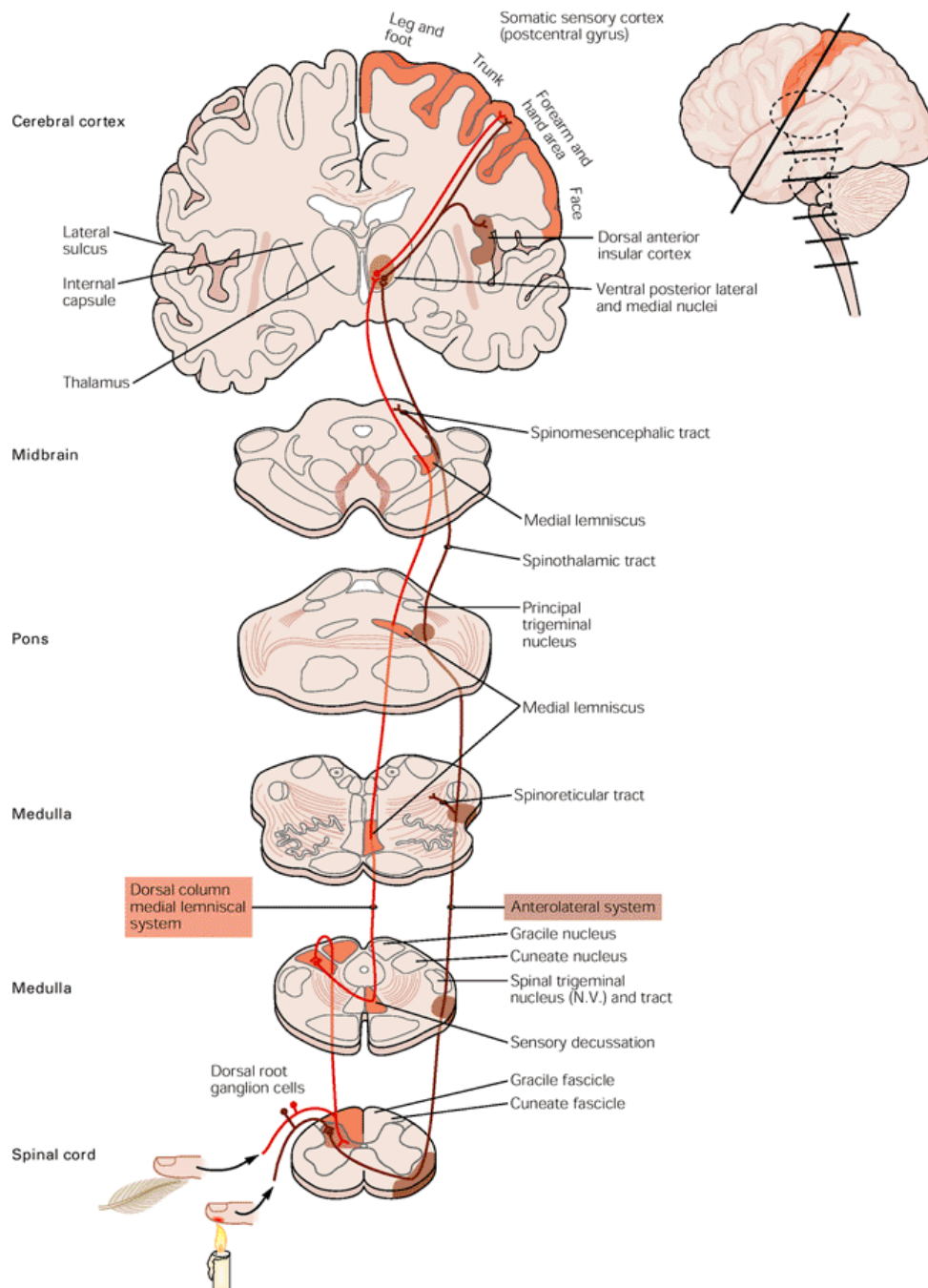
The body is separated in dermatomes, that reflect the surface of the skin that is innervated by each ganglion route (takes the name of the vertebra). Each of these areas corresponds specifically to the neurons located on a specific ganglion structure. Depending on the location of the pain we are actually able to understand where the problem is at level of the ganglion routes, since each of the part of the skin is innervated. (search for examples)

Once the information enters the posterior horn, the main pathway of touch and proprioception is what is called the dorsal column. The information gets in the posterior horn of the spinal cord and the axon that enters progress ipsilaterally up to the brain stem and only at the level of the medulla it crosses the spinal cord and goes up to the thalamus and then to the specific location in the post-central gyrus.

Also, what else tell us about the topographic organization is that axons that enter the cord in the sacral region are found near the midline of the dorsal columns; axons that enter the cord at

successively higher levels are added in progressively more lateral positions. Even in the spinal cord we have very specific topological organization.

So, the tactual component is conveying information on the hemi body ipsilaterally in the spinal cord and then only at the level of the brain stem it crosses to get to the other side. Information for pain stimulation are immediately crossing the spinal cord and they convey information up in the spinal cord contralaterally, to the hemi body that they are actually monitoring.



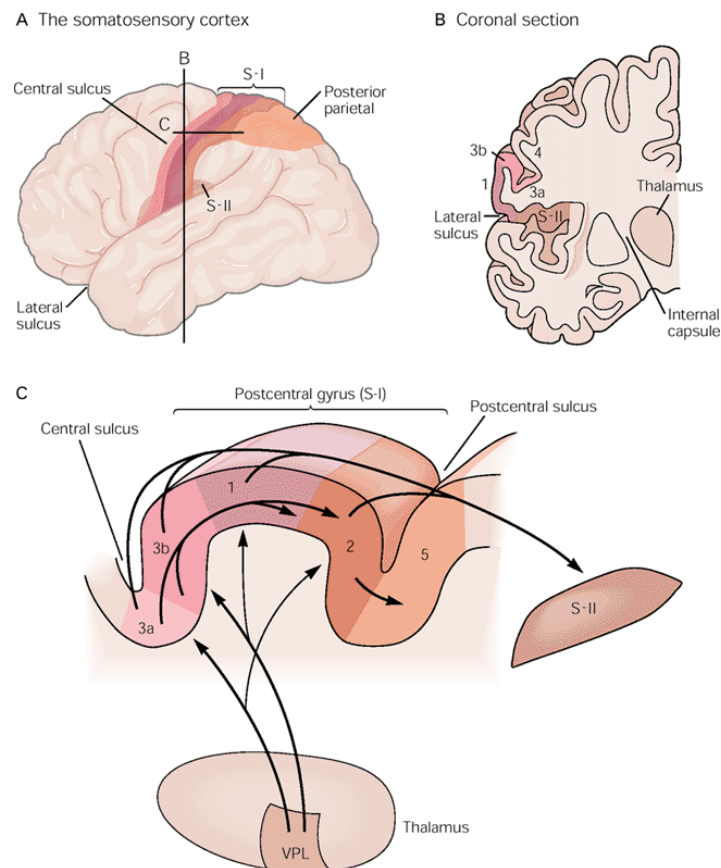
The other aspect is that the somatotopic organization of the axons relaying input from receptors in the skin is maintained throughout the entire ascending somatosensory pathway, through the thalamus, to the somatosensory areas in the postcentral gyrus of the cerebral cortex.

The ability to recognize objects placed in the hand on the basis of touch alone is one of the most important and complex functions of the somatosensory system: *stereognosis*. Many familiar

objects are much larger than the receptive field of any one receptor in the hand. In addition, objects excite more than one kind of receptor.

The brain organization is not just random, shuffled pathways that then get ordered at level of the cortex, but is already ordered immediately from the spinal cord, up to the thalamus. In the spinal cord the fibers that you find close to the midline are the sacral ones, and as soon as you go up and more fibers enter, they go more laterally, so it is already organized.

Once the information gets to the thalamus it then reaches the post-central gyrus, in which there are areas (3a, 3b, 1, 2, 5), looking at the coronal section.



These portion of the cortex (3a, 3b) are responding to tactile and proprioception, and then this information is merged together as soon as we move posteriorly in the brain, so areas 1 and 2 are already merging information from more higher complexity and when it gets to the posterior parietal cortex, we have integrated information between touch and proprioception. Together with the primary somatosensory area we typically describe the second somatosensory area (S-II), and these together are also getting information from other sensory modalities or subcortical structures, and areas 5 and 7 are merging information not just from the somatosensory cortex but also information that gets dorsally. Furthermore, as soon as we move from area 3a and 3b posteriorly, the information of a small part of the hand is merged into a broader area, and this is a typical phenomenon that we are going to see also in the visual cortex: at early stages the sensory pathway have a small receptive field, but once the information is integrated across different steps the receptive field gets broader.

The organization in columns is the same kind of organization that will be seen in the primary sensory visual area. Each chunk of the cortex is made up of six layers, and all neurons that are

contained in a single column are processing the same kind of information, and this is related to a rapidly adapting receptor or slow adapting receptor. So, each column of the sensory cortex specifically contains information from one family of receptors and a specific function. There is different scale of functional organization across these columns, and the principle that rules the organization are: the functional response of mechanoreceptors; the somatotopic organization of the cortex. If I have an information that gets only in the second finger, only his portion of the cortex will light up.

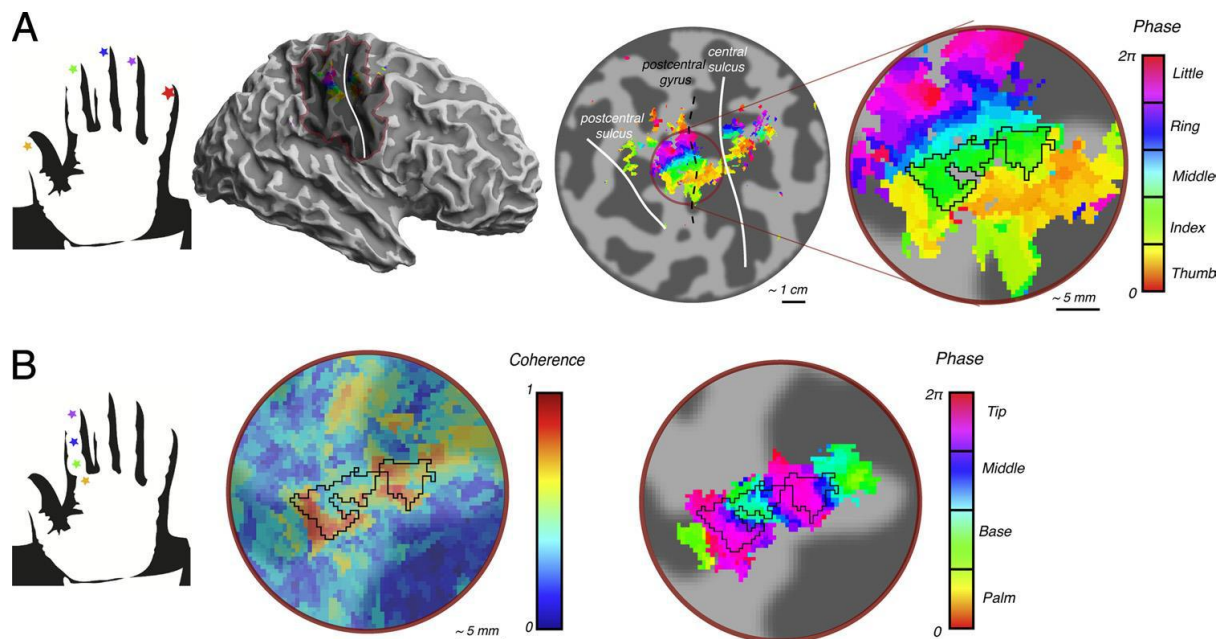
There are already mechanisms at the level of the skin that enhance specific information, the periphery of the sensory organs is already organized in order to provide specific way of discriminating different stimuli.

Haptics

Tactile discrimination normally entails active exploration. Psychophysical evidence indicates that relative movement between the skin and a surface is the single most important requirement for accurate discrimination of texture. Animal experiments confirm the dependence of tactile discrimination on active exploration.

Active touching, which is called ***haptics***, involves the interpretation of complex spatiotemporal patterns of stimuli that are likely to activate many classes of mechanoreceptors. Haptics also requires dynamic interactions between motor and sensory signals (tactile and proprioceptive), which presumably induce sensory responses in central neurons that differ from the responses of the same cells during passive stimulation of the skin. Proprioceptive signals because you have to have clear in mind where your hand is.

Tactual discrimination per se is something that occurs passively, when you have an active exploration and you have to merge several pieces of information (touch, proprioception, motor), this is what entails active exploration, and you call that haptics.



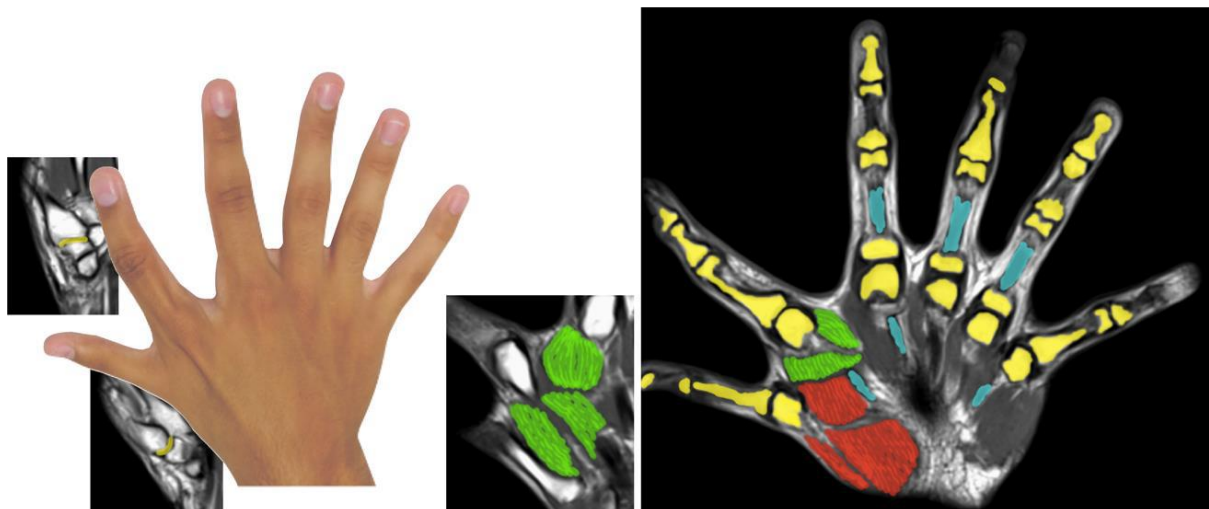
You can discriminate between the different fingers, but also between fingertips or regions closer to the palm.

To study the phenomenon, we can also use data acquired with electrocorticography, in which we have electrodes implanted on the cortical surface and the movie nicely tells how we process tactile information, which gets in the post-central gyrus and the post-central sulcus. Information that gets there is immediately spread to the motor cortex, so information from the tactile modality is mirrored in the motor cortex and then is slowly progressively sent backwards to the posterior parietal areas and then to the secondary tactile regions.

With brain imaging we can study different elements of tactile representation, but in some specific environments is not really easy, for instance MRI. Also, when you do haptic studies and you ask the subject to actively move and explore objects most of the time, he also moves his head, and this doesn't get along with MRI.

It's really difficult to distinguish the contribution that is related to touch from that related to motor, because of the tight coupling between motor processing and tactile information processing.

There are genetic disorders that can increase the number of fingers, and you still maintain the five finger organization, but you are also able to adapt a sixth finger representation in the cortex of the subject (augmented manipulation ability).



In the case of amputees, you have to cut the nerve, and the left other fingers detect the missing hand, so subject may have sensation about the limb that they don't have anymore, also with pain.

Once that you play with this kind of extra limb in the body and provide a new route of tactile stimulation, then you can map at brain level how this information is actually re-organized.

Pain and nociception

Pain is something that could be described in different ways, somehow is very distinctive among the different sensor modalities, and unlike other somatic and sensor modalities the quality of information that is conveyed by pain is less specific but has the characteristics of urgency. It is a sensory modality that primarily has to alert our brain, and this kind of alertness describes how the brain is actually the site that captures these elements of urgency.

The sensations we call pain are the most distinctive of all the sensory modalities. Pain is a sub modality of somatic sensation and serves an important protective function: It warns of injury that should be avoided or treated. Pain has an urgent and primitive quality, a quality responsible for the affective and emotional aspect (I am able to get aware of this emotional aspect later) of pain

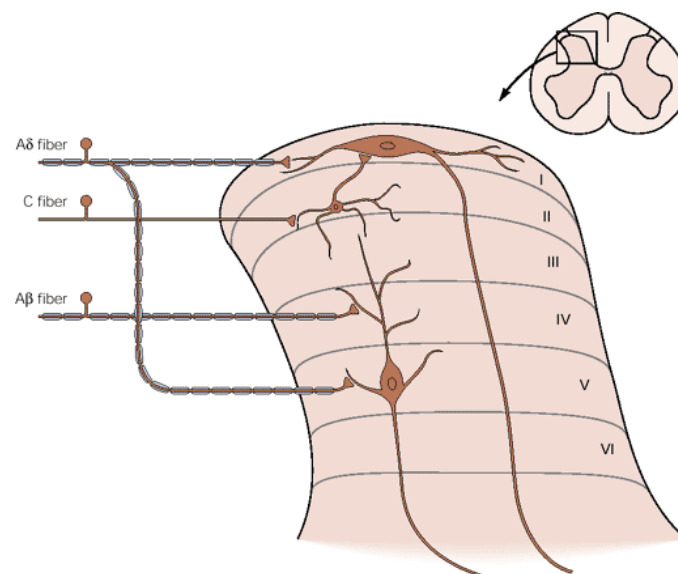
perception. Moreover, the intensity with which pain is felt is affected by surrounding conditions, and the same stimulus can produce different responses in different individuals under similar conditions. The ability to reduce the perception of pain is called analgesia, but we can also exaggerate it or anticipate it.

That is why we separate pain itself from the perception of it.

Pain is a percept; it is an unpleasant sensory and emotional experience associated with actual or potential tissue damage. Although pain is mediated by the nervous system, a distinction between pain and the neural mechanisms of nociception - the response to perceived or actual tissue damage - is important both clinically and experimentally. Certain tissues have specialized sensory receptors, called *nociceptors*, that are activated by noxious insults to peripheral tissues. Nociception does not necessarily lead to the experience of pain.

There are thermal nociceptors, mechanical nociceptors, polymodal nociceptors, that respond to different stimuli (mechanical, thermal) but are innervated by small diameters no-myelinated C fibers (velocity of less than 1 m/s).

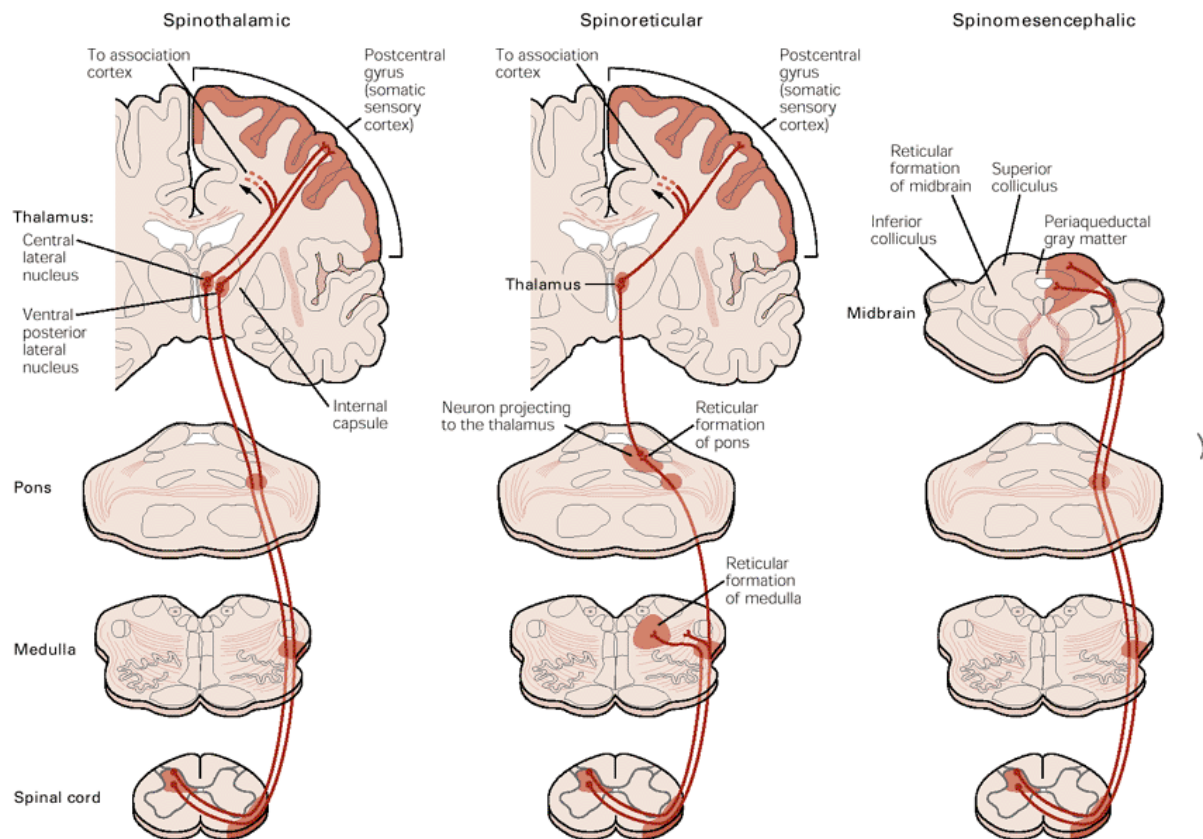
Having two fibers that convey information from thermal and mechanical nociceptors and polymodal, allow to get generally what is called two ways of pain. These fibers are getting this idea of the first very brief input of pain and then the second longer wave of pain, you can notice it when you cut yourself, an immediate pain but it also continues later.



In the posterior horn of the spinal cord, you can see that the Ad fibers get to lamina 1 and 2 with the C fibers, while the tactile Ab fibers get to the lamina 4. Clearly the painful stimulation is somehow getting up to the brain also modulated by a tactile component. For Ad and C fibers that convey two stimuli, in order to reduce the pain stimulation either you provide a tactile stimulation to the side of pain or you move because you want to increase the proprioceptive stimulation, and these elements are actually gating and blocking the information that goes up to the brain.

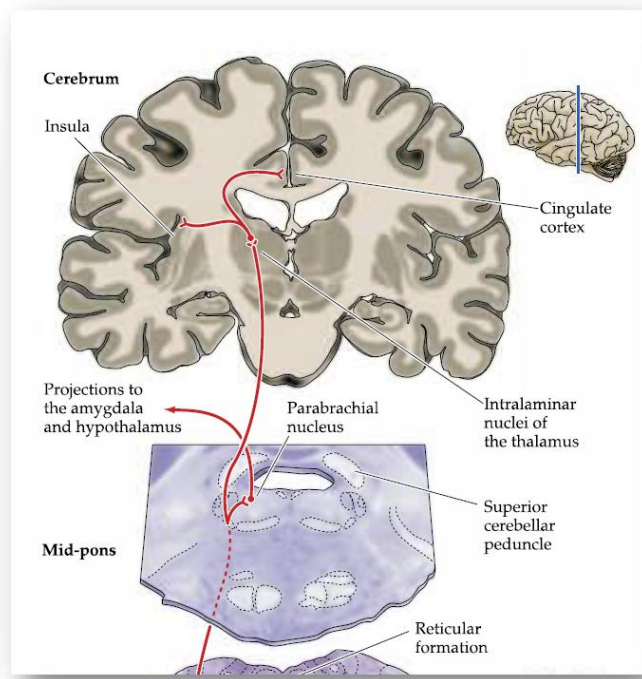
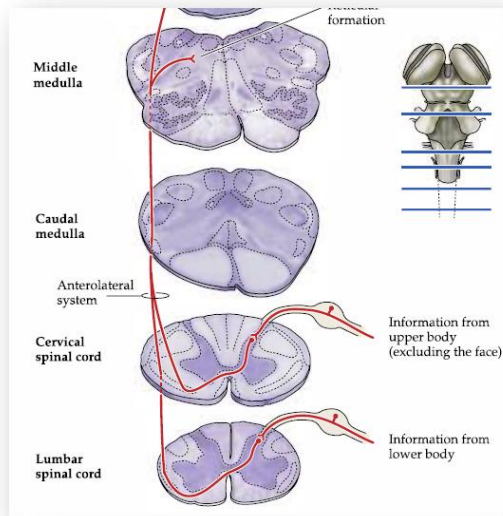
Painful stimuli are so important that the brain, from lower part of the body up to the head, has five different ascending pathways that convey information up to it. The most important one is the spinothalamic. As for painful stimulation the information enter the posterior horn and cross immediately to the contralateral part and goes up to the brain stem, and this is very important, because even if painful stimulation gets to a different pathway that ascend parallel to the

posterior one for the tactile component, also this information gets immediately to the post-central gyrus, because it is important, beside this information of urgency, that the brain is aware where is the source of pain.



Some fibers of the same pathway are actually going to this specific portion of a nucleus (bundles neurons), the reticular formation, that is very important for arousal, if you have a pain stimulus you wake up, your alertness is increased, and your attention is verted towards the painful stimulus. It is the reticular formation in the brain stem that regulates this aspect.

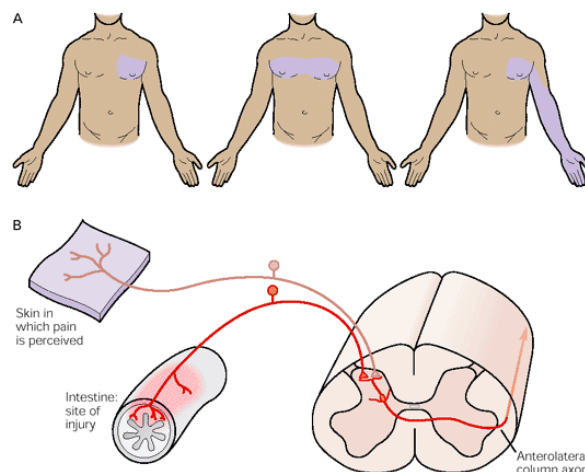
There is also another pathway other than the spinothalamic (important to localize and characterize the source of the pain) and the spinoreticular (important to alert the brain about the painful stimulus). We have the sinomesencephalic, that is actually getting information from very deep structure that is called periaqueductal gray matter. The mesencephalo has this role of activating and regulating all the body structure, so together with the spinoreticular elements, this alertness is also reflected in several changes at the peripheral part of the body (heart rate increasing, sympathetic system increased). But more importantly this periaqueductal gray matter is a sort of an hub that modulates the incoming and outcoming painful information that goes to the brain, a gate that modulates the access of painful stimulation.



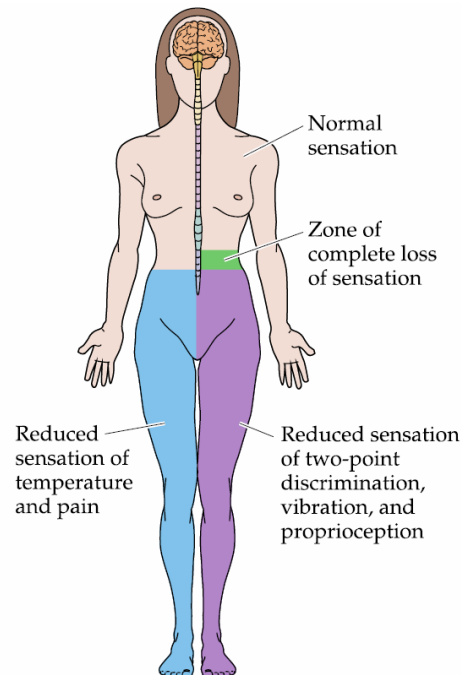
Then there are two other pathways that are primarily related to two systems getting to cervical components that may actually send information from the cervical and the face and the neck up to the brain. One feature of these elements is that they relay to the reticular formation, but then they start to get other branches of this information, to the amygdala and the hypothalamus, but also the insula (primarily considered as the central pain area) and the cingulate cortex, and this final is important because it seems to be fundamental to monitor conflict or painful situation.

This idea of having a primary area for pain stimuli has never been confirmed, and typically we describe that all brain areas involved in the process of pain have a pain matrix. Typically pain is a modality that does not rely on a primary area, it relies on more cortical areas that actually elaborate different elements of the painful information.

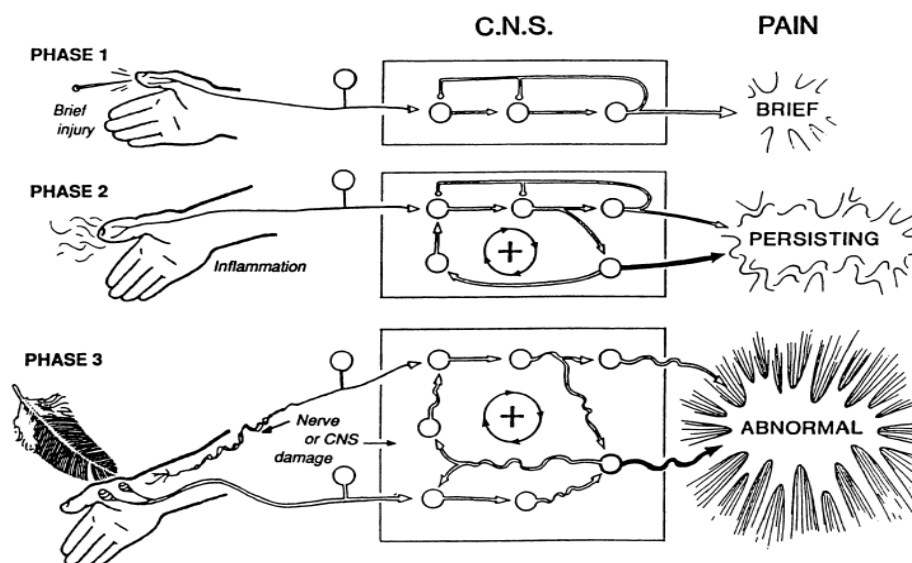
When we have afference that innervate internal organs of the body, they also have fibers that monitor painful situation and rely on the posterior horn of the spinal cord. What happens is that when information from the internal organ gets to the posterior horn, the brain is just getting information from that neuron but is not actually aware that the information is coming from the body organ or the skin. What our brain is actually getting is therefore a representation of pain on the skin surface, and this is called referred pain.



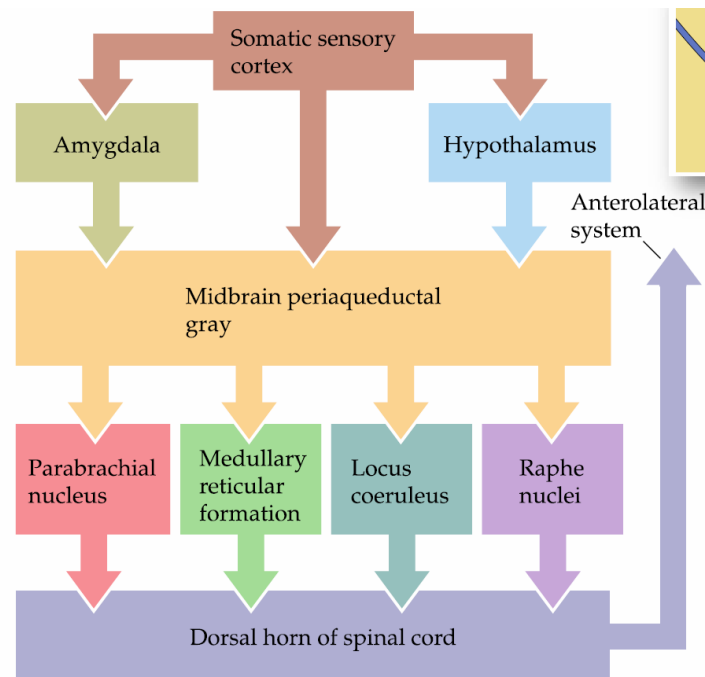
The information is going up to the brain ascending to the antero-laterally portion contralaterally, but ipsilaterally we have tactile information, so in one half of the spinal cord we are containing tactile information from one side of the body and pain information from the opposite side of the body. In those cases where we have lesions of just half of the spinal cord, you have a situation that is called dissociated sensory loss, so you may have in one side reduced sensation in touch and on the other part a reduced sensation of temperature and pain, and that is why the thermal and painful information are crossing immediately the spinal cord and the tactile information crosses at the level of the brain stem, but it ascends ipsilaterally.



Pain is a very tough process. One of the main issues that goes with pain is that typically initially you have a very brief stimuli, but most of the time the pain can persist and then have a persist painful stimulation (inflammation, formation). When it persists at the level of the periphery and the brain you have a sort of sensitization, a circuit of pain perception enhanced. You have very subtle stimuli that due to this reinforcement at the brain level you can perceive as harmless and painless.



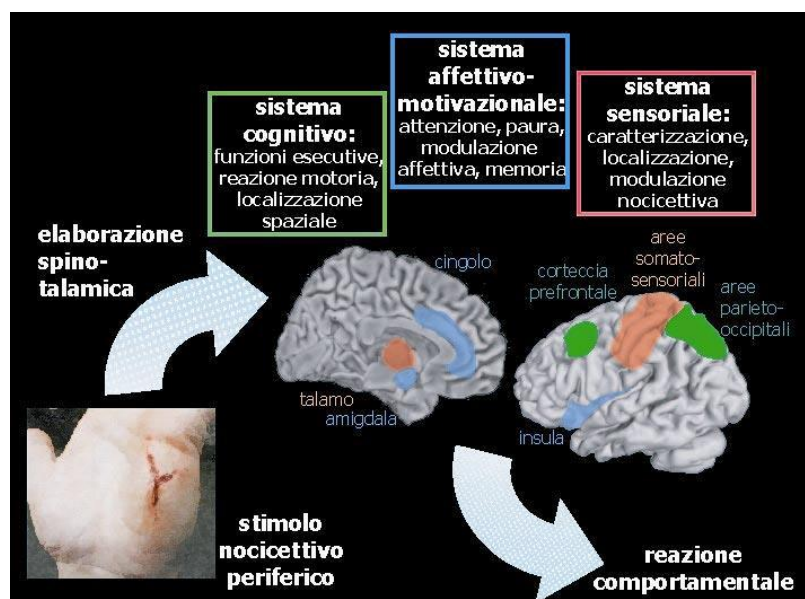
What is interesting about pain modulation is that the information doesn't just go up to the brain but is also modulating many different aspects. And this is what we call pain matrix.



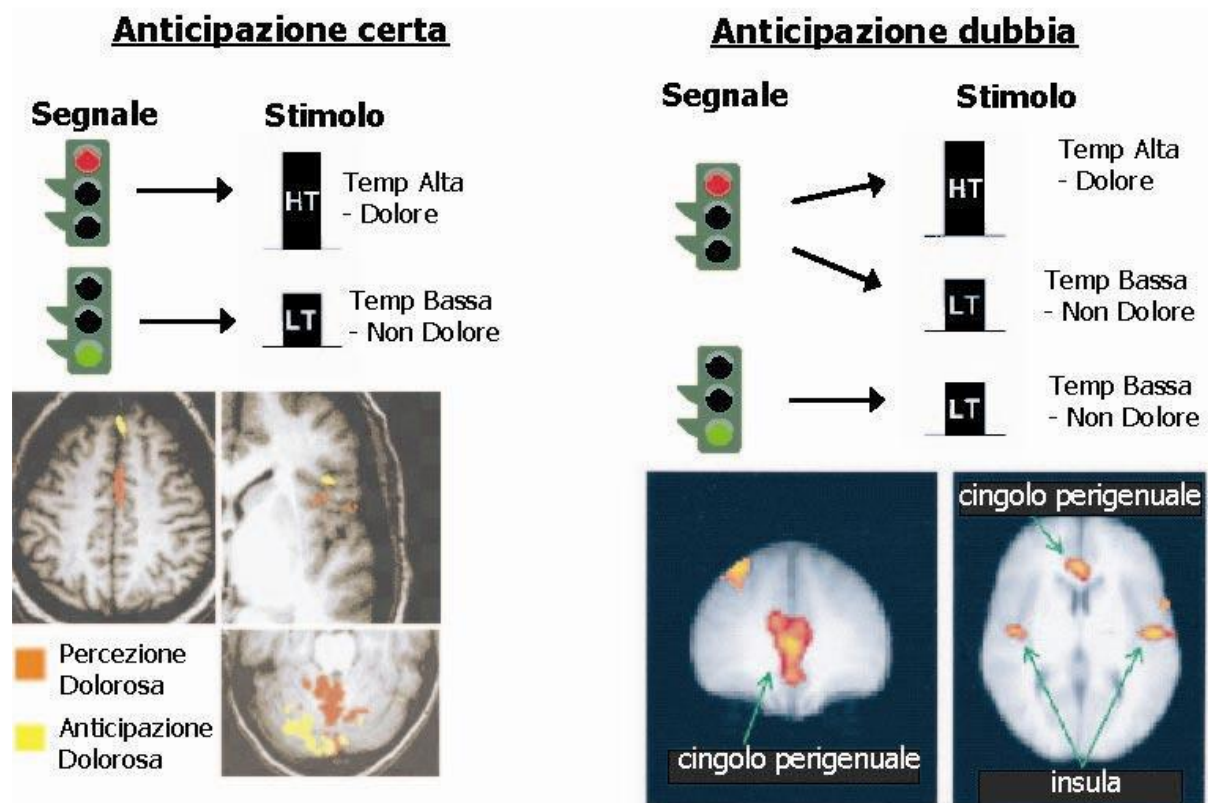
When we have a painful stimulation, we should consider three components:

- A sensory perceptive component, that primarily is useful to characterize intensity and localization of the stimulus, and this is task of the post-central gyrus.
- An affective component of pain, primarily related to the cingulate cortex, the amygdala and the insula, that modulate the affective components of pain (alertness, fear).
- A cognitive component, primarily relying in frontal-parietal regions. A primarily modulated attention, we have to localize spatially the source of the painful stimuli and act a motor reaction in response to the stimulus.

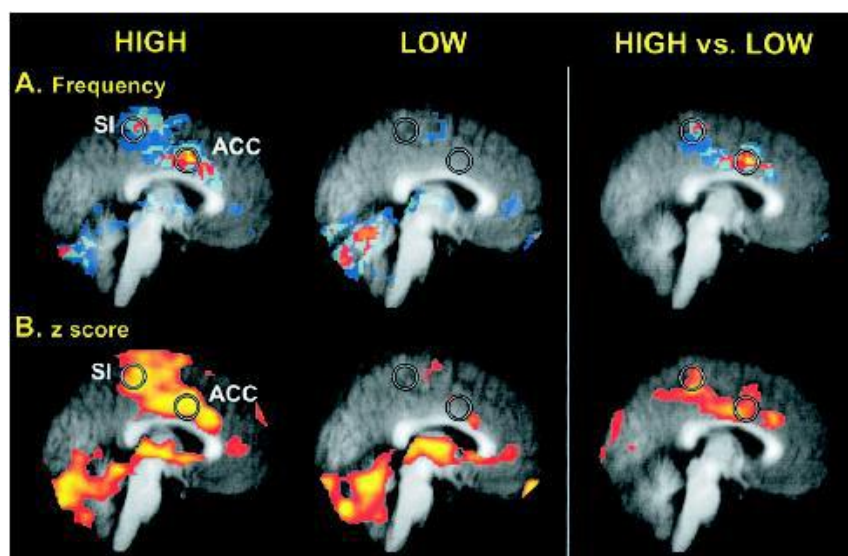
Imaging studies are also trying to visualize the antero-lateral system, so we have also functional responses in other regions.



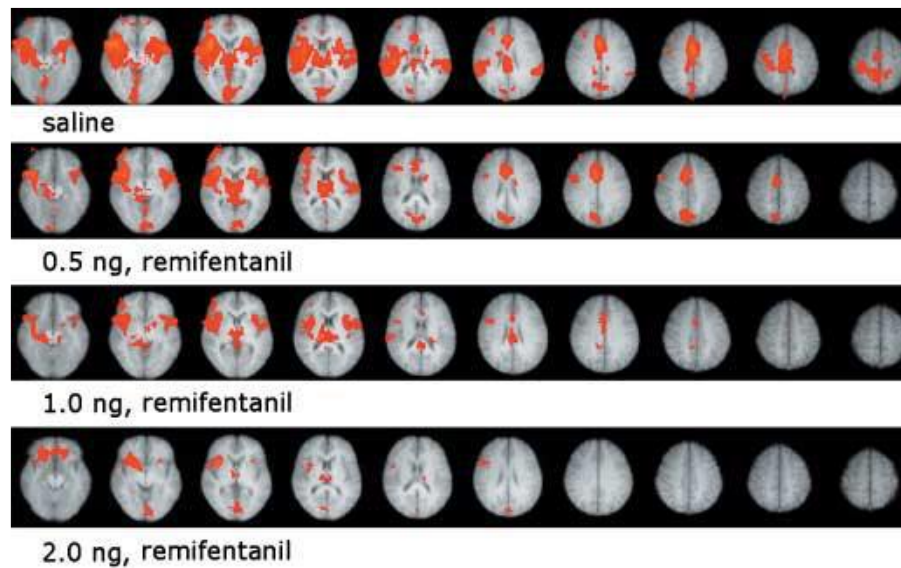
It is possible to have very simple experimental design where we play what is called anticipation of pain. This is typically used in experiments of fear conditions, you have green light that is no harmful stimulus and red light the painful stimulus. When the brain is already expecting a painful stimulus, it happens. When you anticipate pain, you have an affective component that already is having you to expect pain.



These aspects of sensitivity to pain are quite intriguing, because each of us has a different sensitivity to. If you compare high vs low response, there is no specific difference in the post central gyrus, their true difference is in the anterior cingulate cortex. The same exact stimulus is processed at the thalamic level in the same way in people in high or low sensitivity to pain. But then you can see in the anterior cingulate that you have an exaggerated response to pain for high sensitivity to pain persons.



The higher is the concentration of analgesics the lower is the response, primarily in the anterior cingulate cortex. This is what you can study, the effects of opioids. If you give placebo, 30% of people improve.



The interesting part for us is to have a modulation of pain in the anterior cingulate. There were several examples of PET studies measuring blood flow with patients susceptible to achnesia? and through hypnosis they were having them to perceive as less or more unpleasant their handling of cold water. Only saying that they were going to get a painful stimulus, in the anterior cingulate will appear a higher response.

For most of these studies you could claim that the anterior cingulate is not really the pain processing affective component, but sort of like decisional aspect on whether to react to painful stimuli. This portion of the brain is actually getting information directly from the spinal cord, the thalamus, to modulate our behaviour on pain perception.

Vision

Here we would see a much finer degree of description and complexity of representation, as vision is the most well extended studied modality across all our sense.

The information is seen in two dimensions by the two eyes and then processed 3D mentally in our brain. There are at least two parallel and interacting pathways that process information. More than half of the cortex is devoted to visual processing. The whole occipital lobe is subserving visual processing; there is no other region that does the same for a single function.

What we actually can say is that the visual system transforms transient like patterns (even from low level features we can reconstruct the image) on the retina in to stable representation of the tri dimensional world.

The eyes adapt and change their shape to get the light getting properly to the posterior side of the eye. But when the shape of the eyes changes and the ability to modulate this accommodation is lost, we typically have disorders that need correction to improve our ability to see. The eyes have a protective surface that is mostly transparent (in the cornea), and then the rest of the eyes is the sclera and is mostly peripheral to this central transparent part. Then we have two chambers (anterior and posterior) divided by the lens, and these are filled with two aqueous solutions, the

aqueous humor is much more fluid, and the vitreous humor in the back has much more of a gel-like structure. The opening between the two chambers is called the pupil, and it has the specific role of modulating the number of photons that can get to the retina, in a light environment the pupil shrinks.

The retina is the internal layer of the chamber of the eye, and on the posterior part there is a pigmented epithelium, a dark layer full of melanin that works like the black screen of the movie theatre, since it avoids the scattering of photons on retina. These layers are also full of blood vessels that feed the retina, that is made up of three layers of cells, an external layer that is made up of photoreceptors, a second layer that is made up of bipolar cells, and finally a third layer that is made up of ganglion cells. The axons of the ganglion cells are bundled together into what we call the optic nerve, and it is typically entering the back part of the eye and drags the spot without photoreceptors, and we typically call that blind spot (the finger disappears at the centre).

Light photons have to cross all three layers of cells. It appears to be an efficient system, because they reach a region full of blood vessels, they are more fed.

There are also two horizontal layers, the horizontal cells stand between the photoreceptors layer and the bipolar cells; the amacrine cells stand between the bipolar and the ganglion cells. You understand immediately that this number of layers is telling you that there is a lot of processing, yet at this level.

There are photoreceptors that have a one-to-one relationship with bipolar cells, so their information should be very important, but this information also interacts with neighbour photoreceptors, so there should be some population coding at the level of the retina that captures information and starts doing some specific processing per se.

Photoreceptors in the retina do not exhibit action potential, rather light activation causes a changing in membrane potential, and it comports the release of a neurotransmitter. This regulation of the closure of sodium and calcium channels are telling that the darker the situation and the least intense is the depolarization, the higher is the light, more negative is the response.

One important feature is that this biochemical cascade initiated by capturing photons, provides an enormous amplification of the signal. One single molecule of rhodopsin activates 800 molecules of transducin. These molecules initiate a cascade of events that modulates the opening and closing of the channels.

We have two types of photoreceptors, rods and cones, that are not just established by the shape, but by the photopigment they contain. They have different distribution density along the retina, and also a different pattern of connection to the bipolar and the ganglion cells.

- The rod system has a very low spatial resolution, but is extremely sensitive to light, they are primarily distributed at the periphery of the retina and are primarily used when we refer to mesopic and scotopic visual conditions (reduction of light). An example of when you use rods is when you look at stars, you see them with the periphery and not concentrating on them.
- The cone system has a very high spatial resolution, is relatively insensitive to light and are much more sensitive to acuity, they are concentrated in a specific portion of the retina.

Cones have very low presence at the periphery, are all concentrated in a very small portion, corresponding to the visual field, and it is called fovea. Rods are almost absent in the fovea; they have higher representation in the periphery. In the fovea you also have a mechanism that opens

the different layers and create an area called little fovea, where light doesn't have to cross the two internal layers (even blood vessels), but goes directly in the photo receptors (no distortion or scattering).

In terms of pattern transmission, cones (with a higher acuity) typically are projecting to the polar and ganglion cells directly. This information is also mediated with other cones and rods, but this kind of response is much more mediated by more than one cell.

When the stimulation gets to the ganglion cell, whose receptive field gets the light spot, it responds to it with a slowly adapting response but the whole centre is not responding. There are different patterns of response, that allow the retina to discriminate contrast and contours.

The cone release glutamate, but it can be excitatory and inhibitory. The different cones are already allowing us to process information and get it to the brain. The issue is that this kind of processing is done by our retina, our brain has no impact, it just receives information, that could be distorted by disorders at the level of the eye.

Information from retina is sent to the posterior part of the brain through the optic nerve, that merges into what is called the optic chiasm. Information is typically divided into two elements, the nasal part of the retina and the temporal part of the retina.

- The nasal part is primarily providing information on the ipsilateral visual field.
- The temporal part provides conveys information on the contralateral hemi field.

When the two optic nerves merge into the chiasm, the nasal part is still processing. On the left brain you will get information from the left temporal and the right nasal retina, conversely the right hemisphere receives information from the right temporal and the left nasal. After passing in the optic chiasm, there are two portions that are called the optic tracts, that send information to thalamus (primarily the lateral geniculate nucleus), and this information is all related to the opposite hemi field. This information from the thalamus is then projected to the optic radiations (brain fibers) back to the occipital cortex.

The ganglion cells are providing very specific information or gross information; there are ones that only say if the light is on or off. This is something very useful for the hypothalamus, with the regulation of circadian rhythm. There are blind people (born blind) that has no light perception, and their circadian rhythm is all altered.

The muscle that regulates the movement of the pupil is called the pupillary constrictor muscle, and it relies on the changing in light. To see if there is damage at the level of the brain stem, you can get light to the pupil and it has to shrink, if there is no damage this reflex is very simple direct and is a good way to demonstrate that that part of the brain is really working.

The fourth pathway goes to the superior colliculus, a deep structure in the diencephalon, which has a huge importance, because it orients the movement with coordination with the eyes. It doesn't only rely on vision but also with other sensory modalities.

When we have a visual field, we can divide it in horizontal and vertical axis, and we can select four portions of it. If the two axes are defined with a fixation point you have a portion of the visual field that is actually captured by both eyes, but you have also portion of the peripheral visual field that is captured only by the nasal portion of one eye, and this information is sent contralaterally.

This very specific and defined representation is shared with the cortex, and we have a specific representation at the level of the occipital cortex. The right occipital lobe receives information of

the left visual field, with the upper portion that is represented in inferior dent of the calcarine sulcus, and the inferior visual field is represented with the upper dent of the visual cortex. The most central portion of the visual field is represented posteriorly in the occipital pole, while moving anteriorly I'm going from the foveal to the periphery of the brain.

Visual pathway is really complex. Any damage along the path is creating different elements of loss of the visual field, something that is called scotoma, something that is hidden.

V1 (visual cortex) is divided in columns. Columnar organization is primarily referring to functional organization across all six layers of the cortex, and vertically you can see how a single column responds to the same orientation, while if I'm looking to a larger chunk of the cortex we have orientation selective columns. Together with orientational columns, we have (orthogonally) ocular dominants columns, all together we have hyper column.

Information from the retina is getting first to the lateral geniculate nucleus, that is made up of six layers, which are made up by 2-3 families of neurons, 4 layers are made up of parvocellular neurons and these are indeed connected with ganglion cells in the retina that are already smaller, so in the retina we have small ganglion cells that project to the four layers of the lgn. Then we have large ganglion cells (magnia) that project to the layers 1 and 2 of the thalamus, finally in the middle of these layers we have a layer made up of koniocellular neurons.

This p and m ganglion cells are so important because they have different functional features, magnocellular layers in the lgn are not sensitive to colour contrast but have a very high temporal frequency, the p cells have a very high spatial frequency. The information they are offering is actually processed across two different pathways, the parvocellular pathway is processed more ventrally, along what is called the inferior temporal (ventral pathway), that is primarily devoted to process informational identity (colours, shape, texture). The m cells are going to project to regions after V1, devoted to motion, and then to the posterior parietal cortex, this is called the dorsal parietal pathway, that processes much more information about movement and space.

We don't have two pathways that process different pieces of information. The novelty of this is that already from the retina there is a functional organized information that get to the thalamus and then to the brain. The feature of the cells that are present in the retina and then in the thalamus are already constraining and impacting the information that we are able to process from the external world. This information then will be projected and processed to two different pathways, one whose information for us is much more important for movements and action (dynamic), while the ventral pathway is much more related to semantic, language, knowledge organization, related to the identity and the shape of the object.

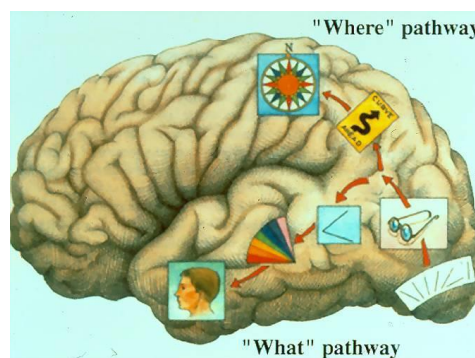
The important aspect is that this kind of functional subdivision is actually starting directly from the retina, the retina itself has sensory component and is already capturing specific features from the visual stimulus, and conveys this to the thalamus (specifically to the lateral geniculate nucleus) and leads backwards to the occipital cortex where, in the striate portion, we have the demographical representation of the visual field.

Subsequently this piece of information follows two different pathways, one moves inferiorly ventrally towards the temporal cortex. These specific pathways appear to convey information primarily on the parvocellular (p ganglion cells) and the magnocellular pathway, that conveys information ventrally towards the parietal cortex.

Besides the two functional aspects and subdivision of the parvocellular ventral stream, that conveys more information with higher spatial resolution, so primarily related to definition and discrimination of shape, texture, orientation and colours. This pathway is actually characterized as a sort of “what” pathway, that is telling me what I’m actually perceiving. This information relative to the consequent recognition of the items I’m perceiving would be therefore related to also understanding a semantic, linguistic interpretation of what I’m seeing. Indeed the information that progressively moves from the early visual areas in the occipital cortex, anteriorly to the temporal lobe, are actually involved in this knowledge organization and semantic processing because these regions are bridging the auditory information with the visual one (the superior-temporal gyrus is not just a region of multisensory integration but is a region involved in speech and language processing). Therefore, we have a functional merging of sensory information (I think in the superior colliculus) coming from the occipital cortex with all the speech component of language.

Every time you have a specific lesion in the anterior region of the temporal lobe, we have a formal neurodegeneration that is more often called semantic dementia.

We have to always keep in mind that a structural and architectural organization of the brain, always mirrors the anatomical features. Similarly, more dynamic information is progressively going dorsally, this piece of information of the magnocellular pathway is much more related to movement perception, dynamical changes of stimuli, and stimuli that change are also changing location in space, and all this piece of information is important to integrate visual information with the information coming from the sensory motor cortex. This pathway is called the “where” or “how” visual pathway and describe and collect all pieces of information that are useful to motor action, or to process objects spatially. That is why when we have a lesion of this pathway, we have the inability to perform actions and motor sequences (ataxia).



These two pathways are strongly bonded in visual perception but appear to process specific content of information independently from the sensory modality, to which this information is conveyed to the brain. This means that if you close your eyes and recognize by touch a bottle you will recruit the parvocellular pathway, so shape is represented in that portion of the cortex if you acquire information with a sensory modality that is nonvisual. Consequently, we also can do motion discrimination by hearing the sound of a car or a bike, or the flow of water in my hands.

It seems like that this fine architecture of the two pathways, nonetheless, looks like it is much more specific to compute and represent some specific piece of information independently of the sensory modality that conveys the information to the brain.

For many years we have been describing these two as the main visual pathways, but we are also thinking that there is a third, going in the middle in the superior temporal cortex, in the supra-

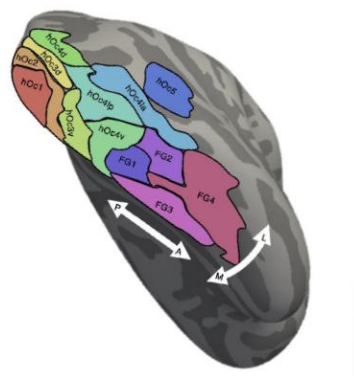
marginal and angular gyri. This third pathway is much more related to social cognition, the interaction with other people.

Vision has been the most studied sensory modality, since it is our most dominant modality. For many years we have been labelling these two pathways as visual associative cortex, it is not only visual. It looks like each portion of the brain computes specific item and piece of information independently of the sensory modality that conveys this information to the brain.

It is impossible to recognize the shape of objects in an auditory way. Furthermore, I'm able to recognize the entity of an object but not the shape, there are pieces of information that could be studied across modalities, like motion.

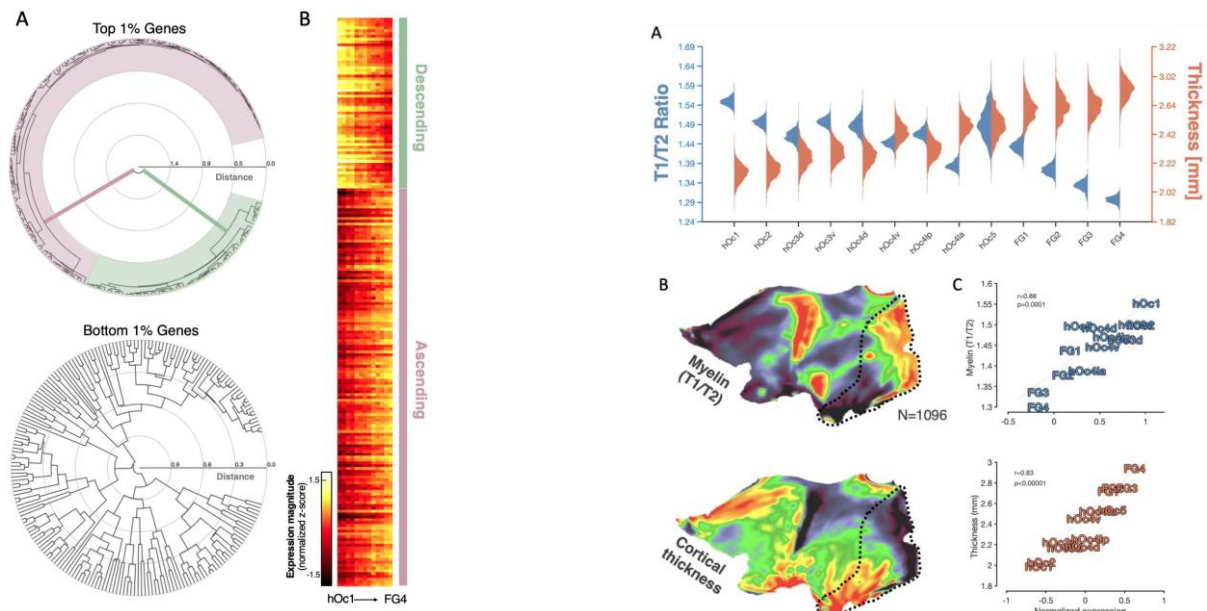
What happens when you close your eyes and you recognize an object by tactile modality? You recall a visual image. This is imagery, and you activate the visual cortex. This activation is actually related to something that is computationally independent from the sensory modality, and it's not related to visual imaging. This is something that happens also in people blind from the birth, when they are asked to recognize an object, they recall the same areas.

In a paper published on top journal, a study relied only on publicly available datasets, they never took a measure. They looked at genes expressed in the tissues of the brain's area in figure and they defined different portion on the cortex (parcellation), and they go from early visual areas to the more anterior temporal regions. If you look at the expression of some genes you can define two gradients: ascending and descending, from the occipital cortex to the temporal cortex.



The whole body relies on these gradients in order to orient the exact position of body parts, organs, and furthermore if there is one body part that should be organized is the brain. This difference of expression in genes is a way to which genes are actually differentiating, for example the occipital from the temporal lobe.

When we look at other measures, for example the amount of myeline in the cortex and the thickness, one thing that we can observe is that the early occipital areas have a high amount of myeline but are very thin, while brain areas on the temporal pole appear to be thicker but have a less amount of myeline.



Then they looked at another dataset, that concentrated in getting different gene expression across different ages and the ages are not just after birth, but also in post-conception weeks. In the newborn brain you can see that the ascending and descending gradients are actually going to develop, then one thing that you observe when the baby is still in the uterus is that the ascending gradient is already quite defined, while the descending gradient is not. The clear maturation of these two gradients is actually occurring around 5 years, so it occurs this amount of time for the visual system to completely form and develop, and between the two gradients the ascending one (related to thickness of the brain) appears to develop earlier, even before birth. What is interesting is that it really takes a lot for the visual system to get a stable organization and maturation, and we should consider this as a sensitive period in which sensory experience is actually impacting the brain.

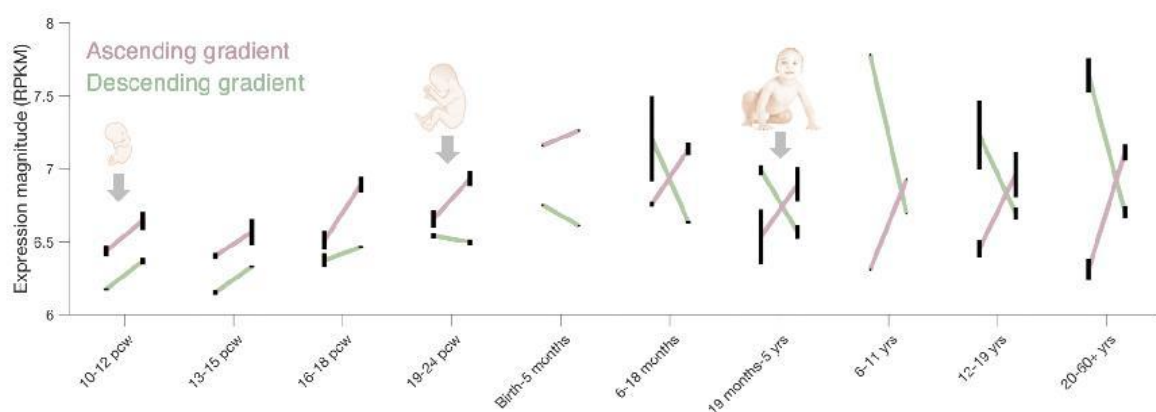


Figure 5: Ascending and descending gene gradients emerge at different points in human development. The two gene gradients

What is quite interesting is that you can compare these genes also in humans e non-human primates, and while the ascending gradient is quite similar, the monkey's visual system has no

descending gradient. So, for sure this descending gradient that is developing during childhood is very important for a more complex and complicated visual representation.

What we get from the analysis of this big dataset is that the visual system undercomes a long period of maturation, during which can be significantly impacted by sensory experience. This post-natal visual experience is fundamental to define the cortical architecture of the visual system. Looking at it from a different perspective, we have a gradient or part of the visual system that is already there few weeks after conception, something intrinsically determined. Indeed, the descending pathway matures with sensory experience. The brain can adapt to the environment, but there is the other ascending component that is intrinsically present in the brain, and this is something you can't change. Scritto un paio di volte perchè dice sia importante.

The retinotopic organization is actually spreading its organization along the ventral and dorsal pathway, so this kind of topography and spatial embedded information is maintaining the organization of the whole brain. Progressively in the human brain we had been starting to define and parcel all the anatomically functional portions of the visual cortex (V1, V2, V3, V4, V5). This is actually derived from studies on the monkey brain, that is probably one of the best, having all the different portions being something different.

We have been describing about this for two reasons, one is that most of we know of the visual system has been studied first in monkeys, and then is the fact that we studied lesions to different regions, seeing the result. For many years we have been trying to describe this hierarchical processing of information, looking at dimension, and the idea that we always got is that we have a hierarchical organization of the visual system, whose information was processed from early visual areas more ventrally or dorsally into something that is going to integrate into a more complex representation. This thing is still valid.

In a study, Paolo created visual kind of stimuli that has been filtered, so no more playing with the models but with a filtered stimulus, seeing how these are actually correlated. You are creating an image of what each portion of the brain is actually viewing, and what you can do is to see how different an image is processed across different regions. V4 for example ignores the background information. Nowadays what you can do is to ask artificial intelligence to reconstruct the best representation that mirrors the specific response in the brain area.



As for the two pathways, these are their different properties:

Core characteristics of the *perception–action* model

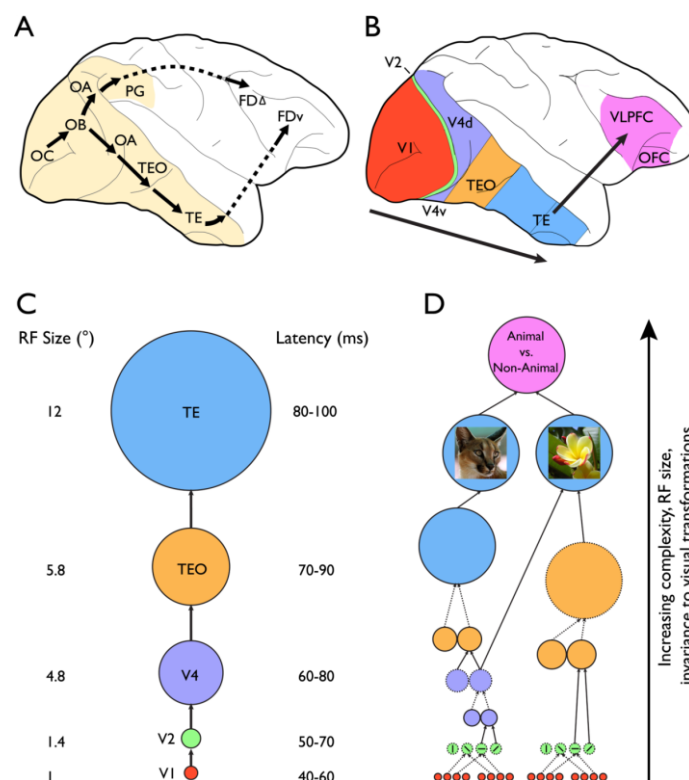
	<i>Ventral stream</i>	<i>Dorsal stream</i>
Behavioral function	Vision for perception	Vision for action
Spatial properties	Allocentric coding/ relative metrics	Egocentric coding/ absolute metrics
Temporal properties	Sustained representations	Transient representations
Visual awareness	Critically linked to awareness	Independent of awareness

Allocentric means that you are looking from outside.

Being linked to awareness means that if you see a bottle, for example, you are able to label it as a bottle.

The dorsal stream has a more egocentric coding, because it's processing action that can be performed on the object, but this is much more independent from awareness. When you are viewing something, you are not actually aware of the action you can perform on it.

If you look at the ventral stream it appears clear that the receptive field size is increasing. It's something that we had already observed for the somatosensory, at the beginning the perception is related to few receptors and the more I gather information from more receptors, I'm integrating information into a broader unique percept, and for vision it's the same. V1 has a very small receptive field size, and if I move more anteriorly the receptive field size is increasing.



For many years we have been debating how the cortex is actually organized, and in order to better understand this, let's focus on a specific item and piece of information so that it's clear.

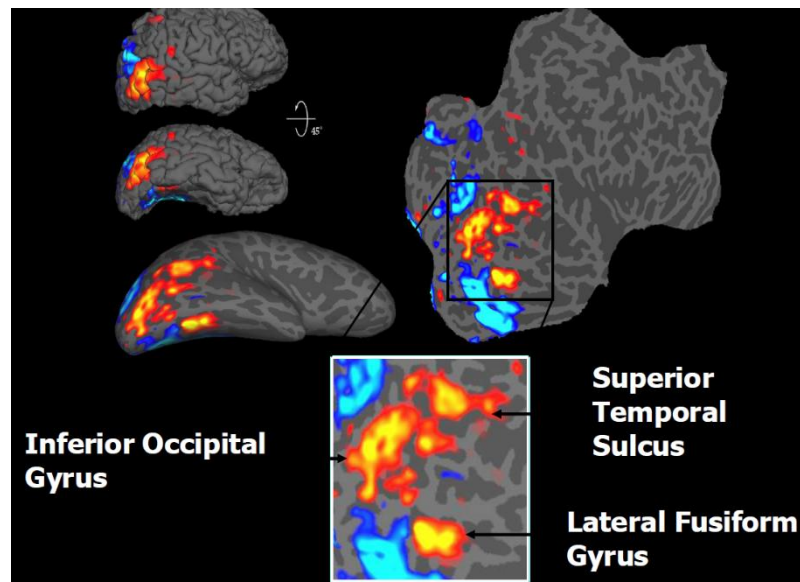
There is a very strong bottom-up component that allows us to capture what is salient in our surrounding environment.

Faces are very special, because there are plenty pieces of evidence that they are telling that one of the early stimuli, that even newborns are able to capture, is the configuration of a face. From many features of the face and face expressions, we are able to change our behaviour and thoughts about a person.

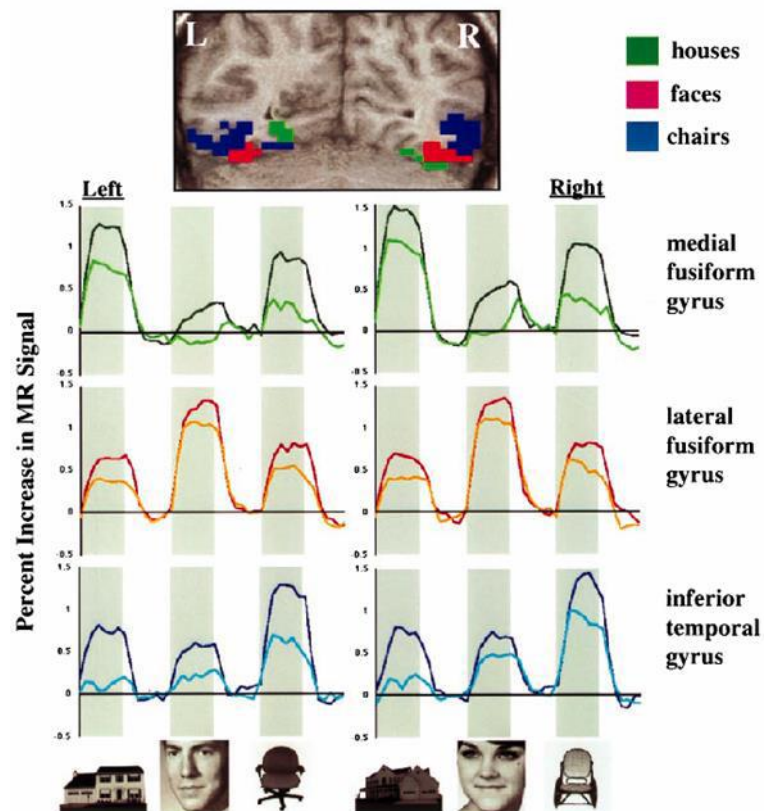
There are features of the face that are stable, and some that are changeable, and there are several additional pieces of information that allows us to say which film an actor worked for. From a face I can extract many autobiographical information.

With a study conducted on macaques it was seen that there are selected neurons that activate when seeing a face.

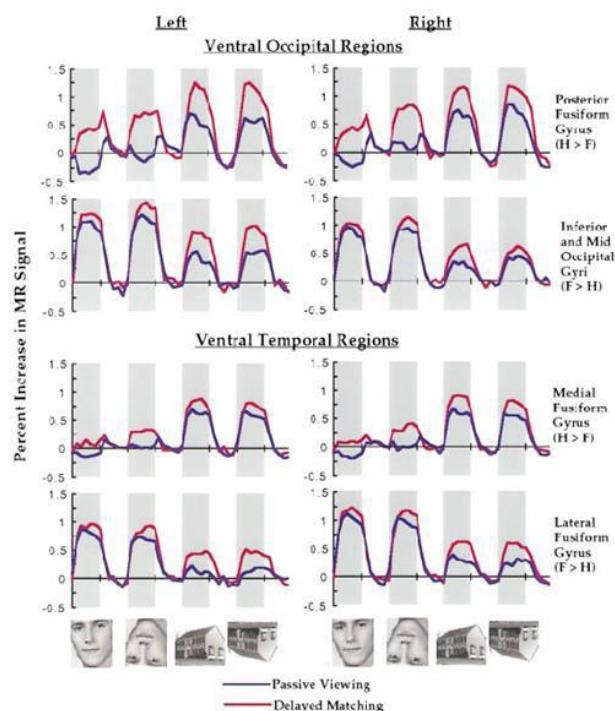
In humans in the fusiform gyrus there is an area that responds more to faces, and this is called the FFA (fusiform face area).



We are also able to detect which part of the brain responds the most to different objects, since there are neurons that activate in certain regions when we are in presence of it.



There are three main groups that create a response, faces, houses and human made objects. The darker line is when you are doing a task in which you are doing perception + attention memory, or passive viewing and no attention (no memory), that represents the lighter line. The response to houses is stronger in the medial fusiform gyrus, para hippocampal gyrus, the lateral fusiform gyrus appears to respond more for faces, and finally the inferior temporal gyrus appears to respond the most for objects.



This made evident that in few years we were able to classify various models in the ventral stream of regions dedicated to the classifying of different things.

The important question is: face is biological relevant, house and body parts also, but how many biological relevant categories can our brain fit? The cortex is not large enough to fit all different categories.

There were researchers that spent decades trying to find the face area. In different studies they tried to compare the responds in brain to face with respect to animals and objects, or living vs. non-living, animate vs. non-animate, it was gradually carried to higher representation.

Studies found that if you have subjects that are birds and cars expert, and you try to make them do a comparison between the two things, the fusiform will respond in car expert when they see a care, while in bird expert they would have a fusiform response only for birds and not cars. But this is not a good stimulus, since they both have a “face”.

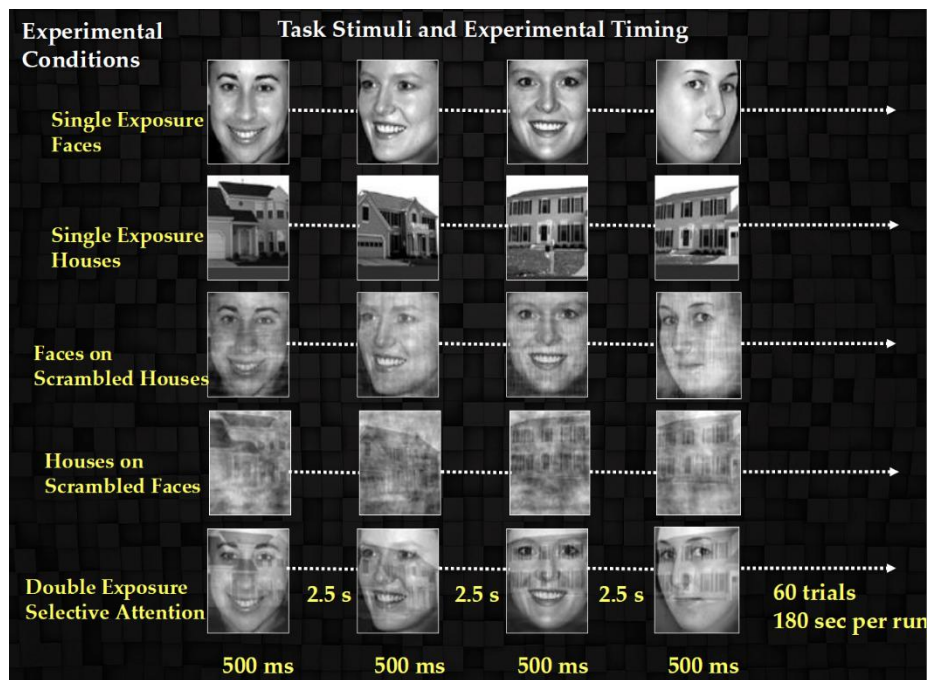
Our brain tries to predict specific shape we are familiar with.

In a study researchers grew some monkeys with the mask on, so that monkeys wouldn't have never seen a face in their life. Consequently, face deprived monkeys were not actually showing specific brain areas or response to the face. They had response for limbs but not face.

Experience is fundamental for this expertise to develop, but if I have the possibility then to study and restore face perception in these monkeys or in a baby, where would be located face perception?

In a study with babies, they were presented different stimuli, and when the baby started responding to faces it was located in the fusiform gyrus, and the same for the PPA. It means that the brain has a proto-organization, architecture, that already scaffolds functionally this ventral stream, and each part of the ventral stream appears to be innately able to compute specific information, for instance face processing. But post-natal sensory experience is necessary for this functional response to develop. If I'm exposed to a face, face processing would occur only in the fusiform face area, in the case I am exposed to places, places representation will be located in the parieto-campal place area, and not in other portions of the brain. That is way we say it is proto, it is genetically determined in the brain.

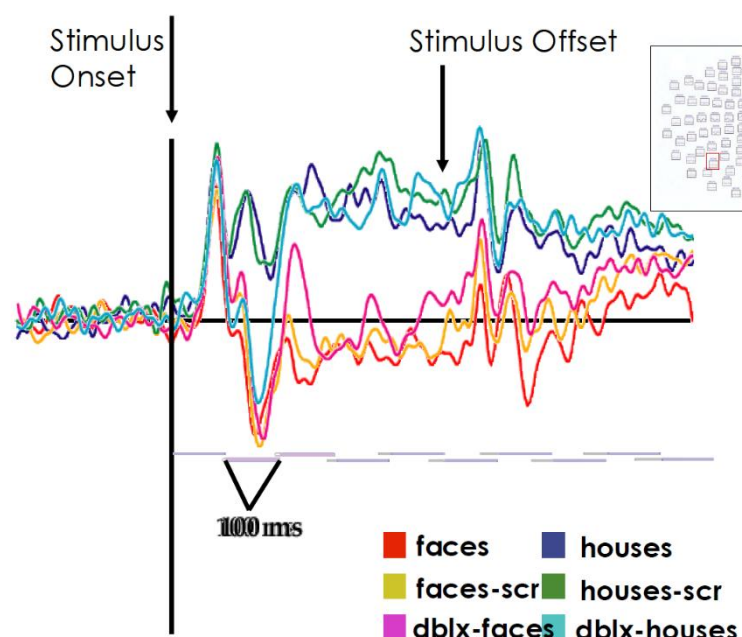
Event-related potential is an average signal that you get from the brain when you are presented several times with the same stimulus, and you record the electrophysiological response, and then you average all of them. For each potential you search for negative and positive peaks, and they called N20 because it is a negative peak after 20 milliseconds (also N17).



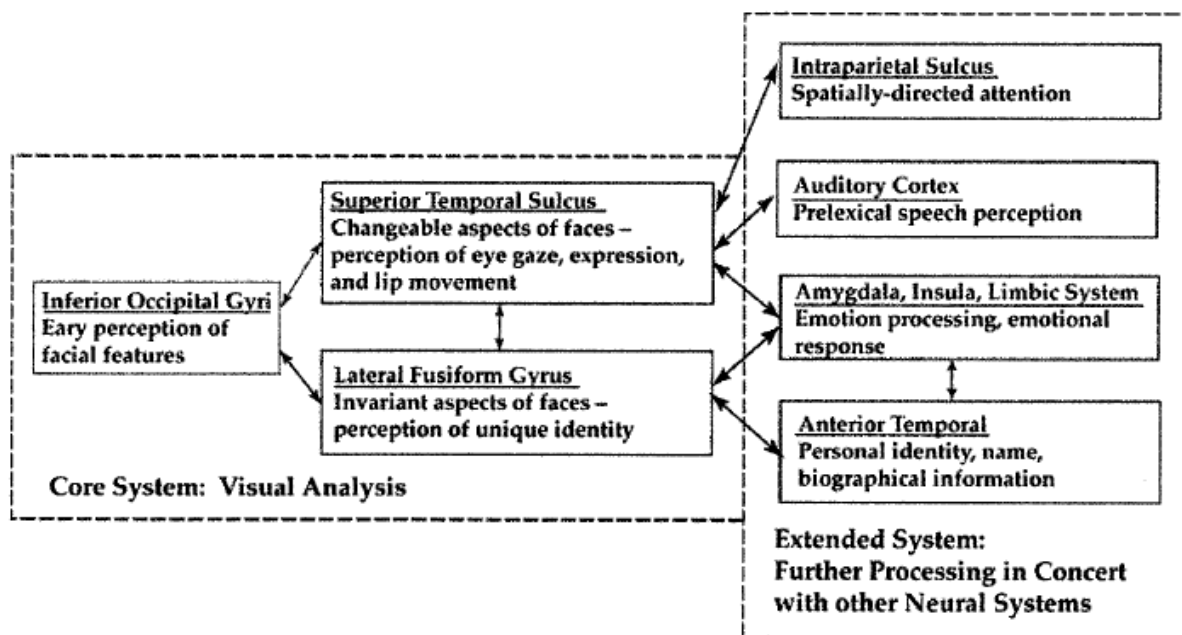
You can have a single exposure to faces or you can ask the patient to say if a face is identical or different from a previous one. And then you can have a check.

The one difficult to see, say to us one thing. When we look at those pictures you do the same task looking at faces or houses, but looking at the house is much more difficult.

The red one is for faces; with a single stimulus you have a negative peak at 100ms. Then you have the blue one for houses, with a positive peak very early that is related to the visual stimulus, and if you look at the temporal occipital cortex you see two pathways, one that is clearly faces and one that is clearly houses. The most interesting part is when it is shown the double exposure figure, asking to perform a recognition of the house, since immediately you have a very bottom-up processing, so at the beginning your brain is captured by the most salient stimulus and it starts processing the face. But then, your top-down modulation enters, because you know that the task is for house.



Getting all together it's clear that we have brain areas that are actually capturing the early features of the face. Then we have the lateral fusiform gyrus that appears to capture the invariant aspects of the brain, like identity, and then we have portion like the superior temporal sulcus, that gets all the changeable aspects of the face, like gaze, eye movements. Finally, we might have other regions that may be involved in integrating further information about face, for instance emotional processing. For face perception we have been defining what is called a core system, that is extracting the basic features of a face, and what is called an extended system (superior temporal sulcus, for example).



Different regions appear to respond more to one category than others, and even if you lower the threshold the response would be different from zero, but you have to see the peak of response to actually know what is exactly responding to. The response is much more distributed broader.

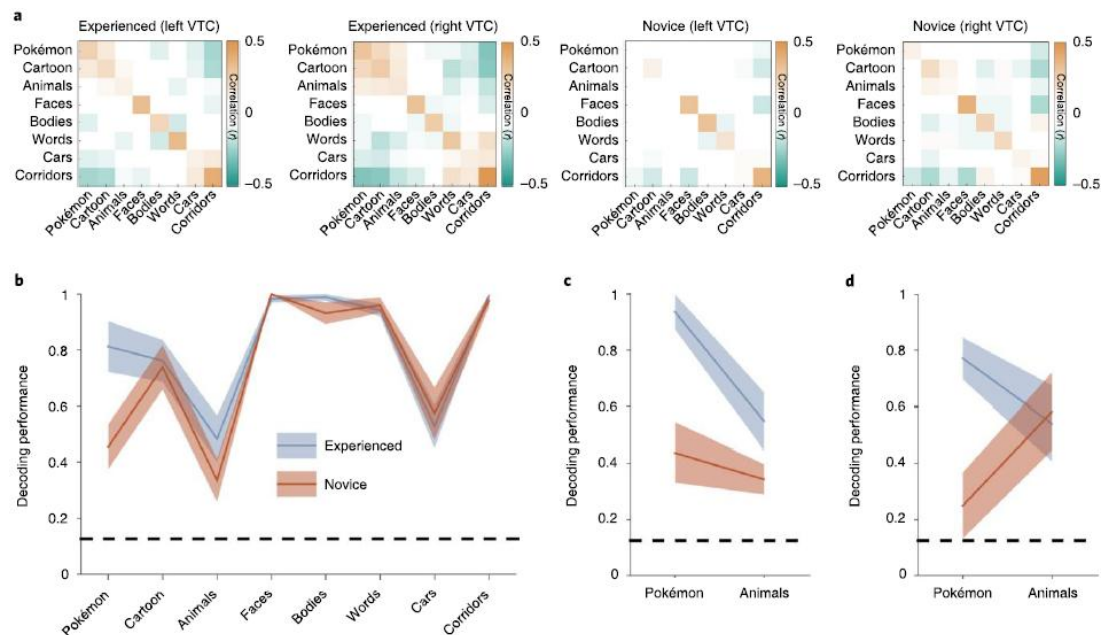
But how can the brain actually represent thousands of different categories of objects?

One issue is related to the importance of where the stimulus is represented. If you look at the ventral pathway, we found several overlapped elements, for instance faces are represented in the lateral fusiform, and places in the medial fusiform of the parieto-campal, but the same kind of discrimination was also possible if we compare animals to tools, and broadly if I contrast animate with inanimate my domain specificity is lost and we are gathering much more an imaging. If you look at objects size, typically faces are smaller, so my discrimination can be real world objects size.

Nonetheless one of the first elements that is important to represent different stimuli, it's the eccentricity bias, that is the idea that the topography organization of retinotopy extends also towards the ventral stream. So, elements that are much more foveal are represented in the lateral fusiform, and elements that are more peripheral are represented more medial. This bias of the retina gets also to the ventral stream. This is why representations overlap, and they probably have something in common; faces are typically represented more foveal (we try to focus on a face), while a place is much more relying on the periphery.

There is a study, relying on Pokémon representation (Pokémon are well distinguished between each other). Experienced people were actually very good in representing Pokémon, while novices

were not good representing them in the brain (also other categories, animals, faces). If we look at the regular image of Pokémon, they are still foveal, and corridors and space had a larger peripheral representation. So, looking at the cortical map, the Pokémon pictures should be able to be more foveal than faces, represented even more laterally.



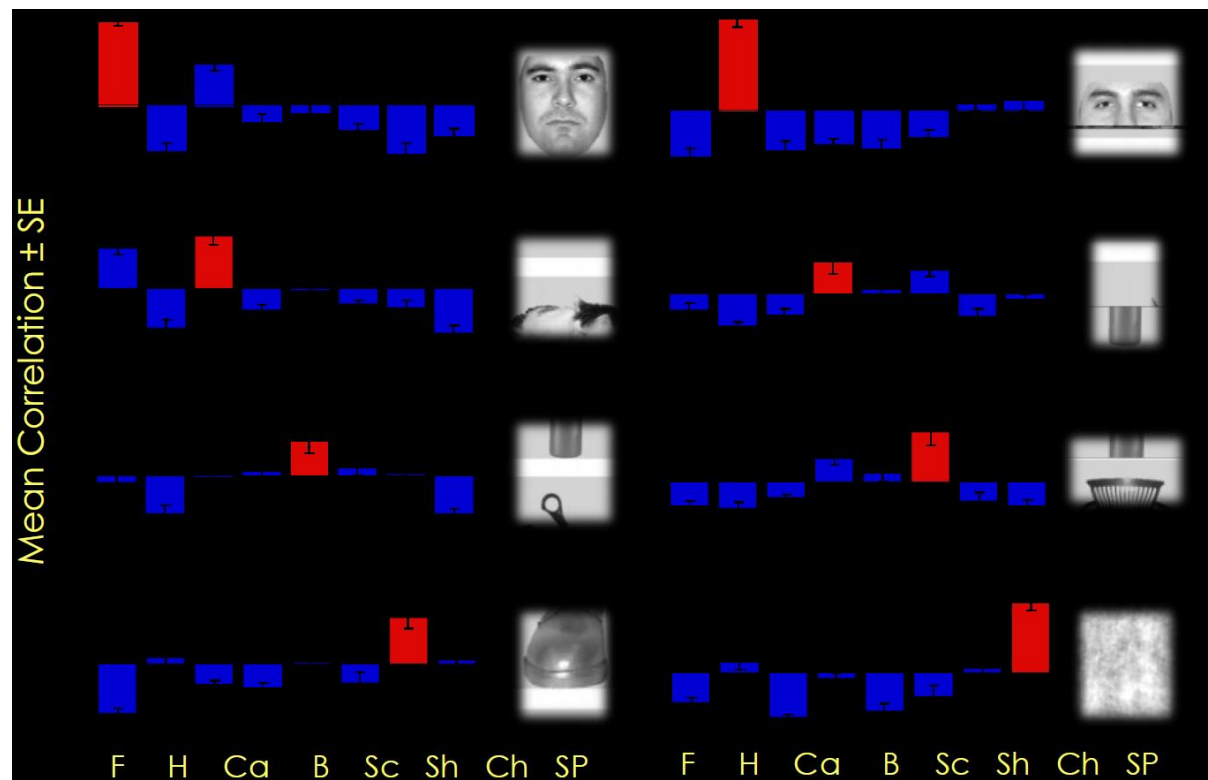
If you are looking at some other elements, like rectilinearity (the amount of linearity of representation), Pokémon should be represented even more medially than face and corridors.

They had to test models of representation of novel categories, so that we are able to know which is the feature that drives my representation. When we look specifically in experts where Pokémon are represented, it seems like that they even represent them more laterally than face, so probably the way also categories are represented and processed at the level of the brain is impacted by the way the eye and the retina (parvocellular pathway) is conveying the information to the ventral stream. This certified that we could accommodate novel categories.

But how can we accommodate thousands of categories? The idea of a researcher was to merely look at responses on a broader chunk of the cortex, look how the whole ventral-temporal cortex was responding to different categories. He used something similar to machine-learning, he divided the response that he recorded in two samples and was using one as a training and the other as a test. He was finally able to describe a specific pattern of response for each category, and then he created an average across categories and then subtracted the average from each category specific response to normalize the response and better highlight the areas with a higher or lower response to the mean. Therefore, you can recognize that for face, the higher response is in the fusiform gyrus, and that for places and houses the higher response is in the medial fusiform parahippocampal gyrus, something that we are already aware of. If you specifically look at correlation between different responses, seems like two responses are correlating higher, so something is likely to be a face. And he did this work for all different categories.

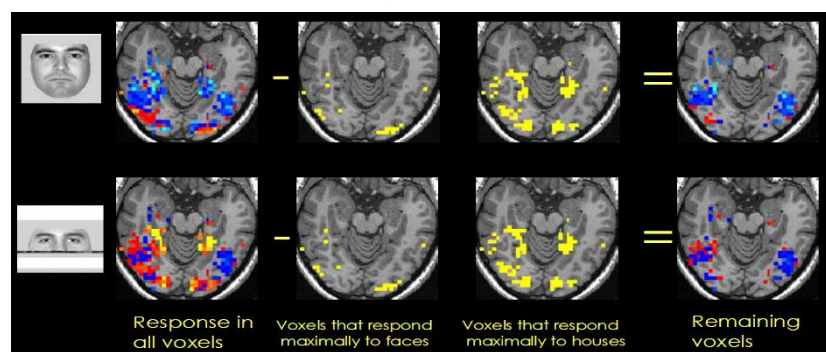
He showed that for a face the higher correlation is for the face, and less for all other categories, and what is interesting is that there is also a slide correlation with animated cats. He specifically defined category specific responses, not as a single module and regions, but as a pattern of response. He didn't limit to a specific representation but he went for the whole pattern, so the

difference was moving from a module to a map, from an area to a pattern of response. It looked like that the pattern was much more able to capture elements of a broader representation of category, and each pattern was specific for a domain for a category.



Criticisms?

These patterns are all distributed, and they are also overlapping. If I want to correlate and to show that a pattern is actually unique for an element, what can drive the correlation? Probably the peak response. Likely, the peak response of the fusiform gyrus could drive the whole correlation, and therefore if this information is really distributed across the whole pattern, so the whole pattern is significant for recognizing a face, I should be able to remove the peak response and get the discrimination across categories. Consequently, what the researcher did was to take each peak response (voxel that responds maximally for the element) and remove them. The remain voxels should be equally able to discriminate each category, if the information is truly distributed, and even if the accuracy of classification is reducing, you are still able to get higher accuracy across categories. If the information is truly distributed, then, it should do also the opposite and limit the classification and discrimination within the maximal responding voxels and be able to classify the other categories. So, you could go in the fusiform face area and rely on those voxels to classify also houses, bottles, cars. And this is actually what we are able to do.



So, the novel concept (in the study called object form topography) is that the information that is present in the ventral stream is probably more distributed than we expected, and also a little bit related to the kind of approach that we adopt. The idea is indeed not focusing anymore on few voxels that respond maximally to a piece of information, to find the category selective regions, but to go to a much more distributed representation of a domain of category. Therefore, I'm no more focusing on the fusiform face area, but I'm actually thinking that a face is represented broadly across the cortex, and this is a huge pro, because the number of maps and patterns that I'm going to have is illimited, and I don't have to concern anymore with which feature are relevant in representing the cortex, because all elements could be represented here and the whole pro is that patterns of categories that are semantically similar (faces and cats) should be somehow similar and a little bit overlapping in some of the features, but still distinguishable between them.

An aspect less clear is, why for many years we had been able to describe disorders that are very selective? If we have a face selective area lesioned, we have prosopon agnosia, and the information is distributed all across the cortex, why a lesion of the fusiform face area is producing the inability to recognize a face. We can claim several things, but probably we are really looking at the same response with two different kinds of analytical tools, highlighting different perspectives of my brain organization. Probably there is something that is in the middle.

The main difference, in what we talked about, is that if we look at the single voxel, you have a single musical instrument playing, if you look at the pattern you are listening the whole orchestra playing. And finally, what this study is telling us is that for sure the information about a specific category is distributed and involves more regions, but probably some regions are contributing more to the representation than others.

In order to merge this kind of view, we should actually think that the ventral stream may continue a sort of like hierarchical and nested information, and just a matter of were I'm actually focusing. So, probably you have in the brain different areas related to the response for single elements, and all this elements are responding to something animate, but this pattern of response is actually integrating different pieces of information, and the more you broaden your map the more is the possibility to represent diversity, but at the same time the risk is to miss the specificity of the response that our brain is able to. Furthermore, linguistically I'm able to organize these domains and categories at different level of complexity and comprehension, and likely this way of organizing knowledge that we express in language is also the same way the brain is organizing knowledge within the cortex, defining a sort of hierarchical and nested representation of different elements, that then gather together in a broader super-ordinate element.

Obviously methodologically I could drive my interpretation; if you decide to look at the connectivity between regions, the theoretical model that you have is that information is connecting different brain areas and pass one area from another. If you look at one small portion of the brain and you ignore the rest, obviously your idea is that that piece of brain contains meaningful information for your task.

There is not a good or a bad approach, if you decide to find a threshold and get the higher response of your voxels, your idea is much more the one of looking at the modules doing different things. If instead, you lower your threshold and you look at broader number of voxels distributed in a wide extent of the cortex you are looking at the map, the distributed representation. The first one is very good for any univariate approach, and the second one for any multivariate approach, from a statistical perspective.

One thing that you should always keep in mind, is how much your methodological approach is impacting on your theoretical model. Or you can have a strong theoretical model that will guide in using on methodological approach or the other. But beware that once you look at the results you should keep in mind that they come from a specific methodological approach, and this could drive the interpretation of data.

There may be features that are representing different elements and you can actually use this according to the different categories (face for humans but also for cats, body is not). It's probably that we have simple different elements that are summing up to represent different elements.

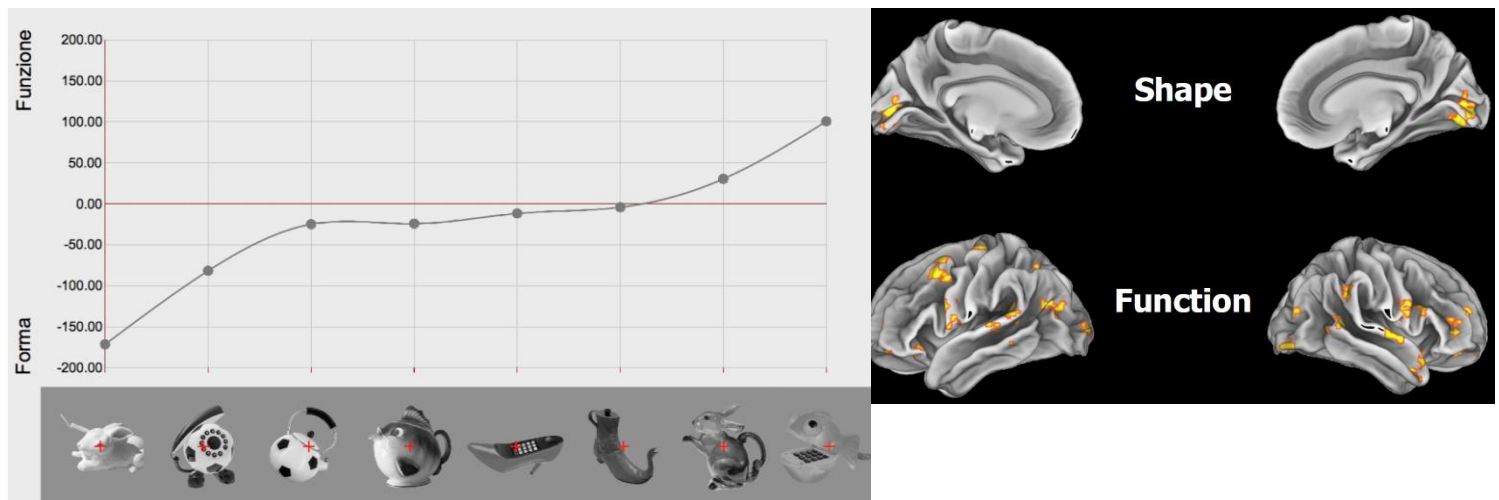
The deep network has different layers that extract different piece of information from the stimulus you are processing, and these layers often overlap with the hierarchical processing of vision, and so V1 is much closer to early layers in our network, while advanced regions are going much deeper, so there are several attempts to module on this way. Other approaches are relying on language, using it as a model to test the complexity of the representation. So, nowadays computation tends to work a lot, we should think about doing the opposite thing. If my representation is not primarily done by different features, but actually what I am going to represent is much more related, for instance, to action and verbs that are much more related to my stimuli, this is much more related to encoding approaches and are equally valid to our purpose.

If we have different verbs (bark, fly, run, eat) and then you see a dog, it will be in different ways correlated with these different verbs (bark 0.6, fly 0.01, run 0.4). If we have a specific voxel that are processing the verbs, just with giving a weight to these voxels you can represent a dog. For an apple we can do the same thing. This kind of approach (encoding of the information) is working on the opposite way, so you define the criteria organization of your ventral cortex and you just weight the representation on the cortex per se, and this kind of thing is impressive, because you can find and replicate the verbs or elements semantically related to house, for example, would be represented much more on the parahippocampal area. This is actually how you can process the information. Is the brain actually operating in terms of these 25-20 verbs? No, but this is how you define and decide to describe semantically this organization of knowledge and is actually impacting your theoretical model of the brain, that is much more related to the events that you are doing with these objects, other than on their shape, texture or colour.



This is not a pipe because you can't use it to smoke, and you can create hybrid objects like this. When you try to understand and describe this element, this is something "pipe-shape class", and the idea is that you try to disentangle the shape and the function. With this type of objects, you can define a behavioural curve, that is telling you whether each object is much more driven by the shape or the function. When you try to look at the response and weight according to these elements, when you see that the shape is something that you can detect easily in the early visual areas and a little bit in the ventral stream, but function is indeed extended more broadly. For a

stimulus, if you look at which features are represented according to the behavioural description, you see that information about shape is just limited to the ventral stream, but when you talk about something more related about function and action, there is a broader engagement of the whole brain.



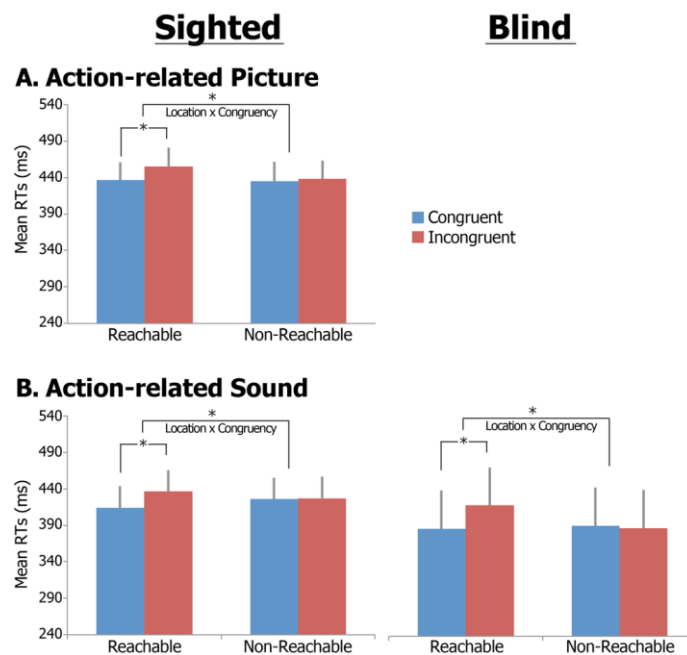
The third pathway is much more related to the superior temporal sulcus involved in social perception (recognizing facial expression, aspects useful for social interaction), a theory of just 1-2 papers. Obviously, all the information of the two pathways is highly integrated constantly, to get more complex and high-level representation.

Supramodality

Neuroscience started to consider other sensory modalities. Do we really need vision to capture all different pieces of information? When we listen to sounds that we know, our ventral stream is processing even auditorily the sound, but without being aware our brain is also processing the action that produced the sound, we do not truly need vision to make sense of the surrounding world. It seems like that several features are not actually depending on visions, sight has always been regarded as the most important sensory modality, and more than 50% of the brain is devoted to process vision, nonetheless all those who are visual deprived since birth appear to show most of their perceptual and social skills similar to those found in other individuals. There are no superior skills, they simply rely on other sensory modalities and some kind of external world representation are almost similar with some difference.

It's important to mention the spatial alignment task, which is something that can be done when you create a sort of discrepancy in congruency between the vision and motor command. You can have a conditional incongruency, so the command is to perform the action with the right hand, but the ball is on the left, this creates a small delay in the response. The ball can also be in a reachable distance or not reachable, if the ball is in a space that is not reachable this effect disappears. When the object is outside our space of action, we are not processing it anymore.

The first attempt was to translate the task in an auditory one, so people were blindfolded and the ball was bouncing on the left or the right. Some individuals could present the same effect on the auditory condition, but also congenitally blind people. So blind people that never experienced vision have a sort of representation of the space around them.



You can also do the same experiment with or without an agent, that is present on the side of the table, and he just helps you reaching the ball, he's making that action possible, and the ball is in his space of action, and my space of action is the same. Important for next week lesson.

In blind people (because of defects of the eyes) we have a reorganization of the visual cortex, being it a little bit thicker and less myelinated with respect to typical individuals. But the major alteration we find are related to the visual pathways; all of them, from the thalamus back to the occipital cortex appear to be reduced and atrophic, since there is no input from the retina. There is obviously a reduction in the visual fibres of the corpus callosum, and also there is a reduction in the lateral geniculate nuclei.

But what happens to V1? Sighted people rely on primary visual cortex; while providing auditory input they respond with auditory primary regions. In blind people the primary visual cortex tends to respond to non-visual modalities, something that historically has been called cross-modal plasticity (one modality is invading the other brain areas). We also found out that the recruitment of the occipital cortex was correlated with the performance, appearing to be independent thus responding both to tactile and auditory, also across different tasks. This obviously made people wonder on the functional meaning of this. The functional significance of these changes remain unclear.

A first hypothesis is that this brain area in blind people is actually starting to do something important in compensating, a hypothesis that is truly compensatory. There is piece of evidence demonstrating that if you inhibit or you have a damage in the occipital cortex, blind people have the ability to read blind, so it looks like a sort of compensatory task could be there. Nonetheless, even this kind of compensatory role appears to be very limited, and the area for language and word representation is indeed very close to primary visual area in occipital cortex.

Other people think that V1 becomes pluripotent, starting to do high level functional and semantic processing, we do not believe it. In nature what is pluripotent is not highly specialized, a cell is pluripotent and is not functionally specialized in tissue, it can develop. It's very difficult that pluripotent is becoming highly productive.

Others think that the recruitment of V1 is a little bit aspecific, that it just reflects top-down (attention) modulation. But V1 is still receiving piece of information from the other sensory modalities, since there is plenty of evidence that V1 is actually perceiving information from visual cortex. All the ventro-temporal regions are conveying information to V1, even if it is not sending anything back, there may be connections with other sensory modalities to V1.

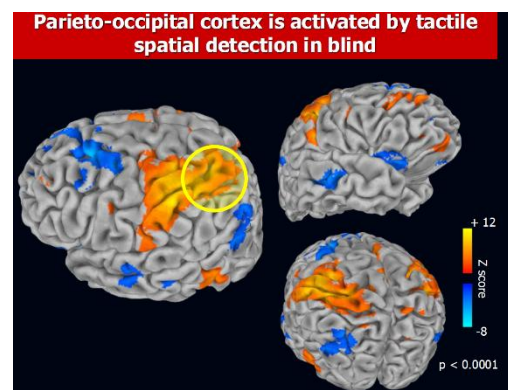
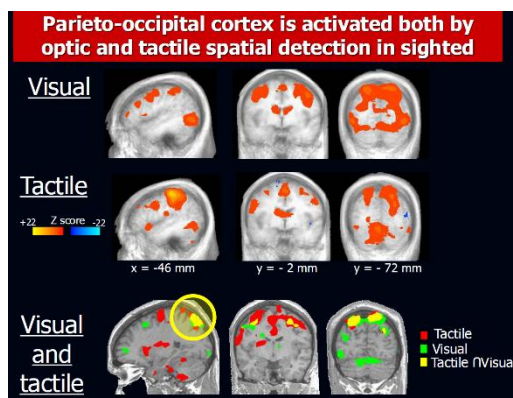
These two perspectives are very interesting, because you are telling us that in a multisensory perspective one region when gets specific sensory input is also sharing this information with other modalities, and in a multisensory world like ours it makes sense that when we get an auditory input the visual system should be synchronized and rephase in order to respond to respond to the multisensory evaluation.

It is likely that V1 is responding somehow in an aspecific way, it is not a compensatory plastic response, but there is no clear demonstration. Also, one other thing that we have to keep in mind that different brains may have different weights of the different modalities (the bat brain is highly representing the auditory area). In rats that have very poor vision, even though it is useless they still keep a small V1 region.

In a study, subjects had to recognize tactilely shoes, bottles and faces. With visual recognition, you can see a nice activation of the occipital region and of the inferior temporal cortex. Blindfolded you recognize all the elements, you have the activation of the sensory motor cortex and also of the inferior temporal cortex; vision is green, tactile is red and yellow is overlapping. So, even if your eyes are closed, when you touch an object, you recall the visual image. Blind people are different, because they are used to recognize objects touching them, so their recruitment of the sensory motor cortex is much smaller. But this area is present here, so clearly it looks like that the ventral extrastriate cortex shows category specific response on the tactile detection of objects, while the cortical area is involved in visual discrimination of objects, but also in tactile discrimination

of

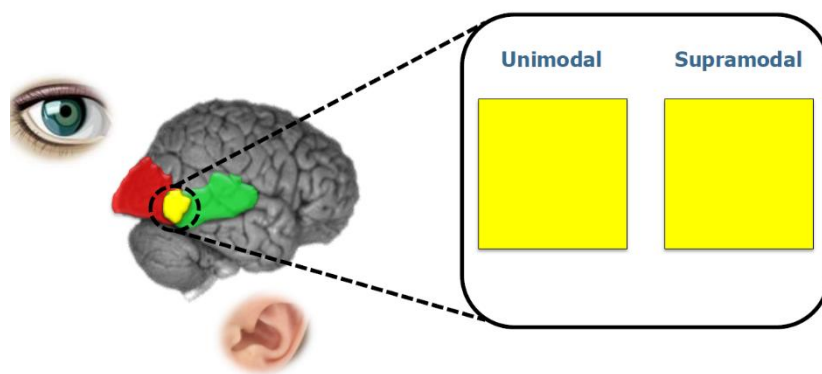
it.



Moreover, doing the analysis showed before of category related patterns even in blind individuals we are able to get bottle, shoes and face response, but they can't really recognize face that they don't know from the touch. One of the issues when you try to recognize a face tactilely is that you are not able to reconstruct the configuration of it, but only the separate elements, and this is called gestalt (the ability to bind together elements into unique percept). While you have specific category patterns for objects, it is a little bit ambiguous stimulus, nonetheless to our purpose the fact that blind people, that never got visual experience, have this type of response, is telling us that this response is not due to visual imagery but more important that visual based experience is not necessary for the development of the functional architecture of the ventral stream.

After few years, there has been several experiments about all different brain areas. And f-MRI studies in blind people told us that certain regions appear to be specifically related to capturing, for example, motion discrimination either tactilely or visually. Visual and tactile modalities project on the dorsal stream, so not just the ventral but also the dorsal, and we can see in blind the same spot.

Blind people are an example of these overlapping elements, and this is actually something that we have been calling supramodal. For some individuals that for instance perceive motion for objects either visually or tactilely, there is an overlapping region activated. In blind people you have the same type of recruitment, so if cross modal is something that happens because of the lack of vision, supramodal is something that takes place despite the lack of vision, something that appears to be there even without sensory input and experience.



Most of the brain appears to be supramodal for several aspects, not just motion discrimination in green or object sight in red, but also for some other like spatial orientation, mathematics. So, all the ventral and the dorsal pathway and several regions on the pre-frontal cortex, appear to be independent of the sensory modality, overlapping. Task specific but modality independent.

It looks like the retinotopic organization in blind is preserved as spontaneous activity, retinotopic organization is actually maintained in blind people.

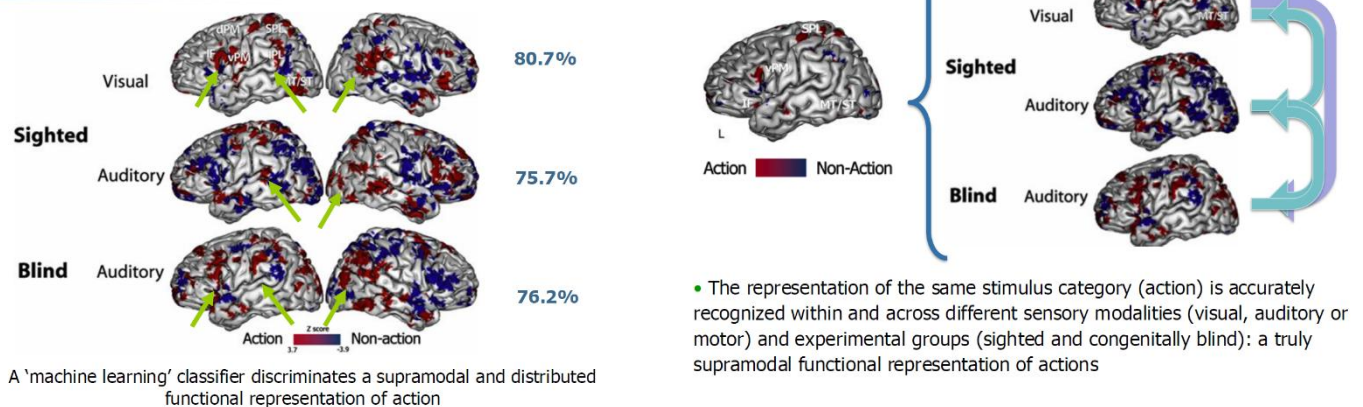
So, it seems like that because of the lack of the sensory modality individuals undergo a plastic organization, something that is called crossmodal plasticity, even if we still don't know how this occurs, and brain organization is primarily driven by task specific but modality invariant computation, something that can be referred to as supramodal. The idea is that the kind of information is shared across typical developed and sensory deprived individuals, that is that when we say that blind people are differentially able to process information is true from a neuroscientific perspective, so they actually are able to differentially perceive the world, but represent the external world in the same way as a sighted individual does.

The main idea is that truly these aspects are overlapping. One main criticism that could exist is that being an overlapping region of different voxels, the voxels can be formed of different neurons of two unimodal, either tactile or visual, or there are neurons that are processing information conjointly? If we are looking at discrimination of action vs. inaction and we have 3 experimental conditions:

- Sighted individuals doing the task visually.
- Sighted individuals doing the task auditorily.
- Blind people doing the task auditorily.

The ability to discriminate is well above the 50% chance and what you can do is really get a classifier into a unique independently of the 3 modalities and what you are doing is to define regions, that are typically useful for discriminating action, with which you can train your machine learning tool on the visual stimuli and then test on the auditory stimuli and then get significant ability to discriminate action vs. non-action. And this is a sort of demonstration that this data is truly representative and there is plenty piece of evidence that are showing this aspect.

Beyond motor scheme: A supramodal distributed representation in the action-observation network



Even if you go on other topics, like conceptual representation, and you have like 8 categories, and you ask blind people to produce semantic description of those, you find a very high correlation between the description of sighted and blind individuals and yet characterized by domain specificity. If you give them a banana the first thing they say is that is a fruit, so the domain is the first aspect. Blind people rely a lot on synesthetic aspects.

What happens if you restore vision? There may be function that we are able to restore, and others that still need time to develop. People whose vision is restored (kids in Africa), for many weeks are not able to recognize face, they need experience, they rely in voice to recognize a person. So, there are features (low-level features of perception) that are already present in the brain, like shape and biological motion, and this is something that can be easily restored, complex stimuli indeed, like faces, require experience. Our brain is a mixture of experience and innate features, and there is no one dominant aspect. One important thing is that clearly the brain is able to process and to develop its functional architecture independently of a sensory experience, like blind people.

Motor and action

We should depict the brain as a system that has a specific aim, collecting piece of information from the periphery in order to enact some sort of behaviour, and obviously motor behaviour action is a sort of main outcoming of brain processing. We have been actually extending our idea of how our brain get information from the external world and try to make sense and interpret this. Our brain specifically selects information that has to be collected, and the way it interprets it is always something that relates to somehow how they can modulate consequent behaviour.

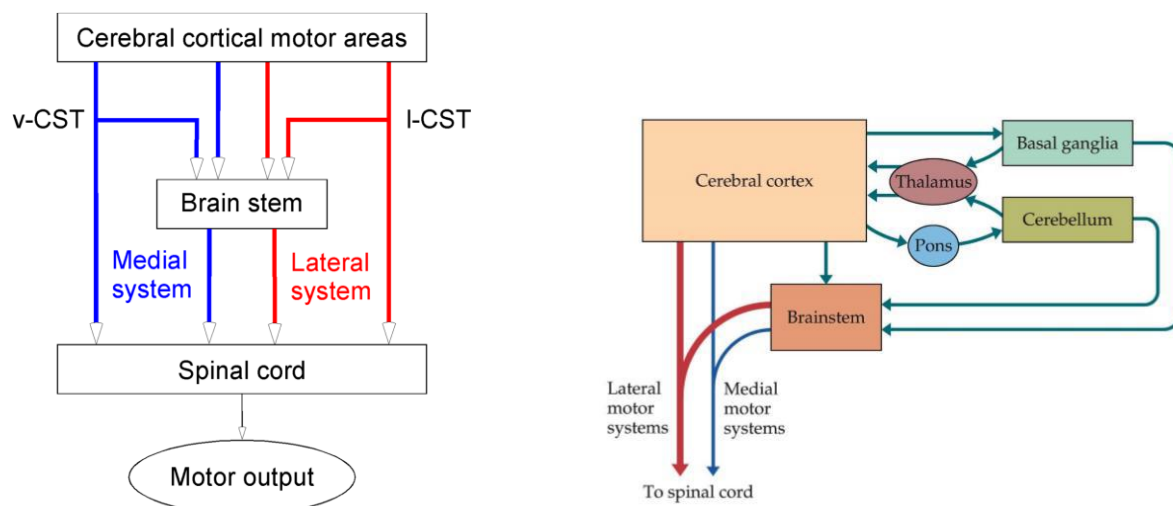
One thing that is still a question, is how the brain organizes and represents simple motor element into a finalized behaviour. The way the brain represents action is something that is struggling several disciplines. If you think on how to provide commands, we have several redundant piece of information that represents most of the time noise for our purpose, and even we try to classify

and define a taxonomy or features that represent and characterize action, we fail. This is because when we consider action per se there are plenty of features to be considered, and most of the time when we consider action as a whole we have other cognitive functions that are involved (if I have to grasp something I first have to perceive the object and understand the features and how part of my body, what we define an effector, is actually interacting with the object, and even the way we are going to interact with it). One other message that is a sort of feedback message we got from this, is that all different aspects of our cognition (including perception) are influenced by the motor activity that can be acted upon a specific object.

If we focus on the effector, the way our body interacts with an object depends also in the way I represent it in my body, and at the same time the possibility of actions that my body could act upon the specific object depends on its position. So, it's really difficult to comprehend a classification of actions, based on the different levels to which you can describe it (action can be transitive or intransitive). This is something that we can describe semantically, but if we go on a clinical perspective, you find a different classification, and this is really getting different levels of taxonomy that make hard to understand how we really process with action and also tells you how why AI based systems fail classifying action. It's very difficult to deal with an unpredictable event.

We are going to go briefly to describe different levels of representation and just see a flavoured glimpse of how we can start from motor commands going to action representation and processing according to different levels of representation and primarily figuring out the whole brain is involved in action processing.

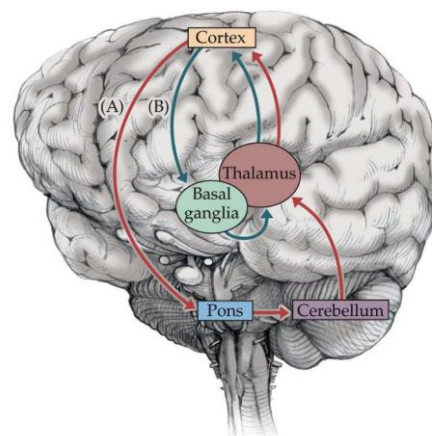
When we talk about the general organization of the motor system, we could obviously get different levels of representation. Our first aim is to have some specific piece of information that originates from the motor cortex and has to be forwarded to the peripheral of the body, and primarily to muscles that allow us to enact specific behaviour. This process is offered very effortful and time consuming, and indeed we have also motor components that are enacted independently of the commands from the central nervous system, there are specific behaviour that for the sake of survival are enacted without being conscious of that at the level of the spinal cord (central pattern generators) that regulates the simple movements. When there is a damage in this pathway, the command cannot be fostered from the cortex down to the muscle, but if there are connection there are still valid at the level of the spinal cord, for relatively simple commands (like locomotion), you can stimulate spinal components to induce very simple motor elements, and the sequential stimulation can allow to put together different movements and components into a simple finalized act.



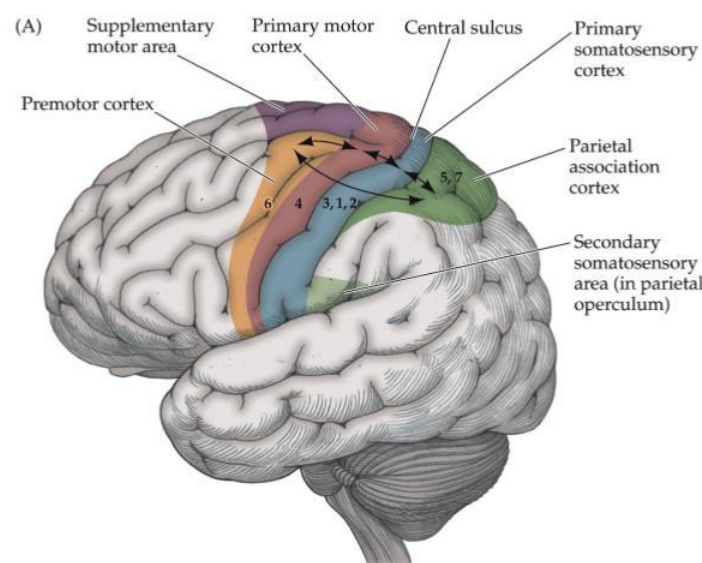
The most difficult part of our motor system, is that we have not just to operate as a sort of like simple element, but for instance in order to still fruit towards something I have also to use other muscles to stand up, so there should be some processes that should be come somehow automatic, and there are plenty of movements that we are able to perform that are not actually under voluntarily control, but are performed without really conscious processing or representation of those, and the more you become skilled in a specific movement the more effortless the movement becomes.

One thing clear is that in this chain of acts controlling motor functioning, there are two things that we have to highlight:

- There are different levels, and this is hierarchical (cortical motor areas -> brain stem -> spinal cord, directly having specific motoneurons having an acetylcholine synapsis with the muscle).
- At all these stages there are specific feed forward and feedback processes.



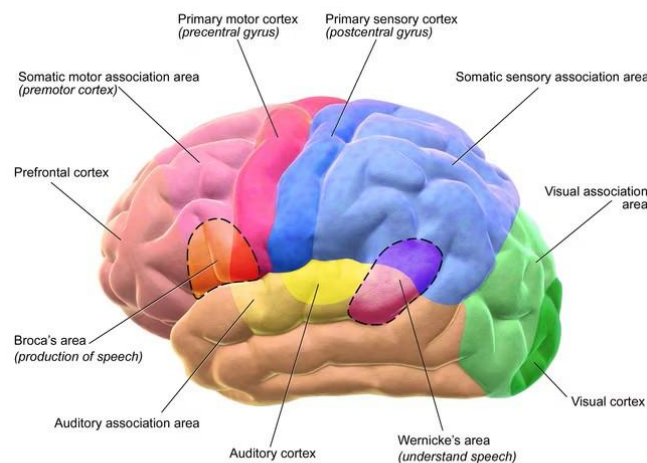
These refined movements are actually under the control of different acts, that have constantly this loop of control, that primarily guarantees that when you decide to perform a movement it's fluid and is constantly updating. The most skilled person are athletes that are able to predict in advanced some specific behaviours.



The primary motor cortex is where we have these pyramidal neurons, also the motoneuron α , that is the biggest one in the cortex with its long axon, and they project downwards primarily to the spinal cord. But some of these neurons are actually going and arriving just at the brain stem, so if

we look at the cortical level, most of descending pathways that start from the cortex get from the Brodmann area 4 (the pre-central gyrus), but this is not the only region that is able to send projection down to the spinal cord, because 80% come from the Brodmann area four, but the other 20% also come from other regions. More anteriorly we have the pre-frontal cortex, and anterior to the pre-central gyrus we have the pre-motor cortex, primarily divided into a dorsal and a ventral portion, and the ventral portion is almost w shaped and this where we have the inferior frontal gyrus, that has this m or w shape, and here there is the Broca area of language but also the ventral pre-motor cortex.

Motor and Sensory Regions of the Cerebral Cortex



More immediately we have the supplementary motor area, that appears to be fundamental to process bimanual coordinated processing. The pre-motor cortex (both ventral and dorsal component) appears to be much more fundamental for processing the goal of the action.

The precentral gyrus is also mirroring the postcentral gyrus, that is where the primary somatosensory cortex is representing different body parts, location and mechanical stimuli getting to the skin. This piece of information is projecting posteriorly to the Brodmann area 5 and 7 of the parietal cortex, where we have several portions of the cortex that are very interesting in order to represent action, but here we got information from the motor component with the visual component about space, so in the parietal cortex we'll have actions primarily representing according to their location in space.

The eccentricity bias is the idea that the retinotopic organization is projecting beyond the early visual area not just ventrally but also dorsally, so clearly the way I'm representing actions depends on the way it's located in my visual field, and there is this tight coupling between the motor representation of an action with what is typically the representation and location in the visual field.

If some of these regions appear to be sending very simple commands about motor planning, these other pre-frontal and parietal regions appear to be much more related to a sort of higher order representation of motor planning.

In the portion of the dorsal and the superior frontal gyrus there is another area called the frontal eyes field that primarily regulates the movement of the eyes, and this region in the frontal portion is actually related in moving the eyes accordingly to perform some specific actions.

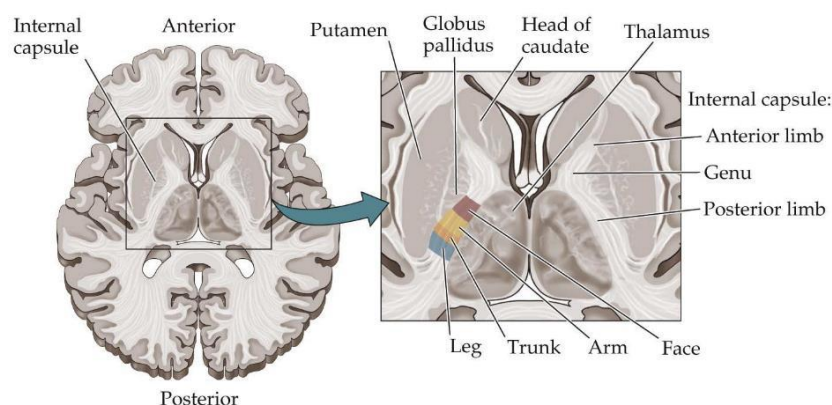
Here in the superior parietal cortex there are single patches dealing with different elements, so if I have a disorder in a certain point I have a disorder called agnosia (inability to recognize object identity), and when I'm going dorsally I may have two kind of lesions; one that is called ataxia but much more optic ataxia (I'm able to recognize the object the object and decide which action to perform upon it, but I'm unable to perform a precise reaching space).

One of the features that is very similar between the somatosensory cortex and the motor cortex, is the representation as the homunculus, even with some differences. The motor component is much more enhancing the effectors, the representations of feet and fingers are much broader in the motor homunculus as compared to the somatosensory one, so it's mirroring physically some aspects of our body, how we are effectively able to interact with the surrounding world. This kind of somatotopic organization at the beginning has been described by a neurosurgeon (most famous one Penfield), putting electrodes and trying mirroring the somatotopic organization, but this is mostly true for the most posterior portion of the motor cortex, because the more anterior one is actually already trying to process action much less as organized as from a somatotopic perspective but much more relative to a sort of aim of the action.

The information from the cortex is in a constant feed forward and feed back loop with the thalamus and the basal ganglia, together with the brain stem, that is typically then forwarding information at the level of the pons of the cerebellum, that typically receives information at the level of the cortex and it goes to the nuclei of the cerebellum, and finally the information is projected back both to the brain stem and then to the basal ganglia. This loop of motor control highlights two aspects:

- Motor commands are redundant, we have to be confident that if we have a specific goal, I am able to perform that.
- There are several mechanisms that control that the whole process is controlled smoothly and uneventfully.

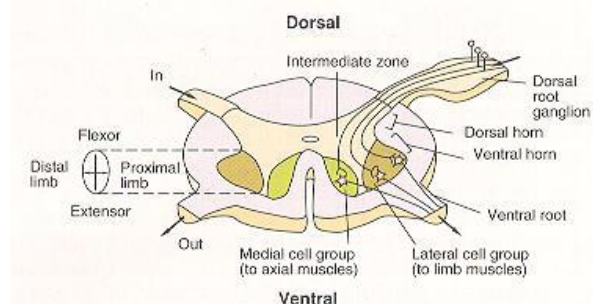
The first kind of control is in the basal ganglia, where we have several structures, like the caudate and the putamen (together they form the striatum, that releases dopamine, and disorders at the level of this substance are causing Parkinson syndrome), the thalamus and the globus pallidus. It's important to know is that all fibres that from the cortex are going down to the spinal cord, are actually passing in what is called the internal capsule, that is separating the caudate from the putamen and is also separating the pallidus from the thalamus, and we can also separate what is called the anterior limb, the genu and then the posterior limb. According to the organization of the fibres we have a specific representation that is maintained also somatotopically, so there is a topography also of the fibres that from the cortex progresses downstreet to the spinal cord (we have the leg, the trunk, arm and face).



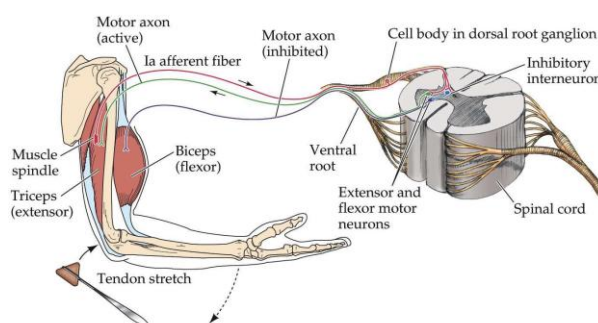
The information from the cortex to the internal capsule is then getting down along the spinal cord and we have also fibres that progress from the spinal ganglia down to the spinal cord. We are going to separate two systems, the lateral motor system and the medial motor system. The first one is primarily fed from the cerebral cortex, but there are also fibres from the basal ganglia and the brain stem, and this system is also called the pyramidal motor system and is primarily related to the regulation of voluntary movements. The medial motor system is also primarily fed by the basal ganglia and the brain stem and is primarily related to regulating all the muscle chain of the median part of the body, the central part of the body; these are movements that are not voluntary and are primarily dealing with regulating automatic movements, cyclic movements, and maintain posture and balance. Any disorder that affects this pathway (called extrapyramidal or medial pathway), either because of a lesion or of the source, impacts posture, gait, so the fluidity of movements, even voluntary ones.

At the beginning they were primarily described as pyramidal or extra-pyramidal, pyramidal because pyramidal neurons are the motoneurons that are present in the primary motor cortex, but since the cortex is contributed to both pathways, we rather describe them as lateral and medial motor system, that mirror their position in the spinal cord.

The dorsal horn of the spinal cord is primarily sensitive, with information getting from the periphery to be forwarded to the post-central cortex as part of the somatosensory cortex, and the anterior portion of the spinal cord (the grey matter of the spinal cord) is instead related to motor, and the medial pathway (in yellow) is primarily related to the axial muscles, while the lateral portion (in orange) is primarily sending information to the limbs, the effector, to the periphery.



Here typically we have a motoneuron that gets information from the spinal cord to the ventral roots, to the muscles. Here we have a very simple stretching reflex loop, because information at the level of the spinal cord is not just afferent to muscles, but the muscles spindle is sending projections and information to the spinal cord about the position of the muscle and the posterior sensitive elements is connected to the anterior one, through the inhibitory interneuron that connects the two neurons. So, the ventral nerve roots convey efferent motor signals, and the dorsal roots are getting the afferent sensitive signal.



Typically, the muscles spindles (sensory receptors that detect the amount and weight of stretch in muscles) transmit information to the distal processing of sensory neurons and convey to the spinal grey matter through the dorsal root. Here we have the inhibitory interneuron that connects to the anterior motoneuron, and this project to the muscle. So, this local circuit controls what you typically experience, if you want to extend your arm you have to use one group of muscle, but you should inhibit the antagonist group of muscle that is contrasting the action. When the neurologist hit the tendon on the knee, it stretches the spindle and you are performing a stretch and providing information that that part of the muscle is stretching, and this is going to induce an opposite reaction (the extension of the leg). So, what you are actually testing is this type of circuit, that has to inhibit or excite agonist and antagonist muscles of our limbs. These kind of stretching reflexes, are also useful when the stimulus is not starting from the spindle but is starting from the skin, just as with pain, where you have a painful stimulation, and before the information gets to the cortex you just remove the limb from the possible source of damage.

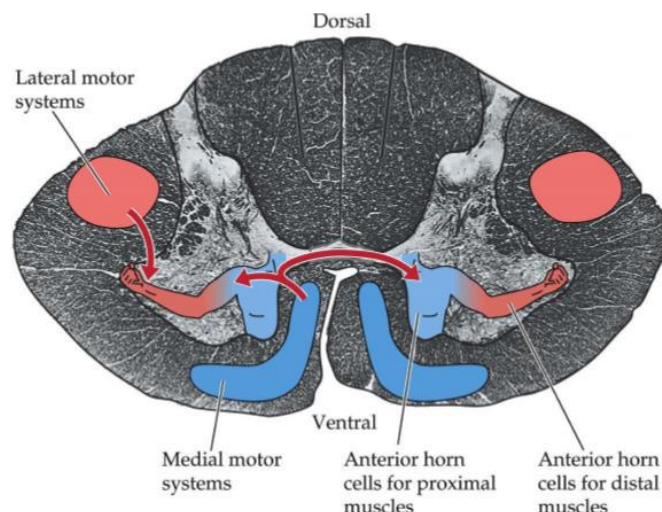
The control effect that the cortical motoneurons are performing to the spinal cord is very important, because is regulating the excitatory inhibitory balance. Most of the time, when you have a lesion at the level of the spinal cord, and this is no more able to get information from the brain, muscles are going in a sort of like conflicting presence of stimuli of excitatory and inhibition, and without this element of control you see that the muscle is primarily stretching and detecting the position that we call plasticity, is excessing excitatory response.

This kind of complete pathway from the cortex down to the spinal cord is fundamental to maintain the proper functions of the muscles.

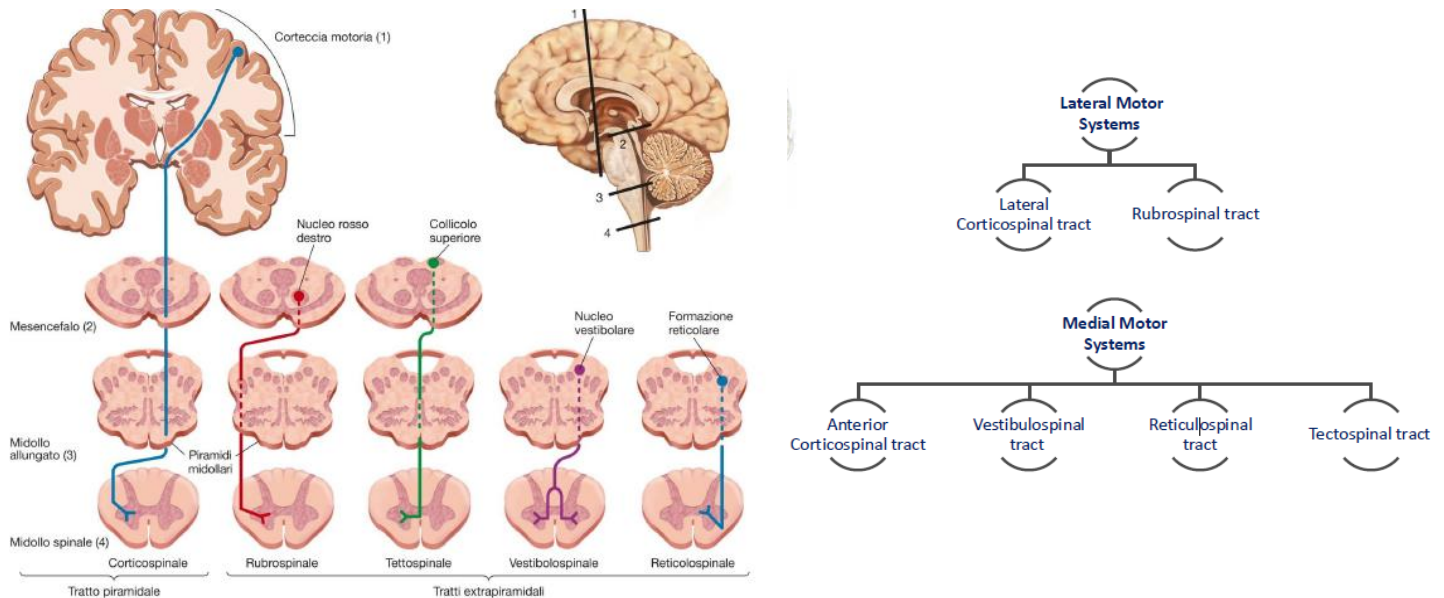
Motor pathways

At different levels the lateral and the medial pathways are not just maintaining the somatotopic organization, but at the same time they cross the spinal cord, and they reach the opposite body. So, similarly to what we have seen describing the tactile component, the right hemisphere is controlling the left part of the body, and the left part of the cortex is controlling the right part of the body.

The majority of the fibres of the cortex (85%) is creating what we have been calling the lateral pyramidal projecting system, but the 15% of the cortical fibres are also going to form the medial extra-pyramidal tract, and this is typically how we do not just enhance the somatotopic organization per se, the more anterior component pathway is processing the medial part of the body, while the lateral motor system is precisely representing the limbs.



We have the lateral motor system and medial motor system, the first one is made up by the cortical spinal tract and what is called the rubrospinal tract (originating in the brain stem, in the mesencephalon, and is called the red nucleus), that immediately crosses fibres and goes down. Then we have other four tracts, that are actually making up the medial motor system. The cortical spinal tract (15% percent of fibres originating from the cortex), the vestibular spinal tract, the reticulospinal and tectospinal. These are all brain stem nuclei that project down to the spinal cord. See the slides to see better each one of them.

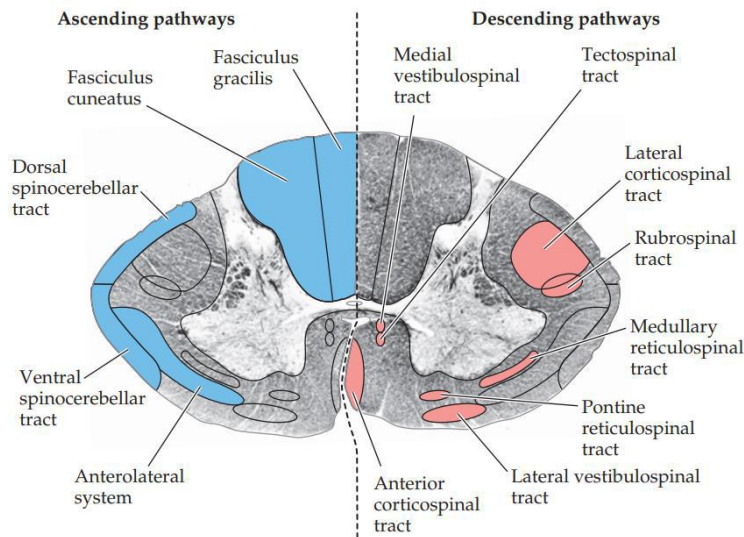


	Site of origin	Site of decussation	Levels of termination	Function
LATERAL MOTOR SYSTEMS				
Lateral corticospinal tract	Primary motor cortex (precentral gyrus) and other frontal and parietal areas	Pyramidal decussation, at the cervicomedullary junction	Entire cord (mainly at cervical and lumbosacral enlargements)	Movement of contralateral limbs
Rubrospinal tract	Red nucleus, magnocellular division	Ventral tegmental decussation, in the midbrain	Cervical cord	Movement of contralateral limbs (function uncertain in humans)
MEDIAL MOTOR SYSTEMS				
Anterior corticospinal tract	Primary motor cortex and supplementary motor area		Cervical and upper thoracic cord	Control of bilateral axial and gridle muscles
Vestibulospinal tract	Medial VST: medial and inferior vestibular nuclei Lateral VST: lateral vestibular nucleus		Medial VST: cervical an upper thoracic cord Lateral VST: entire cord	Medial VST: positioning of head and neck Lateral VST: balance
Reticulospinal tract	Pontine and medullary reticular formation		Entire cord	Automatic posture and gait-related movements
Tectospinal tract	Superior colliculus	Dorsal tegmental decussation, in the midbrain	Cervical cord	Coordination of head and eye movement (uncertain in humans)

The reticular system is primarily involved in regulating arousal, and the superior colliculus is receiving information from the retina but is also receiving information from others sensory modality and is fundamental as a first hub of multisensory integration. More importantly, is also sending commands to very simple elements of motor effector, but this explains why coordinates end in the eye movement, because the superior colliculus is primarily related to perception and

integrating perception from different modalities. So, this small nucleus in the brain stem is coupling perception with movement.

So, the main subdivision is between a system that appears to be regulating the most peripheral part of the body and voluntary movements, and another system (the medial one) much more regulating sensory motor coordination, gait, posture and balance.



Action is really made up of several parameters and if we have to look at different action features, and neuroscientists have always been describing action as a sort of hierarchical processing of information and commands, we can describe the kinematics, the effector, the effector target interactions, the target features, the call of the action, the outcome and several others. So, there are several pieces of information that we could try to represent and process at the action level.

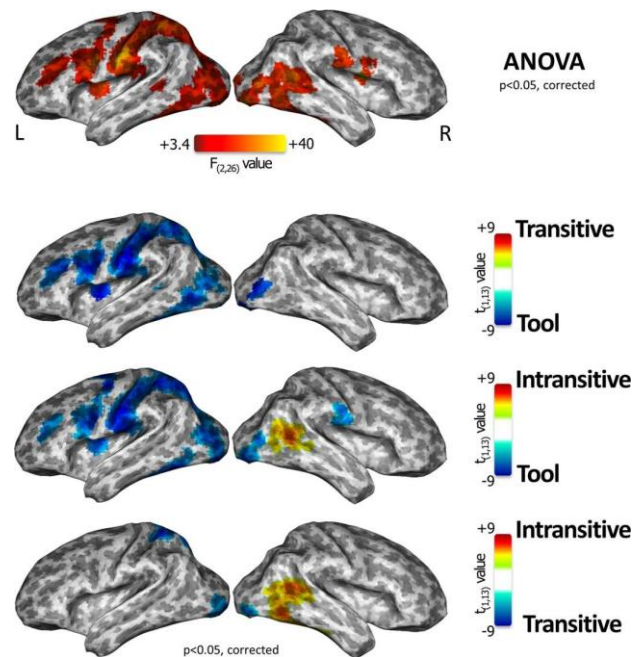
If you look at all the different kind of experiments, this definition is really telling us that is really difficult to have right comparisons in the tools that are available in neuroimaging, trying to disentangle regions specific for trajectory, for a goal, an object, because these regions are actually overlapping, so one thing that you should imagine is that these regions appear to represent all piece of information relative to action.

They tried to use recognition of three different kind of actions, tool-mediated, intransitive (action more social) and transitive (like grasping). If you look at which action is represented, you find that motor cortex most of the time is hardly representing, so when we are perceiving action we are not relying on motor cortex per se, but there is a wide gamut of brain areas that appear to be involved, and several of those are actually involved in representing actions and are the posterior parietal cortex, the supplementary motor area, dorsal and ventral premotor cortex, and there are also portion of this superior temporal vision cortex.

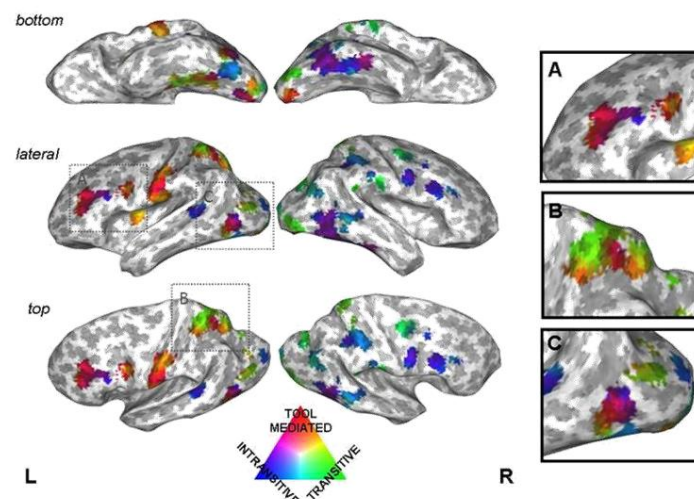
If we look at the comparisons of two mediated vs. transitive, etc., clearly, we are unable to find which regions are specific for each class of action, the whole brain is processing information.

You can see a sort of map that in a better way represents the different representations of action classes, and the same idea that we have been presenting about the topography of object form (pattern of response for different categories) was done here on action representation. This kind of separation allows to capture specific elements that are of interest; first of all, when you have a

representation of intransitive action and you have to represent objects, you see that the ventral pathway is deeply involved, because obviously we need to have information about objects here. The same thing happens for tool-mediated actions.



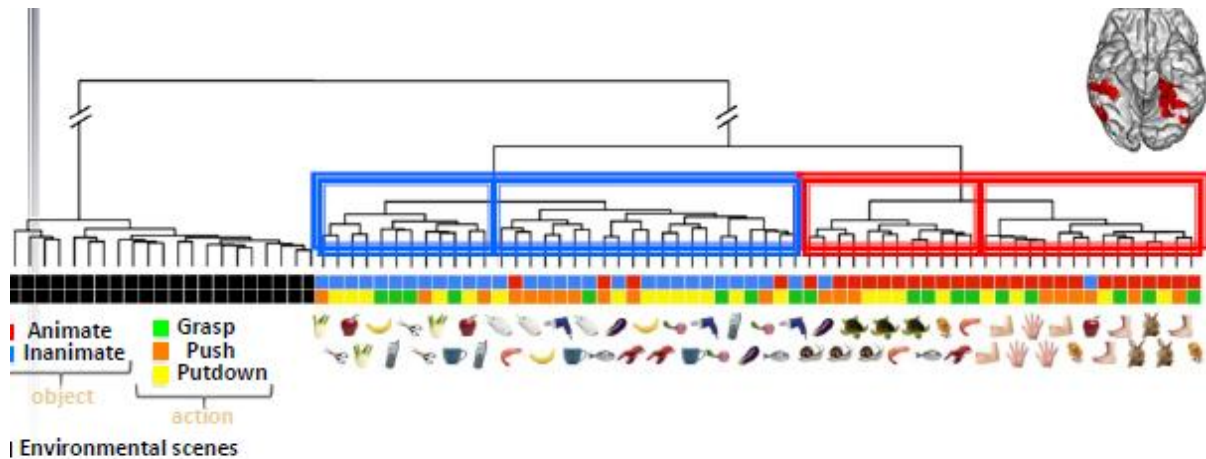
In the p-pattern in the superior parietal cortex we recognize a sort of gradient, that mirrors the representation the more transitive in the periphery and the tool-mediated, so there is some kind of idea of how we can represent action differentially. And the left premotor cortex, the left inferior parietal cortex, is actually processing here tool-mediated action.



So, for sure this kind of representation of the action, and here we are just performing the perceptual component of that, is telling you that gathering together different features involved in action representation, we can separate different classes of motor behaviours, and this is just telling us that action representation, as a distributed pattern of the cortex, is actually representing all these features into a single unique representation, as we were observing for the ventral stream in visual cortex. The brain is actually integrating different piece of information into a unique representation distributed on a broader network of action representation, something that we can call as action representation network, that involves frontal, parietal and occipital temporal area, and this broad network appears to allow the brain to separate and distinguish interaction classes

and then each specific action. For example, the ventral temporal cortex will process the identity of the object on which we are performing the action.

In a study three actions were performed, and each action had to be performed either on an animate (in red) or inanimate (in blue) object.



- Actions vs. environmental scenes (Accuracy 90.1%, $p < 0.0025$, chance level 50%)
- Animate vs. inanimate objects (Accuracy 68.3%, $p < 0.0025$, chance level 50%)
- Bodyparts vs. animals vs. vegetables vs. tools (Accuracy 37.7%, $p < 0.0025$, chance level 25%)
- No classification of motor acts (Accuracy 35.2%, $p < 0.8$, chance level 33.3%)

The first kind of classification tried to have was to separate action stimuli (a hand performing the three actions) vs environmental scene, so the classifier has a huge accuracy. Looking at a short stimulus, how brain is able to separate action vs. environmental stimulus.

One thing that was clear, was that the ventral temporal cortex recognizes objects, so we can recognize animate vs. inanimate. Objects were identified irrespective of the actions they were acted upon. So, the ventral stream is focused only on recognizing the object, the goal of the action. Also, one thing interesting is that there is no classification of motor acts from the ventral stream.

The parietal cortex is actually able to classify action vs. environmental scene and is also able to classify the different action classes, but it separates the grasping but is not able to separate the push from the putdown. The parietal cortex acts and relies on a special component, grasping is something that approach the object to me, the other two are putting the object away from me, so there is a spatial component that changes. The parietal cortex is also able to separate animate from inanimate and this is because information from the ventral cortex about identity are also conveyed to the dorsal pathway. So, while the ventral stream is actually able to discriminate several pieces of information about the object classes but no motor component, the parietal cortex starts processing and separating some action classes, maintains some high level discrimination, but is losing the detail of information that the ventral temporal cortex has, it doesn't discriminate body parts from animals or vegetables, the information in the parietal cortex is primarily related to the motor action represented in space.

The prefrontal cortex is doing something similar, but even more detailed. The pre motor cortex should really get represented the goal of the action, and is able to separate with detail all the

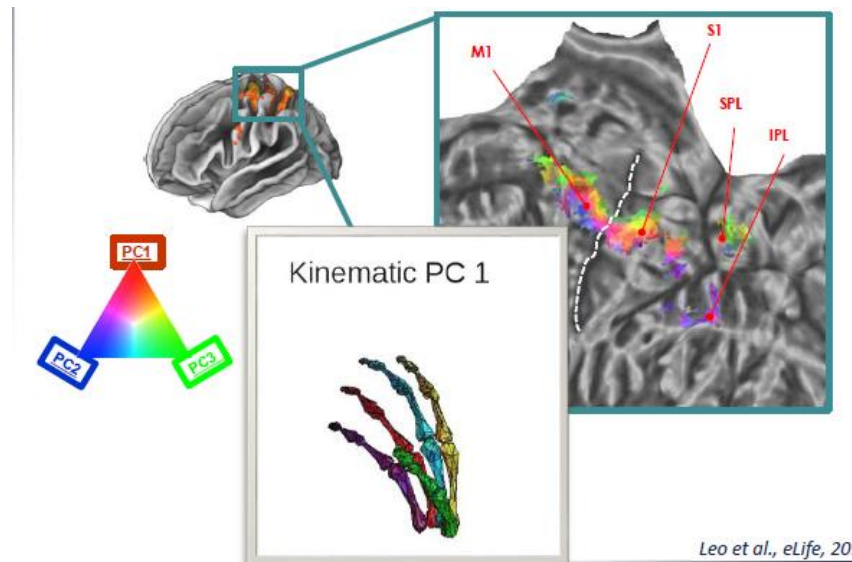
comparisons, so is much more detailed about the identity of the object to which the action is acted upon, but at the same time is able to classify much better grasping from putting down or pushing, so we have more information about the details of action. We are also able to discriminate whether is grasping an inanimate or animate object, so the premotor cortex is getting together the information relative to the identity of the object and the aim of the action, because representation is getting more meaningful in the premotor cortex.

The different portions of this circuit appear to subserve different pieces of information of action processing. So, the motor representation of an action appears to be more distributed broadly at the level of the cortex and specifically depicting information of the action as a whole. This actually relates to something we were saying before, action is processed as a whole and the different components also have unique representation according to action class and identity, and this is also reconciling with the idea that the whole brain is working to represent the different actions.

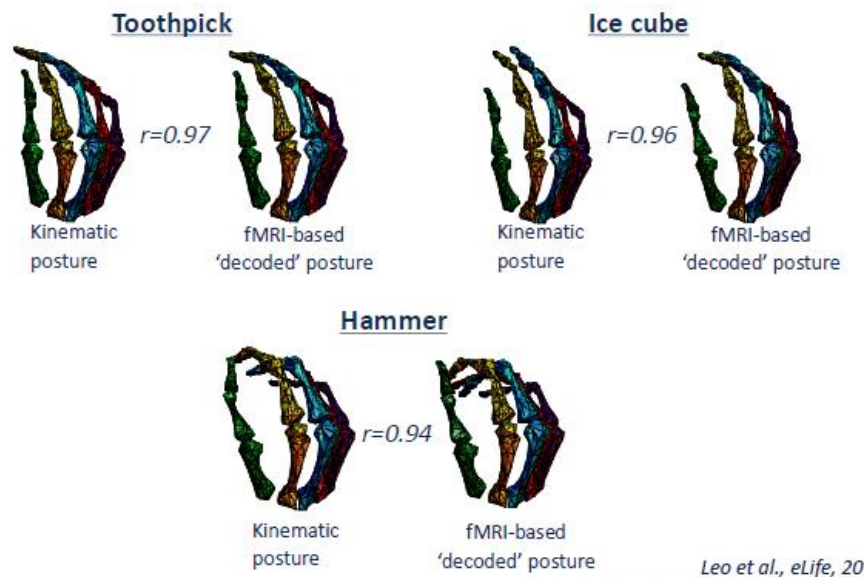
But how we really jump from grasping an object to someone else doing it?

One element that we have to consider is how much important is the kinematic and the motor information, so how does the human brain control movements to carry out different tasks. The message from the cortex to the effector is redundant and has a huge cost transmitting the information since our effector is a very complex tool. The central nervous system manages and controls movement of individuals muscles and joints with 23 degrees of freedom. Synergies represent functional modules that includes cells of hand muscles and joints which are recruited simultaneously and act as a unit. Typically, the idea of reducing these 23 degrees of freedom to a lower number of dimensions is telling that probably every kind of grasping could be reduced to a combination of specific joint angular movements and muscles recruitment, and this is also something that typically has been suspected with the first neurosurgeon dealing with the electrostimulation of the cortex. You may have to try and move you little finger without moving the ring finger (difficult), obviously if you pay attention and you exercise you can adopt some strategies, but nonetheless there is some kind of synergy and this because they are the less used fingers in order to perform any kind of action. The idea of synergy is reducing a lot the number of information, and the combination of different synergies can adapt to any type of grasping movement. When you decide to reduce from the principal components analysis, only the first 4-5 components allow you to describe about 95% of all your final hand postures.

It was tried to record 20 different gestures with optical tracking of muscular and neural components. The images don't tell us which brain area actually activates, but in which brain area this kinematic information is actually represented. You are doing what is called encoding, you take the matrix that you create according to your kinematic data and define hand posture, and you use this to encode whether during grasping these components are represented at the brain level. What you find is that these regions are all actually involved during the performance and you see that the kinematic component is present in the precentral cortex, in the postcentral cortex, in the posterior parietal, in the supplementary motor area and in the dorsal and ventral premotor cortex. All the regions are representing these kinematic loadings and if you try to map this you see that the three component that are reproducing 80-85% of the brain are actually reflecting; the different components are moving the five fingers together, they are nicely and topographically represented at the brain level. This explain how in the hand area we have a very nice representation at the level of synergistic movement have that appears to be topographically represented in the cortex. It's interesting because the motor component appears to be present at the level of the representation.



You can also try to use brain activity to reconstruct specific postures. If you are able to recreate a specific gesture from brain activity you can convey this information to any kind of external device.

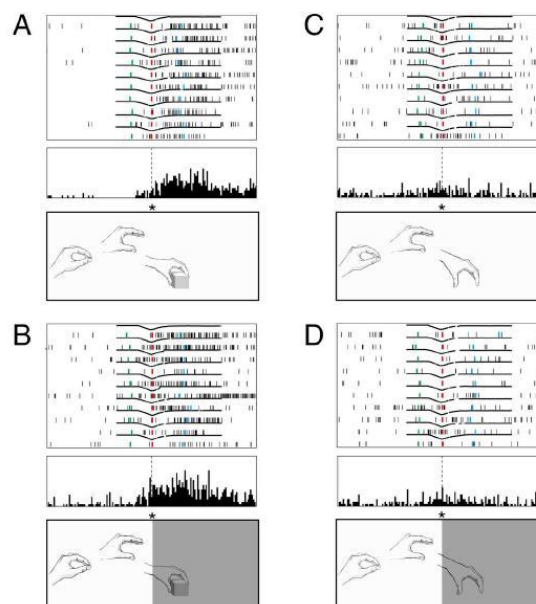


So, if kinematic synergies appear to be topographically organized within motor cortical area, what about perception? They tried to reproduce the same kind of experiment also during motor imagery, so subjects weren't performing any actions. You can see similarity in the pattern primarily in the premotor and in the posterior parietal cortex. This is telling us that there is a common language, a motor information, that share between motor execution, performance, planning and recognition.

There is motor information that links and associate motor performance to perception, and this is fundamental because all this organization is broadly impacting our representation of the external world and the interpretation of other people behaviour. There is a strong demonstration that motor information is mirrored in performance and representation of an action.

Mirror neurons (I don't have the first part)

If I would record eye movement, as soon as I'm going to pre-shape properly, your eye movement would direct with the touch from my head much in advance before I'm reaching the object and would go for either the bottle or the pencil. The same did the monkey, you show a cube, you show the hand resting, pre shaping and grasping, with the monkey that recognized the movement preparation and grasping. Is that related to object per se? No, the same thing could be done like in another experiment, the monkey is showed a cube, then the cube is obscured, the monkey sees the hand pre shaping and then is not actually seeing the last grasping, but the monkey know that the cube is there and it sees the pre shape, so that is going to grasp, the mirror neurons discharge. So, here is not related to the presence of the object or the kinematics. You can perform the same thing but without the object, and this is telling you that the action is not a finalized one, so the mirror neuron is not discharging, as for the control condition.



Therefore, mirror neurons appear to represent the aim of a finalized action, the meaning of the action, but not its specificity. Together with mirror neurons, the researchers were studying another family of neurons, canonical neurons, that are slightly different from the first ones, because, in the same region where mirror neurons are, canonical neurons appear to be activated when the monkey motorically grasps the object, or when the monkey sees the object without any action. These neurons are something that we have been describing in the dorsal pathway, they are neurons that respond to the object and when I have to specifically grasp the object.

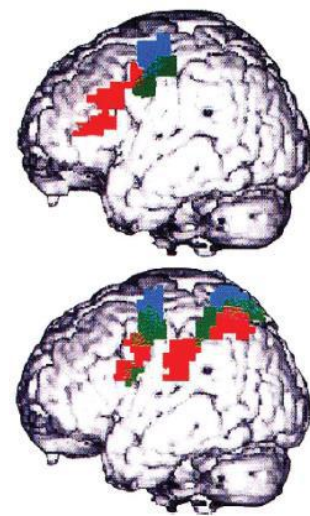
Mirror neurons appear to be activated when you perform an action and you see another agent performing the same action, canonical neurons appear when you perform an action on an object or when you see the object without any action, So, canonical neurons appear to be more primitive, simpler as a conceptual representation, mirror neurons appear to be conceptually little higher in the hierarchy of representation, they represent the finalized action per se. But, how we really 100% confident that it doesn't relate with the kinematic? Because kinematic matter is really widespread in that portion of the cortex.

In another research pliers were utilized, where in order to grasp you have to close the fingers to press an to use this tool, but there are also reverse pliers, and here in order to grasp you have to open your hand. Kinematically these two actions are opposite. The same neurons that are telling you to grasp the plier is responding only when you grasp the plier independently of the kinematic

of closing or opening. So, mirror neurons appear to be totally unrelated from the kinematic motor aspect, appear to be a different way to processing it. When the monkey observes someone else opening the peanut and grasping the fruit, when you just see without even hearing the cracking of the peanut, but since this representation is independent of the kinematic aspect you also see that the monkey is not actually seeing the movement of the peanut breaking but it's hearing only the noise. So, specifically this representation is so abstract that even the sound is actually working to represent the action. For example, if you hear a ring your brain is also processing how to ring the bell.

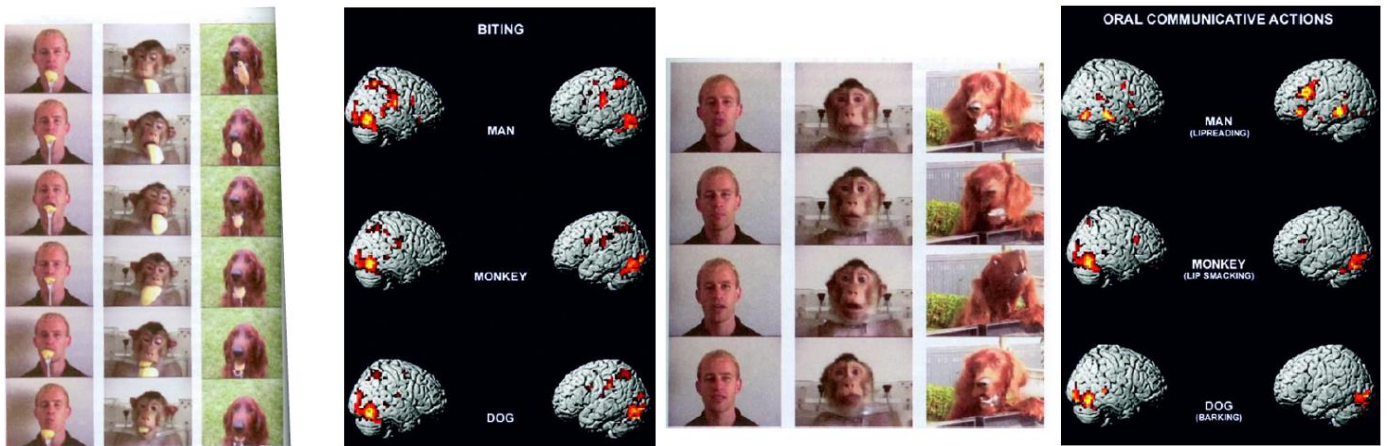
When describing mirror neurons, we were finding mirror neurons that were called as broadly or narrowly congruent. There were mirror neurons that were responding to any kind of grasping (broad mirror neurons), or very specific grasping. This should remind us of the representation in the premotor that we were describing before, the ability to classify a label of grasping animate or inanimate objects.

Mirror neurons have been extensively described in monkeys, but we soon got to humans. Here the goal was to describe what is called the mirror system, and it is also interesting because the mirror system in humans is also somatotopically organized, here we have mirror neurons response of the feet, of the hand (in green), and of the mouth and lips (in red). These mirror neurons are really closed to the inferior frontal gyrus, where the broca area is. Mirror neurons in humans have a much broader network then in monkeys, in humans we are able to code not only transitive actions but also intransitive, because I have language development. Typically, it codes the sequence of movement of an action, so most of the time these actions in the mirror neurons are very complex and, off note, you can also activate mirror neurons when you mimic an action you are familiar with.



Mirror neurons are also important because we learn through imitation, and it's the first way we actually learn motorically, and our brain is tuned and tailored. We tend to imitate and recognize other people, but the idea of imitation is also what allows us to enter into other people shoes and understand their intentions, and this is not just motorically, is also creating what is called empathic, the idea of sharing feelings with other people. This idea is important because it allows people to share common sensory motor repertoire.

In a study there were shown 3 movies, and while viewing them people were recruiting bilaterally parietal and premotor regions, the same action of biting food is shared between man, monkey and dog. We can see without listening a man talking and read lips, the monkey opens and closes its mouth, and the dog bark. While doing these actions, only while viewing people talking, we were recruiting premotor mirror neurons, but not for the dog barking, because barking is not in our motor repertoire and this is what is clear with this idea of having something experienced sensory motorically. In order to understand other people intentions you need to have experienced that.



In a comparison between capoeira dancers and classical ballet dancers both were danced while presented both some video, but mirror neurons were responding higher when each of them were shown videos of people dancing their style.

This is very important in order to actually understand how motor information is really rooted in my perception, and how it allows me to predict in advance, and also the least I know some motor condition the less I'm able to predict that specific behaviour. When you see and are familiar with an action, while I'm doing that and someone is watching if I would record their muscle, they would be ready to enact the movement, and this is also a strategy used for therapy, exposing you to movements that you are not able to perform anymore, in order to have your motor system to get ready to represent and retrieve that kind of motor action. Also, the context of an action can tell us how we can modify.

Mirror neurons too are present in blind people, so people that have never seen are able to have mirror neurons system responding to the recognition of action, and even their mirror neurons system was responding more to action they were familiar with.

Mirror neurons appear to be a sort of like of empathic too; if I'm able to enter other people shoes and understand their intentions, I'm also able to understand their feelings and when you look at regions that are involved in empathic you see that is plenty of regions that are present in the mirror neurons system.

Anthropomorphism is a feature you can play with in your device; in order to build something that your brain is able to embody in your body schema, because we recognize it most.

These aspects will be fundamental if you are going to look at this in terms of embodied cognition; the idea that every time you have to perform an action, every time we have to process a stimulus or try to have a broader conceptual representation, this is rooted somehow in a sensory motor experience. How can I work with a tool or device to be embodied with the patient, to be truly part of his body representation, part of his motor repertoire.

Plasticity

Plasticity has been defined as an intrinsic property of the human brain, representing a sort of invention of evolution that enables the nervous system to overcome the constraints imposed by its genome, allowing it to adapt to the environment or physiological changes and experience.

We often think of plasticity as something that is beneficial, something that even in the condition of damage and lesion could help recovering, but this is not often the case; we also have the case of maladaptive plasticity, some kind of adaptation to a lesion that is not perfectly fitted with the means of the brain, and more importantly how is the brain able to adapt to an equation in which we add multiple limbs?

Plasticity is the ability of neural circuits to modify their structure and function in response to stimuli, both during development and throughout adult life. During the early stages of brain development, plasticity is extremely high: some neuronal circuits are selected while others are eliminated; in the first months of life how brain matures, it takes 10-18 months to learn how to talk and walk and we are mostly blind at the birth, so our brain has to adapt. In adulthood, many circuits remain largely stable, yet neuronal populations continue to exhibit a certain degree of dynamism, reorganizing under the influence of the external environment to meet specific motor, sensory, cognitive, or emotional demands. There is a period of life, until 30-33 years, where the brain is adapting to the effect of hormonal changes (pruning). Our brain remains stable until 60-65 years and then will start facing issue of cognitive decline, nonetheless the ability to change in time is not the same for all cognitive motor perceptual domains, so we have different windows through which this kind of changes can occur more easily.

Development of the embryonic brain:

- CNS as “a fluid filled tube”.
- Proliferation and synaptogenesis.
- Migration.
- Differentiation and maturation.
- Myelination.
- Excess production of neurons.
- Development of connectivity.
- NGF and other neurotrophins.
- Pruning.

The migration of cells gets to the development of the brain birth; the brain of a newborn is not proportional with the rest of the body. This is all regulated at molecular level since we have molecules called neurotrophic, that are feeding the neurons and the synapsis to grow, and this phase is really important. But there is an intrinsic organization that is determined at the genetic level, the brain is already defined in a specific way innately (even blind brain has a large scale of organization very similar to the sighted ones, it's not just the motor experience).

Following several cell migration steps, three main layers emerge:

- **Mesoderm** cells give rise to the structures including muscles, bones, cartilage and the vascular system.
- **Epidermal ectoderm** cells give rise to structures including the skin, nails and sweat glands.
- **Neuroectoderm cells** give rise to the brain and the nervous system.

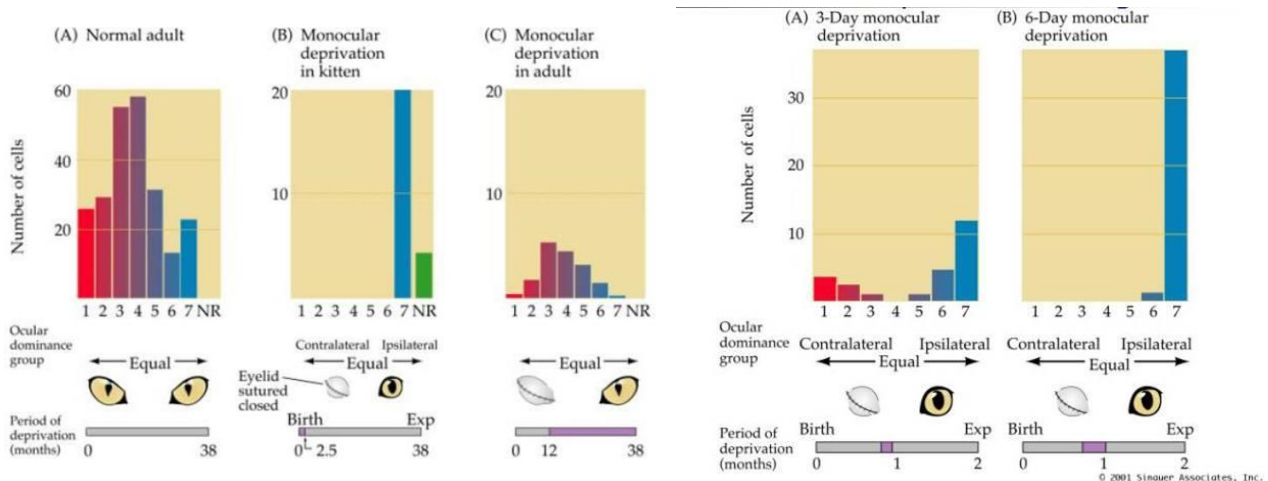
At first it starts as a plate, then it closes into a tube, and then we start having areas and portion that appear to have and rely on these projectile cells that create the whole brain. (NOT GONNA ASK ABOUT BRAIN DEVELOPMENT, TO READ MORE SEE THE SLIDES).

We define as sensitive period, a period that is a limited epoch during development in which the effects of experience on the brain are usually stronger; certain capacities are quickly shaped or altered by experience. We have a specific architecture of the brain, but there are periods in development in which experience shape more the functional and structural organization of the brain, and these are called sensitive periods.

Among the sensitive periods we define a class called critical periods, that results in irreversible changes in brain functions. After the end of the critical period, effects of atypical experience during this period cannot be remediated by restoring typical experience. Critical periods are primarily related to perinatal time, so after conception we have several critical periods, first for sensory motor, olfactory, taste and auditory and vision. During these critical periods, on of these environmental input, for instance a sensory modality like vision is missing, we are not able to fully restore all functions if in the future, for instance, I'm recovering vision if we treat the problem at the base, because some structural functional features cannot be recovered anymore since the critical period is over. This is why most of the ways of restoring vision are typically done in people that get blind late, and also why when we have a baby born deaf, we try to implant the cochlear implant as soon as possible, because we don't want to miss the critical period.

Sensitive critical periods are reflected in behaviour, but they are property of neural circuits. It means that critical periods reflect for instance the ability to develop dominance of the eyes, develop acuity so on, and this is something that we actually reflect behaviourally. But critical period is a period where we have higher development of synapsis and higher metabolism, so there are always structural components that subserve this development properties.

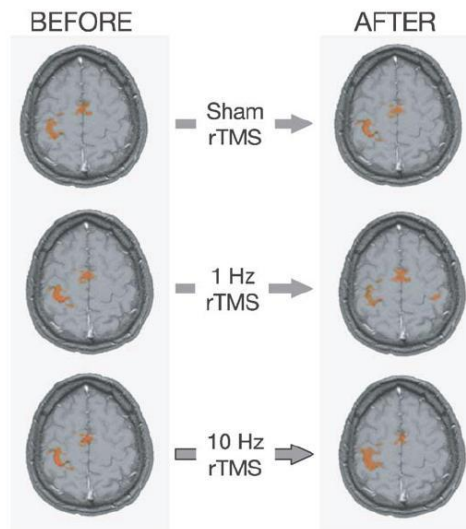
If you deprive the monkey at the beginning of their life from 2 weeks to 18 months of age (first 4-5 years for a monkey), and you look at the visual cortex, you see that the ocular dominance is lost, so we have one eye that appears to dominate, that is the non-deprived eye.



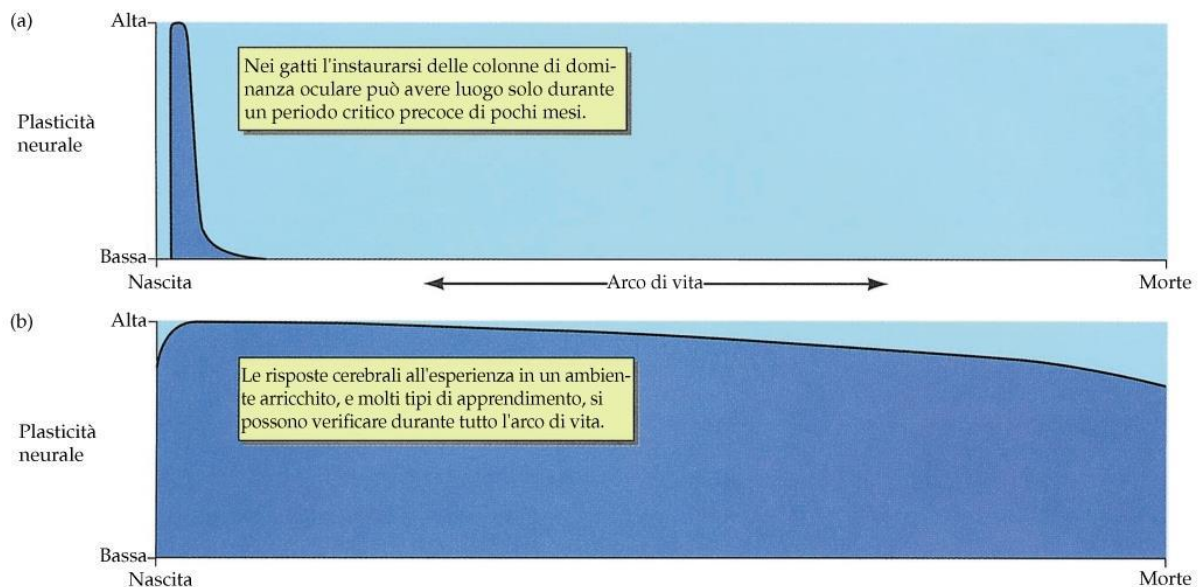
When you have a monocular deprivation, you clearly have a number of cells responding only to the eye that is not been damaged. If you deprive the adult, you see that the change that impacts is reducing the number of cells responding to the stimulus, but not the sensitive distribution of the left or the right eye. So, one of the things is that the impact on changes that could occur in the brain is different according to the age these vision deprivations occur.

This ideal plasticity is also related to time, because if we are not actually defining what we refer to as plasticity, and we refer to plasticity as a wide gamut of phenomena, even the modulation with repetitive transparent magnet stimulation could induce very swift changes in few seconds.

If you stimulate motor activity for different time and frequency, you see in few seconds an increase in lateral connectivity.



Plasticity may take much longer, so time window is an issue, an example is relative to the exogenous modulation of plasticity, thinking about the hormonal effects in brain structure, and the difference between male and female. Male has a much more specific element on one side and female has more from a lateral. So, one thing clearly evident is that there are specific aspects that appear just at a very short period of time, and these are the critical sensitive periods, some other form of plasticity, like learning, last for the whole life, we can just have a slight reduction but we are able to learn things along most of our lives.



We could have aspects of periodic learning, and this is not often present in humans but very common in animal for instance. There are birds that remember up to 80 sites in which they hide food, and each summer they have to change and remember other 80.

There are domains instead in humans too, that have a very specific learning window, and this is typical for language. If you want to learn language early years of life are the best, because we are more able to separate different rules of languages. This ability remains for a long period of life,

and you still use language related brain areas to learn new language, but after that age you can start a new language, but you aren't using anymore this area, but you are using memory.

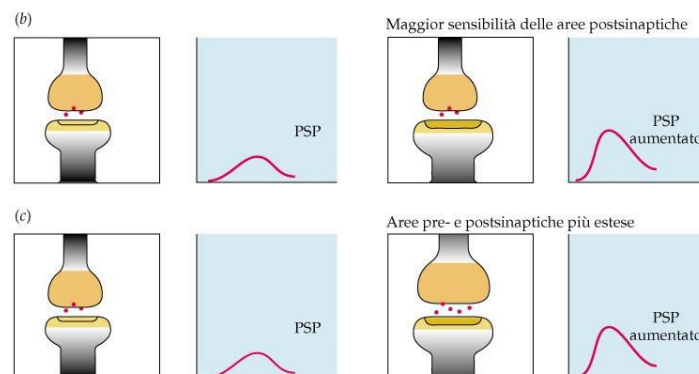
These aspect of learning are something we can define as permanent changes in behaviour produced by experience. Learning involves changes in the nervous system that are produced by experience, the more you experience and the highest the changes in the nervous system. Changes in the nervous system are physical, and learning is what allows us to adapt to the environment.

Our psychology strength decides to define different way of learning, like a perceptual learning, stimulus response learning (conditioning, performing an action in response to a specific stimulus), motor learning (formation of new circuits within the motor system), relational learning (identifying relationships or connections among stimuli).

In musicians the anterior commissure, that relates to connection between the two motor cortices, is thicker and if we have to look specifically at the differences between the two motor cortices you see that while is higher for non-musicians, musicians tend to have a very similar size of the two motor cortices, they have a great coordination of different body parts. Typically, this kind of phenomenon occur somehow twice in life. These changes are higher the earliest in life.

The anatomical differentiation of the brain is clearly tight to experience. Brain maturation has two specific moments, one in early life and the other one is in adolescence; if in early life we have these aspects of synaptogenesis, during adolescence we have a phenomenon that is called pruning. This is a phenomenon that tells us "Use it or lose it", since during adolescence we have actually to select which are our skills and interest, and the brain while is growing modulated by hormones is also adapting to changing environment. Brain cells that are typically strengthen are showing as morphologically as enriched or impoverished, and this kind of thing comes from experiment like the one in the slide with the rats. In an enriched environment you have more animals, it's crowded and not very nice, but at least you have social interaction, very close to real life. These aspects are typically impacting the size of the brain; one of the effects was the one described in the musicians and one other important study is the one conducted on London taxi drivers. In London you have to learn different streets name and indeed their posterior hippocampus, related to long term memory related to navigation, gets broader and bigger compared to ordinary people.

Our idea is also how can neurons actually try to compensate when we have a damage and we may have formation of new synapsis, so we are extending the map of interest. This is called typically sprouting. We can also have a sort of hypersensitivity induced by inactivity of the innervation, and the size of the synapsis increases.



We already talked about cross-modal organization, with early visual cortex in blind people that is responding to non-visual stimuli, so following sensory deprivation primary and secondary sensory areas indicated to that perceptual modality, reorganize to respond to other sensory modalities. We still have a hill defined understanding of the functional role of this; we can just regard that visual cortex starts to respond to non-visual stimuli.

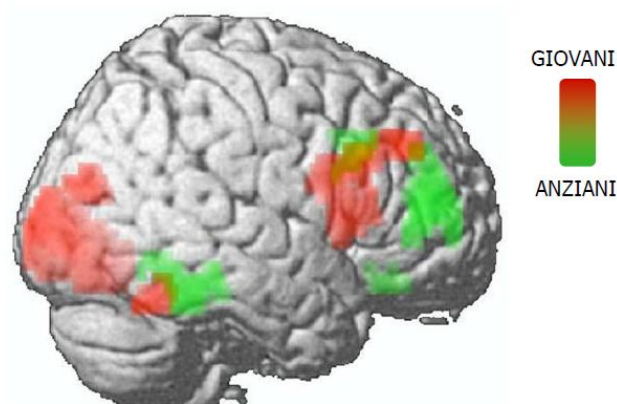
We then have compensatory masking: following injury or learning, an implicit or explicit change in task-execution strategy occurs, allowing a secondary cognitive process—normally playing a minor role—to assume a primary role). The idea is that when there is something difficult, I'm going to rely on other processing in order to achieve that specific task.

Expansion of the functional map: following injury, a functional region adjacent to the damaged area expands its functional/neural territory into neighbouring cortex. Learning can also induce an enlargement of the functional extension of a region into adjacent cortical areas.

Adaptation of homologous regions: following injury, functionally homologous regions in the contralateral hemisphere can compensate for the lost functions. This is something we are actually observing in aging.

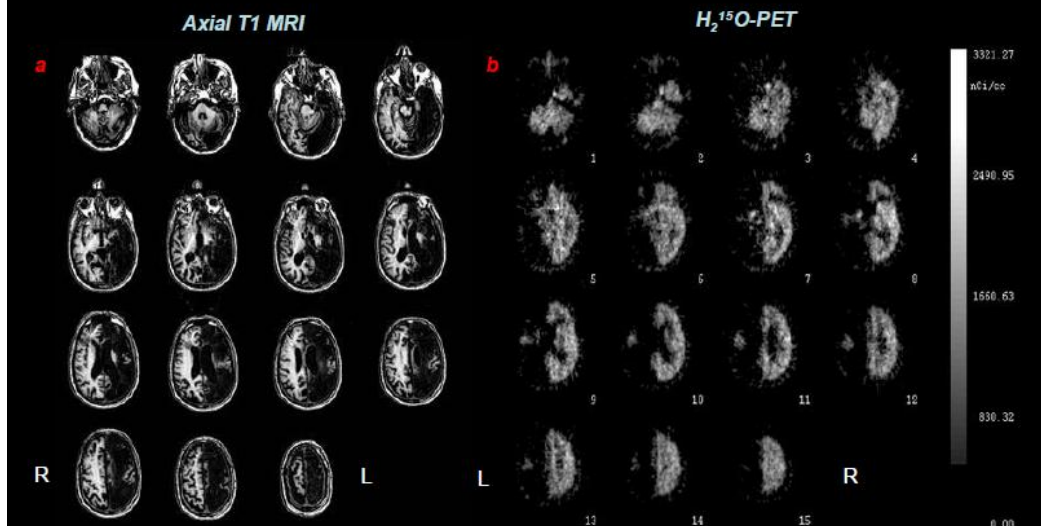
In a task there can be old people performing and old people not performing on it, and they can be called old high and old low. Old low try to extend the map but are not performing well, and in order to compensate they have a recruitment of the homologous regions in the contralateral hemisphere.

During aging you can have different effects also, young and older people are actually even recruiting additional regions involving a single function, and this is called compensatory masking; to do the same thing you recruit other regions. For example, you present faces and then you maintain that information for a few seconds and then you ask to recognize which was the first one; to do that both old and young people are recruiting the visual cortex and the right prefrontal cortex, but also older are actually extending the map for recruiting not just that portion of the prefrontal cortex but also another one, also if you give them drugs that potentiate memory you see that they act in different regions in the two. This modulation tells us that this compensation that is truly effective is the effect of the drug, so younger and older people modulate different portions.



PET and fMRI in the Study of Functional Plasticity in the Human Brain

A 67 y/o right-handed male survived a **massive stroke of the left hemisphere** at age 47 yr. In the left hemisphere, structure (a) and metabolic activity (b) were preserved in limited areas only. We studied **potential compensatory effect** of the R hemisphere (brain plasticity) in carrying out activities normally sustained by the L hemisphere.

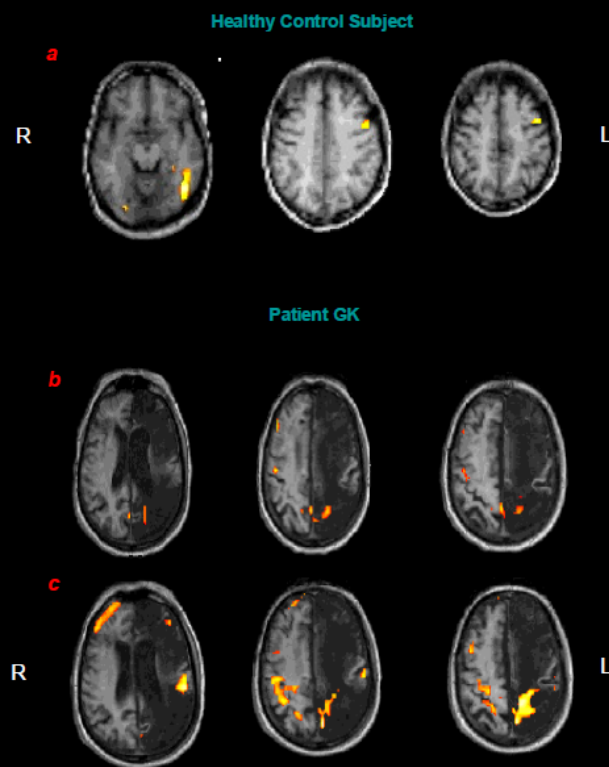


Normally **mathematical functions** rely on parietal and frontal areas of the left hemisphere (a).

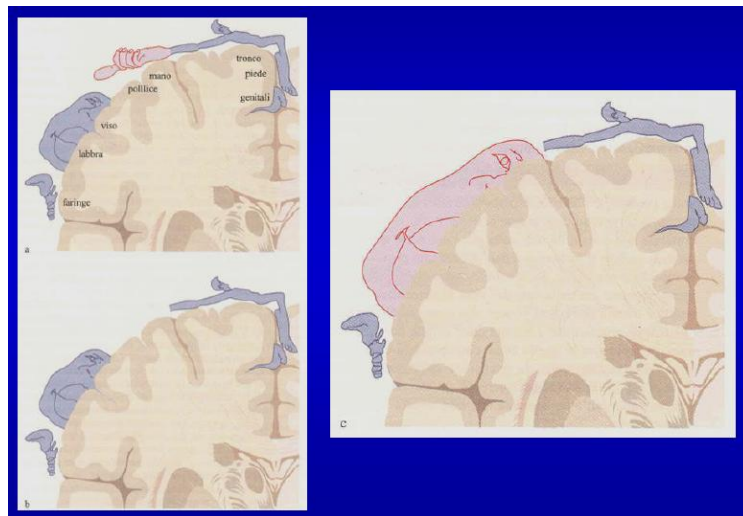
In patient GK performance on mathematical tasks was highly compromised and associated with limited activation of residual parietal areas in the left hemisphere (b).

After a period of training the patient reacquired some ability to perform simple mathematical operations. Improved mathematical performance was associated with activation of **homologous parietal and frontal areas in the right hemisphere** (c).

These findings indicate a compensatory process through functional plasticity in the adult brain.



As for the Ramachandran's procedure, since mostly form a sensorial component, you have the expansion in the sensory homunculus of the face perception area. While we are having novel synopsis, if you touch the patient in the cheek amputees can report that you are touching their head, and this is because brain is not properly localizing cortically the location of the stimulus.



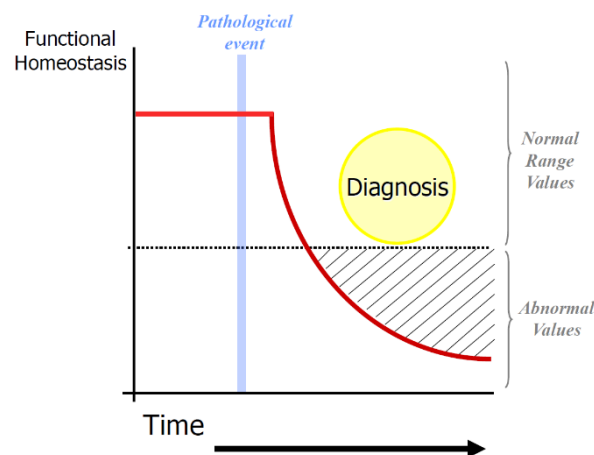
The take on message is that we may describe different forms of the way the brain prefers and tries to accommodate, even if we are not fully understanding the principles, we have understood that this capacity of the brain to adapt to environmental pressure changes in time, changes according to the specific domain that you are exploring, and we have no specific way of classifying how well this mechanism is actually compensatory when there is damage. This is why several researchers are really wondering whether you have plastic phenomena in specific conditions and situations, and this is also primarily relating to the big trade-off between what is innately present in birth in the nervous system, and which are the features that you can modify during life according to environmental experience.

We have to remember that when we change a function often there is a change in the structure of the brain, or you can describe the change in the function but in order to stabilize it you need to have a structural change too. Secondly, with several tools we can describe changes in behaviour or at the broad level but never forget that these changes occur at the molecular level.

We are aware that these maturation skills are impacted by stress, drugs and neurological disorders, and we are yet to understand how we can foster and promote this kind of restoration.

Emotion and behaviour

Humans maintain a healthy status in an active mode; we maintain a sort of equilibrium actively. At a certain point something happens in our body that will change reference values from a normal range to a pathological condition and on this phase, you make the diagnosis, based on something we can measure (symptoms).



What medicine has been found to do is to make diagnosis as close as possible to the moment in which something happens to the body. The moment in which you have a reasonable self-kindness that the patient has dementia or Alzheimer signs, the question is “How long before the dementing course has started?”, the answer is even 25 years.

The question we ask as engineers is “What can I do to detect a dementing process years before that the dementing process has led to a clinically evident proof”. This is called preclinical detection and is a huge challenge in medicine. This brings along important ethical questions, because detecting especially brain disorders does not imply that we have the tools and strategies to stop the onset of them, or even delay it, we have the impossibility to do anything from outside.

An example is the Huntington Chorea, neurodegenerative disorder characterized by the onset of uncontrolled movements with decay of cognitive functions. This disease becomes evident in 4,5,6 decades of life, but it is an autosomal dominant inherited disorder. Autosomal means that the gene is on the autosomal chromosomes, the ones that are not on the X and Y sexual chromosomes; When the disorder is link to an autosomal dominant gene, it means that all you need to manifest that disorder is just one gene. If you inherit the disease from one of your parents you have a 100% chance to develop the disorder, being it autosomal and dominant one is enough. If the gene is recessive, you got to have the two genes.

For disorders like this the ethical questions is whether having children being affected by it is ethic.

This introduction is to make clear why it is important to study the biological bases of brain function.

Searching for the mind

As for the case of Phineas Gage, the change in behaviour after the lesion in the frontal cortex confirmed that the frontal lobe must inevitably be important for controlling behaviour and decision making.

The ability to really enter into the brain and have a sort of true window to observe it in action and also to measure the brain structure, can be done by putting together different techniques. An important factor is molecular biology and genetics, since our genes shape our density.

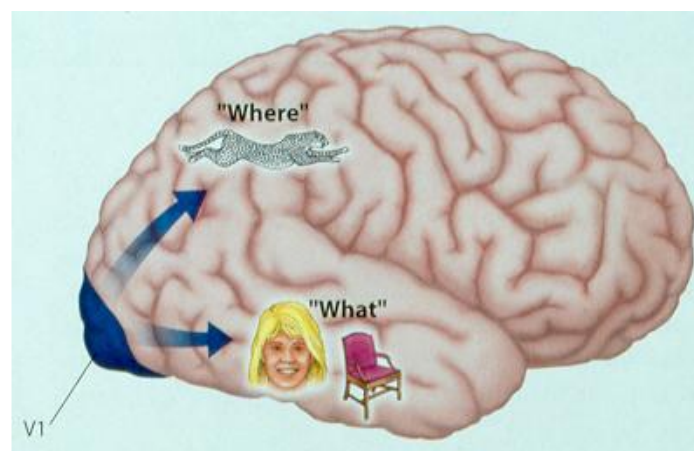
One aspect that in the future we will maybe deal with is drug resistant infections, which means that the ability of bacteria to change overtime is stronger than our ability to find new drugs. Each time they replicate errors happen by chance, the photocopy of their DNA may be with some mistakes developing resistance to drugs. This is the reason why we can't abuse with antibiotics.

Our brain is clearly different from the animals' ones, since our frontal cortex is much more developed.

With the development of MRI, PET scan, and all the techniques that allow us to see in the brain, I can also map what happen in the brain while doing some tasks. For example, in a face recognition task, you can measure accuracy and reaction time. With tasks like these, scientists have shown that once that the information has reached the back of the brain, the information gets divided in two parts, one to the ventral-temporal cortex (the "what" pathway) and the other one goes upwards to the superior part, the parietal-temporal area (the "where" pathway). In order for us to have a coherent representation of the external world we need to have an integration of information across different areas.

In the first stages of dementia, before you have a localized alteration within the brain, what happens is that the ability to dialogue between different areas gets lost.

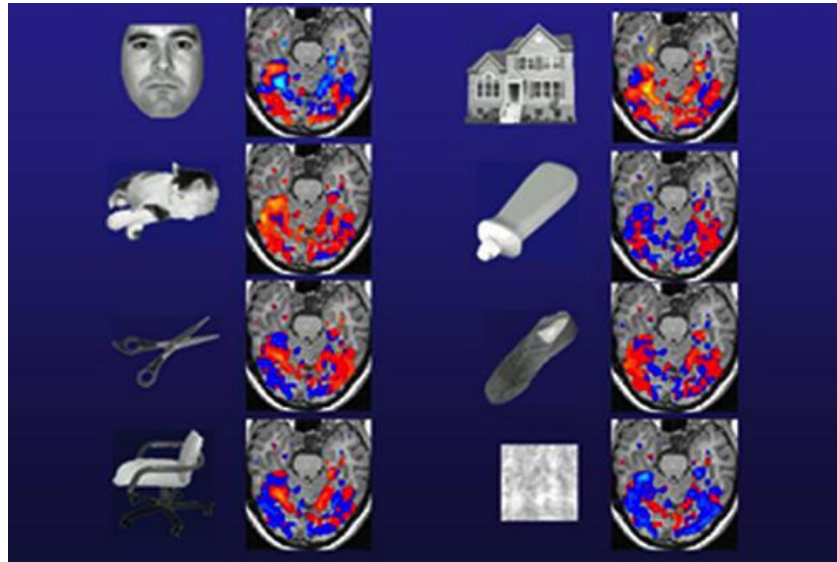
The ventro-temporal cortex is responsible of making us recognize an object from the external world.



But how can we distinguish an infinite number of objects relying on a finite part of the brain? There are different schools of thought, especially for body parts. Somebody says that, for example, for recognizing faces we have specialized groups of neurons. Somebody says also that there is an area for recognizing places, para-hippocampal place area. There can't be neurons for a specific object, there would be no space in the brain and evolutionary time.

MRI allowed to have a very good definition of the structure and see how the brain responds to different categories of object. It was found that you can't really say that for each specific category there is a certain and localized area responding, but there is a distributed engagement of the ventro-temporal cortex with a different pattern of response for the different objects.

If you compare the different brain responses, you don't find a localized response, but a distributed and overlapped response, with a different pattern of response. If you analyse how similar a response to the same category of object as compared to responses across categories you find that between categories that have a really strong correlation, brain responses are highly correlated. If the brain response is not specific, you get correlation within the same category as high as across categories.



There is a small statistical caveat, the correlation is not due to the distributed response but to some subset of area of the brain which drive it. If you have a subgroup with a very strong response, it can drive the correlation. So, what we did was excluding from the analysis the pixels with the highest response and see whether what was left would maintain the correlation or not, the response was yes. The brain does not respond all-or-none to a given category, but you have the ability to produce an infinite number of patterns of brain response, which are so specific for that given category that basically you can predict what an individual is looking at.

Is this pattern of brain response specific for vision or it's related to a more abstract representation and response. Is intended for the specific sensory modality or not? If you study subjects with projectile blindness, who have never seen but have a concept of things, and you ran a tactile study, what you find is that the response to the tactile task in bottle, shoes, mask recognition is a category specific response within the occipital retro temporal area (visual area). If you run in other subject and you do the visual and tactile study you get high correlation across modalities, which together do the supra modal organization of the brain.

We can recognize the external world by producing in our given structures of our brain an input number of patterns. But when we look at the external world, we don't only recognize physical appearance but even before that we perceive and respond to the emotional and affective message that we have from a face.

Emotions

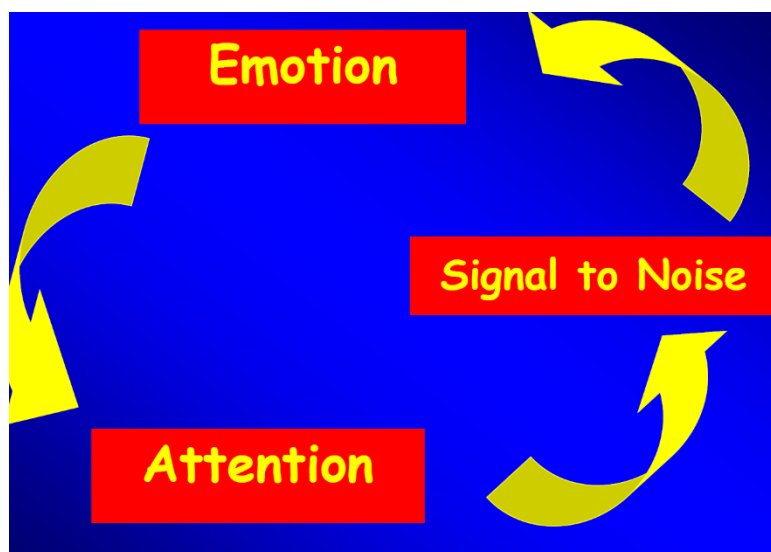
The first thing that researchers did when these techniques of looking at the brain became available was to try to understand what happens when we see a terrified or normal face.

If you subtract brain activity for the neutral face to the one for the terrified one, you see the activation of the amygdalae, two structures in the anterior medial part of the brain. They are the emotional computer of the brain, and if you modify these experiments even a little, by showing

for example only neutral faces and then a scared face is shown for a very short period of time, the subject will say that he saw only neutral faces, but the amygdala shows a response in the exact moment the other face arrives.

The amygdala sends a signal downwards to the brain stem and the hypothalamus to increase blood pressure, the respiration, more oxygen to the muscles. The fear instinct.

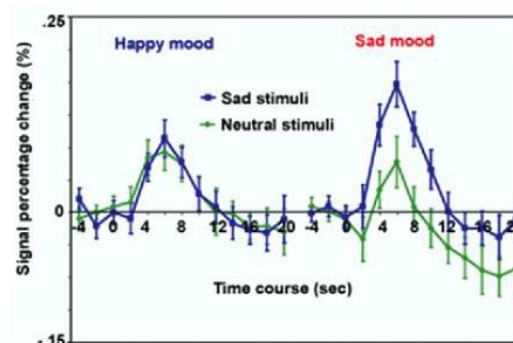
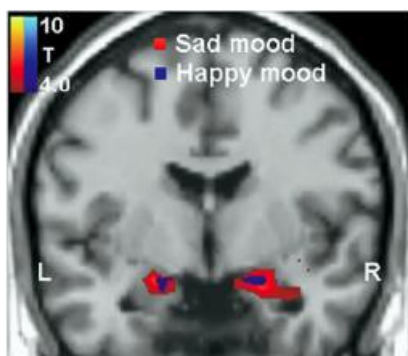
All you need to know is that there may be a danger to activate the chain of fear response, so that you are prepared in the case the danger is real. This health mechanism in the brain is very useful; the emotions increase the signal to noise so that you focus your attention on what you need to know. Emotions are instruments that evolution has equipped us with, in order to improve our survival chances.



Our brain is programmed to filter what is not important. In the case of the photo of the twin tower you don't notice the individual changes in each photogram.

Emotions are also modulated by our affective state. It's important to know how mood modulates brain response to emotional stimuli.

Reaction time task → to people were shown different non-sense images, sad and neutral pictures, and brain activity was measured not only when they pressed the button at the arrival of the stimuli, but at each time. Researchers saw what happened in the brain when they saw a sad stimulus, compared to the neutral stimuli. But also, when subjects were already in the scanner bed, for the 15 minutes before starting the experiment it was projected a movie on the screen (funny or sad). What happened is that if you had been shown a happy movie, when you are shown a sad stimuli you see that you have a response to sadness in the amygdala, but you see that the



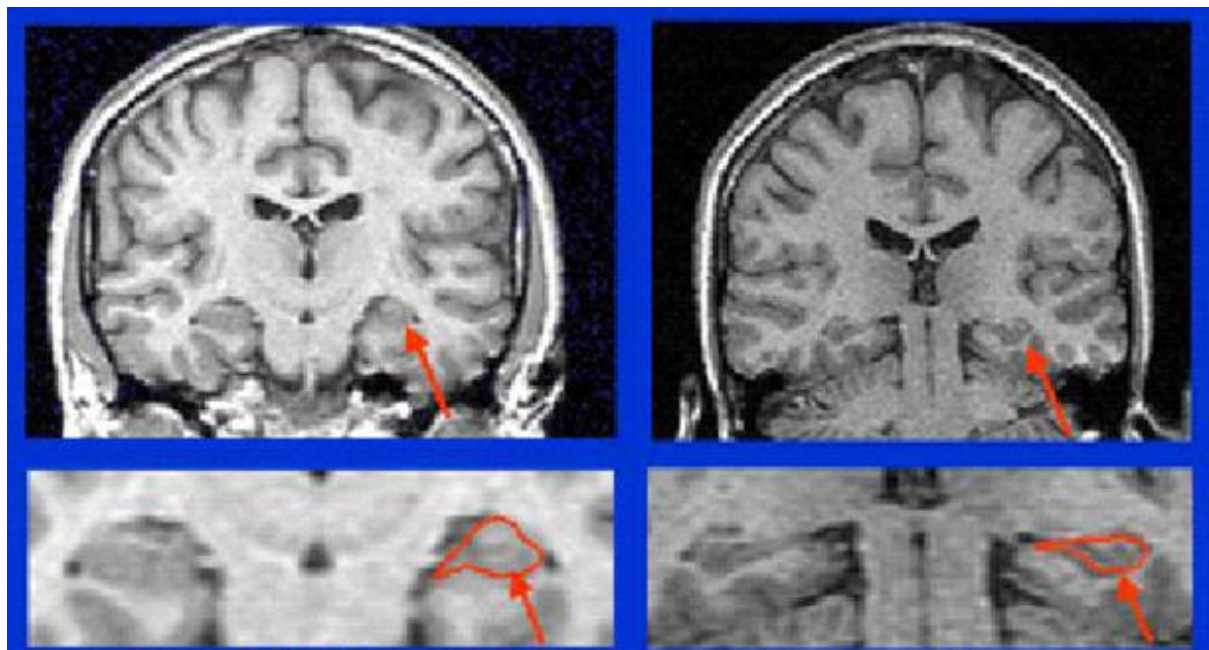
response is very limited in space and intensity. When you are shown the same picture, but you have watched the sad movie, you can see that the response is much larger within the amygdala and is much intense as compared to the other case.

15 minutes can determine a difference in brain response to the identical sad stimuli.

The take on message is that the way we perceive and react to external stimuli is not always the same, it can even change during the day.

The main risk factor for having a depressing episode is having the depressing episode itself.

We have seen over the last couple of decades is that depression is accompanied by alterations in the brain, specifically the atrophy of the hippocampus (is shrinked) in depressed people.



In brain disorders you have atrophy because you lose synapses. Treatment of depression, for example, are associated with a disappearance of the shrinkage of the hippocampus, it's a reversible condition, and can be reversed not only by using drugs or therapies, but also by using meditation. There are better results when you do together psychic therapy and drugs.

If you practice meditation, you can have changes in brain structure.

Behaviour

How do we make a decision?

We can follow an instinct, an emotion, or we can evaluate and modulate that instinctual response by using reason.

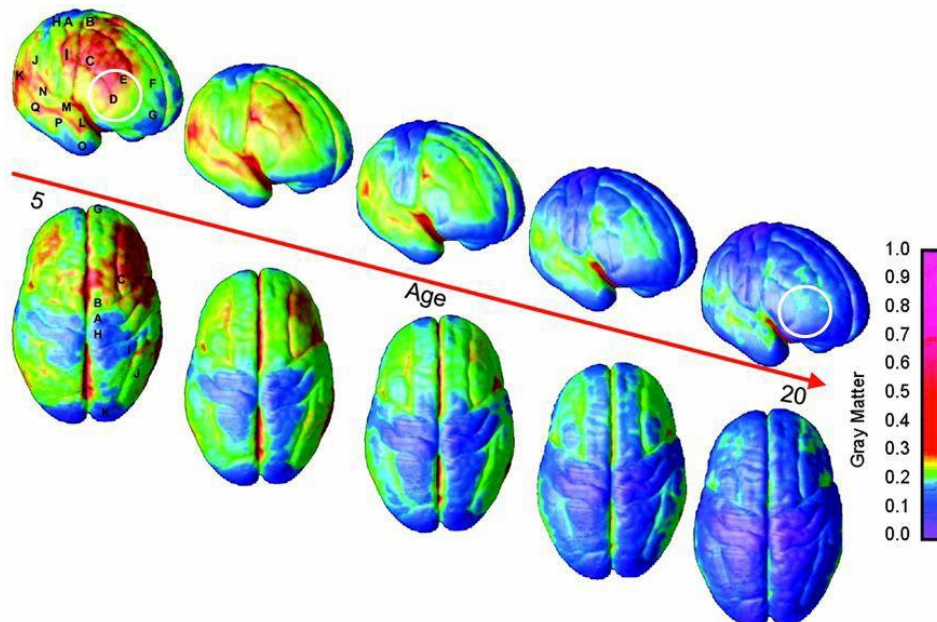
The bear doesn't eat the whole fish, he eats the skin only, because he is in search of the fat beneath the skin that he needs brown fat (it sustains metabolism for months) before hibernating. What he does is not a consequence of reasoning, he simply follows an instinct, an impulse that is genetically rooted into the bear habits.

But this doesn't mean that animals don't have reasoning, they have the ability to modulate the instinct.

How different can we be from bears?

We prefer calories rich food as compared to unfatty food because it's written in our genetic code.

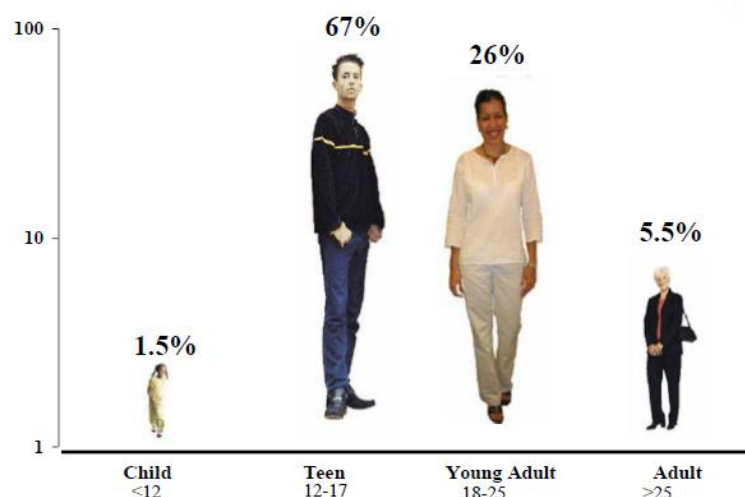
When we have genetically the right imprinted behaviour, we can control it better because of reason. Pre-frontal cortex has evolved in humans much more than in animals, and the pre-frontal cortex is the part of the brain, which is devoted to control behaviour, plan our action, decide what is good and bad, control our impulses. It is also true that pre-frontal cortex matures later in life.



We can see here, in colour scale, that the blue colour indicates the areas that have reached full maturation. At the age of five the primary visual cortex and the sensory and motor cortex are already very mature, the posterior part of the brain also matures very fast, while the anterior part of the brain doesn't mature completely until the third decade of life.

So, our pre-frontal cortex is much slower in getting mature as compared to other parts of the brain, specifically to what we call the ancient part of the brain (limbic system), in which we process emotions, pleasure, sensation. Basically, there is a temporal shift between the

Dependence disorders begins in Adolescence



maturation of the ancient brain and the pre-frontal cortex, where we have behavioural control and decision-making. Adolescence is a period in which brain maturation happens; we still do stupid things.

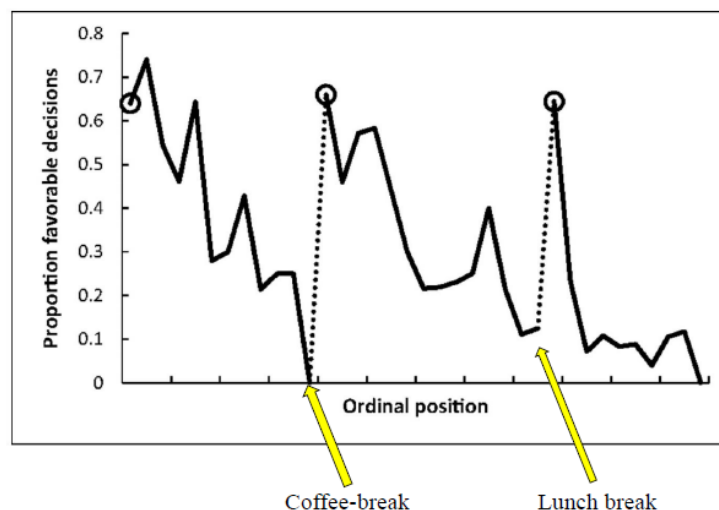
How much are we really in control of our behaviour?

A study tried to see if we can suppress spontaneous emotional responses. We are able to do it because there is a network in the brain which regulates emotional response that includes pre-frontal cortex.

In a study it was asked the subject to keep a neutral facial expression while they were shown a funny movie. What the researcher did was to provide all the subjects an actigraph the night before the experiment, that measures how long and how well you sleep. She tried to verify that the ability to suppress emotion (emotion suppression failures) is inversely related to sleep time. And she found out that the ability to control emotions is impaired when we are tired.

Does fatigue affect the decision-making process?

In a study, the researchers examined 1,112 judicial rulings by eight Jewish-Israeli judges and recorded what they have eaten during the day. For the results, the probability of getting a positive answer to the instance requested is very high in the morning, goes down processively, and reaches zero before the coffee-break. After that there is exactly the same cycle.



What happens in the brain when we are fatigued?

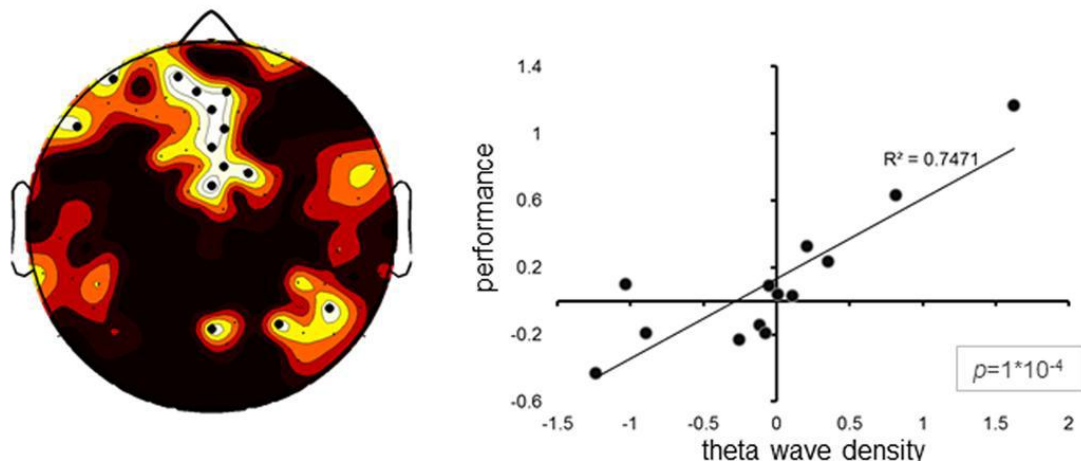
We naturally consider wakefulness and sleep as to mutually exclusive states, but this is not true. There is an intermediate condition between these two, that is called local sleep, which is a condition that happens while the individual is awake, but some regions of the brain go to sleep.

It was discovered that if you put a rat to get some food, at a certain point you can see that the rapid EEG rhythm changes at a certain point, becoming slower. They are phenotypically awake, but they can't reach the food. In coincidence with this missing there is a slow wave activity, not generalized all over the brain, but within the pre-frontal cortex, the area of the brain used the most.

Another study was devoted to see what happens when you do behavioural controlled tasks for a long time.

	Response	No Response
GO	Correct Response (CR)	Lapse (RT>500ms) (LA)
NoGO	Commission Error (CE)	Correct Withhold (CW)

What the researcher did was to get healthy young subjects and putting on them an EEG cap, making them perform the task for 24 hours. What was found is that if the subject presented doing the task significant episodes of slow wave sleep like EEG in the pre-frontal cortex, there was a linear correlation between the number of theta waves and the mistakes done in the task.



A higher number of EEG low-frequency THETA waves was associated with a relative impairment in impulse control.

But returning to the previous study which we talked about (the girl one), in that it was also done an EEG study, and what it was shown that the exact same thing; when the subject failed to suppress the emotional response, they had slow wave episodes within the pre-frontal cortex, Failure to control emotional responses was associated to onset of EEG sleep-like activity in the parieto-temporal frontomedial insula network.

Another study wanted instead to see how much we can control our instinct when we are tired. In what extend brain fatigue may affect social behaviour?

In the Ordini study they had 2 groups of study, doing economic games*, and one of the two went for a series of cognitively tiring tasks. Both then went to do the game, and they all had EEG recording on their head. You see what you would expect, the ones which were tired had a lot of slow wave activity in the pre-frontal cortex, but what is very interesting is how they performed in the game, one which was designed in a way that if you persist choosing the aggressive solution on the long run you lose money and resources. People who performed at this game without being fatigued, choose 86% of the time the pacific solution, on the contrary people cognitively engaged in a massive way for an hour, would choose the pacific solution only 41% of the time. So, if you

make the choice while you are a little tired (1 hour is not so much), this does affect the choice we make, we go towards an aggressive choice even if it is counterproductive.

Hawk-Dove Model: Costs and Benefits of Fighting over Resources

Payoff* to...	...in fights against:	
	hawk	dove
hawk	Hawk wins 50% of fights; is injured in 50% of fights. Payoff: $(V-D)/2$	Hawk always wins; dove flees. Payoff: V
dove	Dove never wins; is never injured. Payoff: 0	Dove wins 50% of fights; is never injured; wastes time. Payoff: $V/2-T$

*V = fitness value of winning resources in fight
D = fitness costs of injury
T = fitness costs of wasting time

*Games which have been used to understand how people behave in social interactions, for example the ultimatum game (there are two players, and you have the opportunity to split or accept an amount of money, etc..). If they offer to split in 9 and 1, from a rational point of view saying no to this is not justified, because you both lose money. What makes you answer in this way is the unfairness you perceive in the personal decision.

Implications for everyday social life and work

- Brain Fatigue appears after a relative short time of cognitive engagement
- Brain Fatigue affects task-related areas and impairs performance
- Prolonged self-control leads to Frontal Fatigue
- Frontal fatigue favors aggressive behavior in social interactions
- Overall, these findings carry implications for scheduling work meetings, school teaching or judicial mediations
- Science appears to provide support to popular wisdom says like “
 - *sleep on it – dormici sopra*
 - *the night brings counsel – la notte porta consiglio*
 - *The early bird catches the worm - la mattina ha l'oro in bocca*

This has implications for everyday life, and also for forensic psychiatric and the law. How do we control impulse and aggressive behaviour?

Mental capacity for the law. In order for someone to behave in a responsible way you need to have:

- Capacity to understand: The ability to understand the nature of one own's actions and their consequences.

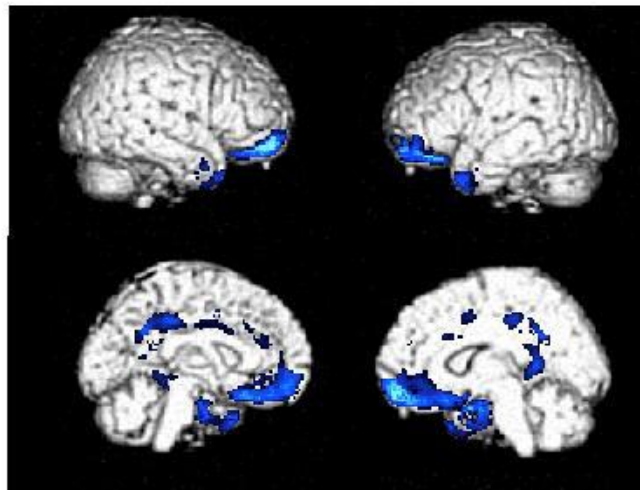
- Capacity to will: The ability to control one own's behaviour.

If either one of these capacities are lost, you are not legally responsible.

Asking individuals to imagine becoming aggressive, you have the same brain response as if you are really being it. In the ventromedial pre-frontal cortex, you have a functional suppression (reduction in blood flow) which accompanies the elicitation at the imagining level of the aggressive behaviour. Even imagining becoming aggressive there is a shutdown of the pre-frontal cortex, which should normally control the release of aggressive behaviour.

Years later, a study demonstrated that if you compare psychopathic criminals with a controlled group in terms of brain structure (neuronal density), you can see two things in the differential graph:

- Differences are not really across the whole brain, is essentially identical with the exception of the ventromedial pre-frontal cortex, in which psychopathic criminals have a fewer number of neurons.
- There is a shut down in the ventromedial cortex.



Light blue indicates the functional shut down, the darker blue indicates a structural difference, with a reduction in the number of neurons.